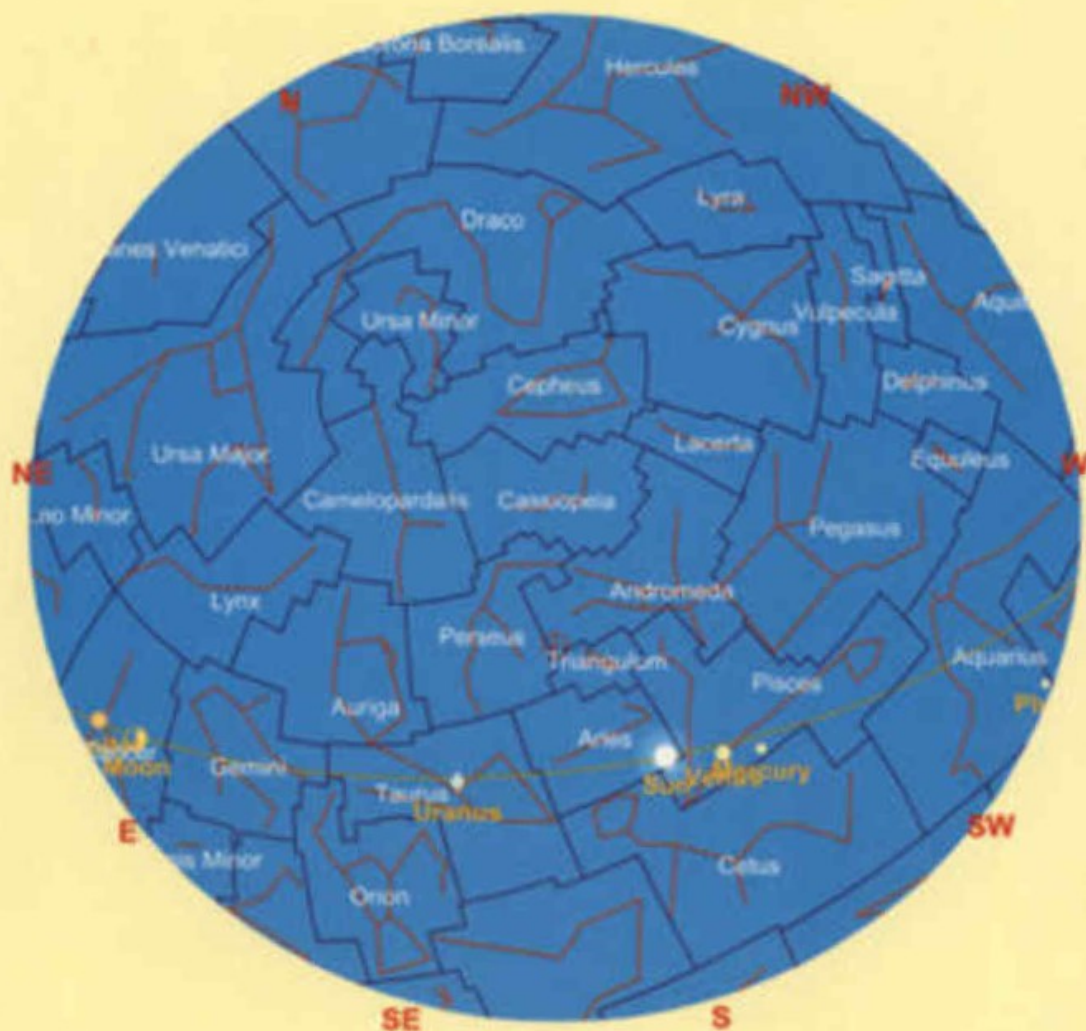


Dating the Era of Lord Ram

*Discover the actual dates of the
lifetime of Lord Ram*



Pushkar Bhatnagar



DATING THE ERA OF

LORD RAM



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LORD RAM

Pushkar Bhatnagar

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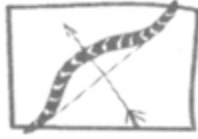
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Preface

I REMEMBER THE month of October 1991 when I was diagnosed with kidney cancer. Within a week, my ailing right kidney was removed under a major surgery. Contrary to normal expectations, that was just the beginning of the agony. Nobody could promise even five years of life to me. Doctors wrote me off on the basis of the biopsy report, which showed that I suffered from a very fast spreading tumor. In those painful and excruciating times, I took refuge at the benign feet of Lord Ram.

Ever since my childhood, I had a lot of interest in the life of Lord Ram. I used to read the *Ramcharitmanas* with great devotion and rapt attention. The more I read, the more my curiosity grew about Lord Ram's actual life. The turmoil of 1991 in my life brought me closer to him. Time passed at its own slow pace. In 1993, one of my colleagues presented me with a copy of a Hindi translation of the Ramayana written by Sage Valmiki. I would, perhaps, never be able to thank that friend enough for that invaluable gift.

Repeated readings of the Ramayana led to the creation of a highly realistic picture of Lord Ram in my mind. The reason was that the book provides the minutest of details about his personality, physique, conversations, surroundings etc. At the same time, another passion of my life - understanding more about astronomy, got a big impetus because of the advent of Internet in India. I kept browsing the web to learn more and more about the celestial phenomenon. Soon I downloaded a number of software that were capable of displaying the positions of planets and stars on the monitor of my PC at the time of making the observation.

A broader knowledge of astronomy and finer readings of the Ramayana soon made me notice that the text of the book is replete with a number of astronomical references, which were comprehensive enough to suggest the exact date when they were actually seen in the sky. I started searching the

software that was capable of generating the sky view on the computer monitor for any given point of time in history. My hunt ended with the purchase of the Planetarium Gold software.

By the year 1999, I had arrived at the date of the birth of Lord Ram and I was thrilled to find that every detail of the planetary configuration given by Valmiki in this regard tallied with the view of the sky generated for that date by the software. The puzzle of finding the nakshatra of King Dashrath took another six months and at least one fresh reading of the Ramayana. The confirmation of the occurrence of the solar eclipse described in the Ramayana to mark the battle between Lord Ram and Khar made me ecstatic. It proved, independently, the accuracy of the findings. Despite being certain that I had discovered the precise dates for the actual lifetime of Lord Ram, I refrained from writing any book on the findings as the motive behind this investigation was contentment and self-satisfaction rather than publicity or business. However, after about three years, the interest generated by a few articles on this subject in some newspapers, persuaded me to get the matter published with a view to share my findings with all those who are interested in it.

Every finding discussed in the book has been arrived at after putting in a lot of effort and spending considerable time. On a number of occasions, I remained stuck for long periods and the solution was nowhere in sight. However, every time intuition, sixth sense or as I believe, divine help, prompted the answer in my mind. I would like to describe one such instance in some detail.

The Ramayana clearly mentions that a lunar eclipse had occurred on the day Hanuman visited Lanka in search of Sita. On examination of the sequence of events narrated in the book, I had concluded that it was the full moon day of either the month of Margashirsha or Paush of the last year of exile (5076 BC). However, I had no means to verify whether a lunar eclipse had actually occurred about 7,000 years or not. My belief and understanding was that the software I was using did not have the capability of showing the occurrence of a lunar eclipse. I browsed the web for any other software, searched the data banks and even enquired from some experts whether there was any algebraic equation or mathematical formula

by which the occurrence of a lunar eclipse, about 7,000 years ago, could be ascertained. These efforts bore no fruit. For almost five years I had desperately wished and prayed that somehow I could verify and demonstrate the occurrence of a lunar eclipse so that my findings become complete in all respects.

Since no solution could be found to this puzzle and being completely helpless in this regard, I submitted the manuscript to my publishers for publication in the month of October 2003. It contained a paragraph mentioning that though it was clear that a lunar eclipse took place on the evening when Hanuman visited Lanka and met Sita, there were no means available to prove or demonstrate that such an eclipse had actually occurred. I also mentioned that as per the sequence of events, the lunar eclipse should have occurred on the full moon day of either the lunar month of Margshirsh or Paush of that year.

After a few days, I read in the newspapers that a lunar eclipse was to occur on the morning of 9 November 2003 and it would be visible from India. Late in the evening of the 8th, without any specific hope of finding a solution, intuitively I felt like examining the position of the sun and the moon during the eclipse due to occur the next day, through the software. My intention was to compare the positions of the sun and the moon during the eclipse on 9 November 2003, with those on the probable dates on which I believed that the lunar eclipse had occurred when Hanuman visited Sita. I thought of reaching a conclusion directly or indirectly in this manner.

However, as soon as I entered the time of the occurrence of the lunar eclipse on 9 November 2003 in the software (since it was known through the newspaper), I was surprised to find that the software was displaying the occurrence of the lunar eclipse very precisely. Without wasting a moment, I entered the date 12 September 5076 BC, (full-moon day of lunar month of Margshirsh), which according to me, was the day Hanuman visited Lanka and the day on which a lunar eclipse must have occurred. With a pounding heart, I started examining the position of the Moon. And then came the stamp of divine confirmation! The software demonstrated that a lunar eclipse had indeed occurred that evening - exactly at the time when it has been described in the Ramayana. I could not believe that a five-year-long

hopelessness was transformed into triumph in just a matter of five minutes. Filled with a sense of blissful delight and gratitude, I bowed before the picture of Lord Ram. This issue had been a quagmire to me for years. Resigned to the fate, I had submitted the manuscript fully aware of this deficiency. If the lunar eclipse had not occurred on 9 November and if someone had not prompted me to examine the position of the sun and the moon on that day, the study would have contained a grave ambiguity forever. This, I believe, is the difference between "our efforts" of five years and "God's wish" of five minutes. Needless to mention, I revised the manuscript by adding a fresh chapter and resubmitted it for publication.

On the completion of this study, I look back at the fading memories of the trauma and despair I suffered about fifteen years ago. What can be a better example of the benign benevolence of Lord Ram? Many well wishers have pointed out that I could do the astronomical dating of Ramayana because of the unique combination of interests I nurture -an incessant quest about the life of Lord Ram and an understanding of astronomy. I, however, believe that I could do it after surviving a bout with cancer because Lord Ram desired me to accomplish this task. Forever indebted, I place this book at the feet of Lord Ram, without whose wish I would not have seen the light of this day.

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10 May 2004



CHAPTER I

Introduction

LORD RAM is aptly worshipped as a legend - timeless, ageless, endless and eternal. Ever since his times, eras have come and gone, epochs have begun and ended, empires have been built and destroyed, yet Ram continues to kindle the lives of millions. No evidence is needed to prove the historical existence of someone who survives in the hearts of masses, but in this age of reasoning and arguments, mere belief does not command any great respect.

Till now, the archaeological findings are considered to be the most convincing evidence in support of any historical fact. It is, perhaps, justified on the grounds that an artifact or a monument of archaeological value is something factually existent, real and available for physical verification. On the contrary, the archaeological evidences suffer from a big shortcoming - the approximations. a broken wall or remains of a building or a dancing figurine can at best be dated to the nearest century. It lacks the preciseness one expects in terms of the year or month or the date. Imagine, even with the best of equipment and technologies available today, if archeology cannot conclusively prove whether a temple existed at Ayodhya or a mosque, just 500 years ago, what help can it render in establishing the historical events, which happened more than 7,000 years ago! The time has come to deliberate whether it is reasonable to be confined to only these kinds of evidences for the purposes of proving the antiquity of an event and construct history. Is it justified to dismiss the historical existence of Ram because of the absence of such 'archaeological evidences'? If the masons of the times of Lord Ram failed to construct such buildings, which could

survive seven millennia, an inference should not be drawn that this country had no inhabitants at that time or Lord Ram was just a myth!

The progress and advancement of science is opening new vistas of exploration everyday. What looked like an impossible calculation, just two decades ago, can now be accurately done in seconds with the help of computers. This study employs the most unconventional, yet most scientific method to fix the dates for the lifetime of Lord Ram by picking up the explicit and unambiguous evidences available in the Ramayana written by his contemporary Maharishi Valmiki. Like any archaeological discovery, the eternal existence and cyclical movement of the astronomical bodies in the sky is a matter of factual existence. Their movements have been unswerving, unchanging and consistent for millions of years. Such systematic and consistent cyclical movements permit us to ascertain when exactly any particular planetary configuration was in existence in history. The biggest hesitation in accepting the dates found on the basis of astronomical dating, so far, was the inaccuracy involved in manually computing the positions of the planets in the historical past. The inaccuracy was inevitable because of the highly complex nature of calculations involved in the process and thus, the dates worked out remained somewhat speculative in nature. However, the advancement of computer technology of the recent past has not only made it possible, but also very convenient, to make such intricate and complicated computations by merely pressing a key on the keyboard. This study uses one of the latest computer software, developed in the western world, which can produce the sky view on the desktop for any given date and time in the history of man. This software is used to find the date and time of the occurrence of those planetary configurations, which are recorded by Valmiki to have occurred at the time of happening of certain specific events during the life of Lord Ram. The most important virtue of the entire study is the objectivity of the investigation and the results. Since the text of Valmiki Ramayana is easily available and the computer software is also available freely, at least in the western countries, the manipulation of any of these two independent objects is just not possible.

The results of the study are, to say the least, startling. Leave aside the series of planetary configurations that tally exactly the way they are

described in the Ramayana, the most amazing finding is the tallying of three eclipses - two solar eclipses and one lunar eclipse that Sage Valmiki has stated to have occurred within a span of just eleven months during the thirteenth and fourteenth year of the exile period of Lord Ram. The software demonstrates that all the eclipses had actually taken place, precisely on the days specified in the book. Not only the eclipses per se, their timings in terms of the part of the day also match with the descriptions. In a remarkable finding, the sky view generated by the software shows that during the occurrence of the first solar eclipse, even the positions of seven visible planets tally exactly with their description in the book. In unequivocal terms, it can be mentioned that reverse calculations of such eclipses is impossible, even by the scientists of today, without the help of computers, particularly when the period involved is about 7,000 years old. Thus, the only inference one can draw is that *someone was actually present to witness the actual happening of these events* (eclipses, and the planetary configurations) and these got recorded in the story of Lord Ram. Such confirmations independently demonstrate the authenticity of the recorded observations, antiquity of those events and accuracy of the findings of this study. What else remains to be proved to establish and demonstrate the existence and antiquity of Lord Ram as well as the Indian civilization, is difficult to argue. The dating of events has been provided in terms of day, month and year, if not hours, in all the results. Every finding of this book is illustrated with the help of slides generated on the computer, so that the readers can see for themselves the position of planetary bodies in the sky and match them with the verses in the Ramayana.

The pioneering method of employing computer software to date the history on the basis of astronomical references should throw fresh ideas and occasions to discover the precise dates and time of composition of the vast ancient literature of our civilization, which contain plenty of such planetary observations. No doubt that the Valmiki Ramayana is an exception, so far as the availability of systematic astronomical evidence in ancient literature is concerned. In other compilations, the references to the astronomical bodies are not so systematic and prolific. In my opinion, this itself is a very strong argument in support of the truthfulness and reliability of the contents of Vedic literature. It indicates that the sages of ancient times were composing hymns with the pure intent of praising and worshipping the Gods rather

than planting some evidences to backdate the literature, as is alleged by some western historians. The astronomical references, which got recorded were purely incidental and ancillary to the main subject matter. One may argue that, if the intention of the authors of these compositions was to leave evidences and back date the literature, nothing prevented them from incorporating false statements about the time of composition of the text. They were also free to incorporate specific, overt but manipulated planetary positions at as many places as they desired in such literature, to give them a back date.

After obtaining the results, it was realized that indeed there were a few archaeological evidences available in the shape of the flowing Saraswati River mentioned in the Ramayana as well as the depth of the 'remains of the bridge' built by Lord Ram under the present sea level. The fact that about 7,000 years ago, the sea level was almost at the level where the 'remains of the bridge' are available today, cannot be discarded as a mere 'coincidence'. While examining the planetary configuration specified to reckon the beginning of the Kali Yuga in the Yuga system, an interesting linkage is found with the dating of the Ramayana. This is discussed at length in a separate chapter. Though the findings or the suggestions made in this book about the counting of Yuga system cannot be the final word on the subject, yet these linkages are worth a consideration. The purpose of highlighting these archaeological evidences and demonstrating the relationship between the dates of the era of Lord Ram with the Yuga system is an attempt to provide the independent interlocking confirmations to the findings of the study. The reader would appreciate that the kind of preciseness and accuracy we obtain through astronomical dating of any historical event cannot, under any circumstances, be obtained through any other method of dating. In fact, looking for such conventional evidences, to support the findings obtained through the astronomical dating, would be as silly as lighting a candle in the day to show the existence of the sun.

This study should supply the answers to all the inquisitive minds who struggle to comprehend whether Lord Ram was just a mythological character or a historical entity. For the followers and devotees of Lord Ram, the findings would only reinforce their ardent belief in the eternal god. For historians and indologists, the findings provide, perhaps, the most

compelling reason, hitherto advanced to rewrite the history of our civilization.





CHAPTER II

The Basic Concepts of Observational Astronomy

It IS A MATTER of common knowledge that stars appear fixed in their places in the sky and they shine due to their own light. Every star has its own planets revolving around it. For example, the sun is a middle-sized star and its nine planets, including the earth, revolve around it. The sun and its nine planets make the solar system. Pluto, the farthest planet of the solar system is about 6,000 million kilometers away from the sun. On the other hand, the nearest star to the sun, Proxima Centauri, is 3,78,43,200 million kilometers away from the sun. This illustrates that in the night sky, the stars, being very, very far away, form the background and the planets appear to be rotating in front of them to any observer on earth.

As the stars are fixed, groups of stars also appear fixed in their positions in the sky. A group of stars is called a constellation and the constellations are named on the basis of the object they resemble. For instance, a group of stars that resemble the figure of a 'sitting lion' is called 'Leo' constellation. So far, eighty-eight such constellations have been identified in the sky and assigned specific names. Generally, the prominent constellations have more than one name since different civilizations have given different names to them. However, with a view to provide uniformity of reference and understanding, scientific names have been assigned to these constellations as well as stars located in them. For instance, the group of stars that we call as 'Saptarishi' is called also termed as 'Big Bear' as well as 'Ursa Major'.

Another important point to understand from observational astronomy's point of view is that although the sun is motionless, it appears to be moving

when observed from the earth. Technically, this is because we make geocentric observations and to us the earth appears to be stationary. Therefore, the relative movement between the earth and the sun gives an impression that the sun is moving. The phenomenon is similar to what we see when travelling in a train. The trees we pass by, though fixed to their place, appear to be moving at a speed equal to that of the train.

We should now understand a bit about the movement of planets around the sun. Out of the nine planets, the farthest three - Uranus, Neptune and Pluto are not observable with the naked eye. The other six planets when named away from the sun are Mercury, Venus, Earth, Mars, Jupiter and Saturn. The earth completes one revolution around the sun in one calendar year. Thus, Mercury and Venus, being nearer to the sun than the earth, complete one revolution in less than one year. On the other hand Mars, Jupiter and Saturn, being farther away from the sun as compared to earth, complete their revolution in more than one year. Saturn (*Shani*) being the farthest - completes one rotation in about 30 years (29.46 years), while Jupiter (*Brihaspati*) does the same in about 12 years (11.86 years). The time period for Mars (*Mangal*) is about 1.88 years while Venus (*Shukra*), takes about 0.615 years for completing a revolution around the sun.

The Astronomical Clock

We all have observed the movement of the sun and the moon in the sky. They appear to move from east to west in a particular path in the sky. The sun rises in the east reaches the highest point in the sky, starts declining and sets in the west. The next day, it rises again in the east in the same place. Not only the sun, the moon also follows the same path in the sky when it rises in the east and sets in the west. Actually, all the five visible planets (Mercury, Venus, Mars, Jupiter and Saturn), the sun and the moon and two imaginary planets called Rahu and Ketu (to be described later) appear to be moving in the same path. This path is called 'ecliptic'. Thus, 'ecliptic' is that imaginary circle around the earth in which all the nine planets (for the sake of simplicity, the sun and the moon will also be termed as planets) appear to be rotating when observed from earth. In geometric terms, it means a circle of 360° around the earth.

The ancient astronomers observed that the position of the planets in the sky keeps changing continuously because of their motion in the ecliptic. If the position of the planets was to be recorded, a reference point was needed in the sky, which should remain fixed in the background. For this purpose, therefore, out of eighty-eight constellations spread all over the sky, twelve constellations were chosen in such a manner that they formed the background of the ecliptic and were located at almost equal distance from one another. Each constellation therefore, occupies about 30° of the background. These 12 constellations are what we call the zodiac signs viz. Libra, Cancer, Leo, Pisces and Gemini etc. Since these constellations formed the background of the path of the planets, it was easily possible to locate and record the position of the nine planets within these 12 zodiacs. This served the purpose of a clock at a time when calendars and watches were not in vogue.

If we compare the astronomical clock of the sky with our present day wall clock, the 12 zodiac constellations would correspond to the numbers 1 to 12 written on the dial and forming the background. The moving planets would correspond to the minute and hour hands of the clock. The difference being that the present day normal clock has only two / three hands but the astronomical clock had nine hands.

Slides 1 and 2 are to acquaint the readers with the 12 zodiac constellations. Slide 3 shows the planets of our solar system.

We do not know whether the ancient astronomers mentioned the position of different astral bodies in the sky, with an intention of recording the time of happening of an event or such planetary positions got recorded incidentally in the text. However, what is important to understand is that if the position of all the important planets is described in a comprehensive manner, then it can refer to one and only point of time. Such comprehensive planetary configurations are not repeated for tens of thousands of years. In such cases, these planetary configurations can be used as an evidence to arrive at the specific date they refer to. Let us understand why a particular planetary combination does not repeat for tens of thousands of years.

Consider that there is only one planet in the sky - Saturn. It completes the full circle of ecliptic in 30 years. Thus, if it is seen in the zodiac of Libra

in AD 2003, it would repeat its position in Libra again after every 30 years or AD 2033, AD 2063, AD 2093 and so on. Now increase the number of planets to two by adding Jupiter, which completes the cycle in 12 years. If Jupiter is seen in Cancer today, it will be seen again in Cancer in AD 2015, AD 2027, AD 2039, AD 2051, AD 2063 and so on. Thus, if these two planets were to repeat their respective positions of the year AD 2003, it will recur only after 60 years or in AD 2063. This is due to the reason that 60 is the common multiple for 30 as well as 12. In 60 years, Saturn will complete two full rounds and come back to the original position while in the same time, Jupiter will make full five rounds and come back to the same position. Hence, although Saturn and Jupiter complete their revolution in 30 years and 12 years respectively, together they will repeat their positions after 60 years.

Now, let us take Mars also in consideration, which completes the circle in 1.88 years. If Mars is in Gemini in AD 2003, in AD 2063 when Saturn and Jupiter return to their respective positions, Mars will not be in Gemini. Thus, we have no option but to move ahead by another 60 years or to AD 2123 and see the location of Mars. Even after 120 years, Mars is out of Gemini. The faster the movement of a planet, the shorter is its stay in a particular constellation and thus rarer becomes the combination. So we move further by another 60 years and see where Mars is. In this manner, the time period goes on increasing.

Bring in Venus that completes its cycle in only 0.615 years. Every condition will add to the interval of time when the combination of planets will repeat itself. And finally, consider the sun which appears to be completing a cycle in about 25,920 years. This is called the precession cycle (described later) and because of the combination of these periodic movements of all the planets, any specific planetary configuration does not repeat for tens of thousands of years.

The Precession Cycle

Actually, after 365.25 days when we count the completion of one calendar year, the earth does not come back exactly to the previous year's position in its elliptical path around the sun. It means that the earth does not cover the

entire 360° of rotation in this time period. Actually, it takes 365 days, five hours, 48 minutes and 46 seconds to complete one full circle. Therefore, when we reckon the completion of one year, the earth falls back in its path by about 0.0136° . In this manner, after our 100 years, the earth would be behind its original position by about 1.36° and in 1,000 years by about 13.6° . Since one zodiac sign refers to 30° , it implies that in about 2,200 years the earth would fall backwards by about one zodiac constellation. Since the observer is located on earth, the sun would appear to be moving backwards due to this precession cycle at the rate of about one full zodiac sign in about 2,200 years. Thus, due to the precession cycle, the sun would repeat its position in the same constellation by completing the 360° of the circle, in about 25,920 years. This is the slowest cycle observable with the naked eye. To illustrate this, we may note that in AD 2001 in the month of January (or say on 1st January) the sun is seen in Sagittarius. It would be in Sagittarius on ever 1st January of 2002 or 2005 or even AD 2100. However, after a span of about 500 years or so it would appear to be shifting out of Sagittarius slowly - due to precession. And if we see its position on 1st January 4200, after 2,200 years it would have moved to Scorpio, the next zodiac constellation located about 30° away. Thus, on 1st January, the sun will come back to Sagittarius again in about AD 28,000 after it leaves this place currently. Else, we can conclude that the last time when the sun was located in Sagittarius on 1st January (before the present cycle) was in about 24,000 BC.

It should therefore be clear that since all these nine heavenly bodies move at different speeds within the ecliptic, any particular position of these 9 bodies in reference to the fixed zodiacs would not repeat for tens of thousands of years again.

Thus, the sky presented to the ancient astronomers an ever-changing picture in which the movement of all the bodies was definite and precise. Similar to our present day clock in which it is certain that the hour hand would move from number 1 to 2 in one hour precisely, the astronomers also knew that Jupiter will move from one zodiac to another in one year and will complete its round of 12 zodiacs in 12 years. We should also note that Jyotish was one of the main subjects to be learnt during the ancient times. The ancient astronomers, sages and seers, therefore, were well aware of the

cyclical movements of the astronomical bodies and were fascinated by the same. Therefore, it is not surprising that their observations of the astronomical bodies got recorded directly or in the form of an allegory while describing the happening of an important event like the birth of a child or a war.

The Concept of Nakshatras

The fastest mover in the sky is the moon. It completes the round of the ecliptic in about 29 days. In other words, it travels about 12.4° in the sky everyday. Since one zodiac occupies about 30° of space in sky, the moon stays in every zodiac for more than two days and actually for these two days the planetary configurations would be the same because there was no change in the position of planets between these two days. If no distinguishing point was available, the so called uniqueness of the planetary configuration for each day would have disappeared. Therefore, with a view to make the astronomical observations more accurate and precise, the numbers on the dial of the astronomical clock were increased from 12 to 27. The background sky of the ecliptic was divided into 27 equal divisions in such a manner that every day the moon moved roughly from one division to another. This was done by identifying 27 prominent and bright stars located equidistantly from one another at about 13.33° in the sky. The moon, which moves about 12.4° per day remained roughly closer to one of these stars everyday and therefore it was possible to distinguish its position from one day to another despite remaining in the same zodiac constellation. These 27 stars were called nakshatras or lunar mansions or houses of the moon in which it stays for one day. These 27 stars were located within the 12 zodiac constellations, and therefore, within one zodiac there were two or sometimes three nakshatras. The easily visible nakshatras, not all 27, are displayed in slides 4 and 5.

The Importance of Nakshatra in Astronomy

The identification of 27 stars as nakshatras gave birth to the lunar calendar in which each day was named after one nakshatra - being the star near which the moon would be located. Of the 27 days, one day would be the full-moon day. That nakshatra, near which the full moon occurred, gave the

name to the month of the lunar calendar. For example, if on the full moon day, the moon was near the star called 'Ashwini', that day will reckon the end of the month of 'Ashwin'. Similarly, the month of 'Vaishakh' would complete on the full moon night in which the moon is near the star called 'Vishakha'.

Since the occurrence of a full moon indicated the completion of a lunar month, a full moon day is traditionally called 'Purna-masi' meaning completion of the month. In the entire lunar year, there will be 12 full moon days (barring Adhik maas or the intercalary month, which is added after even two and a half years, discussed later). On these 12 full moon days, the moon will always be near 12 pre-identified stars or nakshatras. Thus, the 12 months of the lunar year are named after these nakshatras and the names of the months are like - Chaitra, Vaishakh, Jyestha, etc.

The system of nakshatras is not unique to Indian astronomy alone. Same stars have been identified in other ancient civilizations, particularly in the Arab, and assigned specific names. The famous 'Rohini' nakshatra is called 'Aldebaran' by the Arab. For the purposes of uniformity, a scientific name is also assigned to all of them. Table - 1 gives the name of the nakshatras and their corresponding scientific / modern names. This table also provides the names of the 12 lunar months and the corresponding 12 nakshatras near which the full moon would be located.

Table 1

The Nakshatras - Stars in the 12 Zodiac Constellations

S.No.	Name of the Nakshatra (Star)	Modern name	Other name	Zodiac (Constellation)	Name of the lunar month
1.	Ashwini	13 α Aries	Hamal	Aries	Ashwin
2.	Bharani	41 Aries		Aries	
3.	Krittika	25 η Tau	Alycone	Taurus	Kartik
4.	Rohini	87 α Tau	Aldebaran	Taurus	
5.	Mrigashira	112 β Tau	Alnath	Taurus	Margshirsha
6.	Aardhra	24 γ Gem	Alhena	Gemini	
7.	Punarvasu	78 β Gem	Polloux	Gemini	
8.	Pushya	17 β Cnc	Al Tarf	Cancer	Paush
9.	Aashlesha	47 δ Cnc	AsselusAustralis	Cancer	
10.	Maghaa	32 α Leo	Regulus	Leo	Maagh
11.	Porva Phalguni	70 θ Leo	Chort	Leo	
12.	Uttara Phalguni	5 β Vir	Zavijava	Virgo	Phalgun
13.	Hastaa	29 γ Vir	Porrma	Virgo	
14.	Chitra	67 α Vir	Spica	Virgo	Chaitra
15.	Swati	99 ι Vir	Syrma	Virgo	
16.	Vishaka	92 α 2 Lib	Zubenelgenubi	Libra	Vaishakh
17.	Anooradha	7 δ Sco	Dschubba	Scorpius	
18.	Jyesta	21 α Sco	Antares	Scorpius	Jyestha
19.	Moola	42 θ OPH		Sagittarius	
20.	Poorvashada	34 σ SGR	Nunki	Sagittarius	Ashad
21.	Uttarashada	5 α Capr	Giedi	Capricornus	
22.	Shravana	49 δ Capr	Deneb Algeidi	Capricornus	Shravan
23.	Dhanishta	5 ζ 1 Aqr		Aquarius	
24.	Shatabhisha	91 ϕ Aqr		Aquarius	
25.	Poorva Bhadra	28 ω Psci		Pisces	Bhadrapad
26.	Uttara Bhadra	63 δ Psci		Pisces	
27.	Revathi	Zeta Psci		Pisces	

The Concept of the Intercalary Month

In the sky, the moon completes one month in about 29 days and appears to move from one zodiac constellation to another. On the other hand, the sun also moves from one zodiac to another in about 30 days to complete a solar month, as we know of it. A lunar month (or synodic month since it is also relative to the sun) is precisely 29 days, 12 hours, 44 minutes and 3 seconds long. Twelve such months constitute a lunar year of 354 days, 8 hours, 48 minutes and 36 seconds. The solar year, on the other hand, is of about 365.25 days. Thus, after about two and a half-years the lunar year falls behind the solar year by about one full month. To help the lunar months coincide with the solar year, the practice of inserting an intercalary (extra)

month or Adhik-maas was followed by the astronomers. Generally speaking, 60 solar months = 62 lunar months. Therefore, an extra month, called the Adhik Maas, is inserted every 30 solar months, or about every two and a half years. The principle of inserting the intercalary month was also very scientific and it can be observed with the display of the sky by the software used in this study. After about thirty solar months, there will be one solar month in which the moon will appear to move from one zodiac to another while the sun will not move from one zodiac to another. In other words, when the sun remains in one zodiac constellation while the moon completes its one lunar month from full-moon to full-moon, that lunar month is called Adhik-maas.

This practice of adding an intercalary month for synchronisation between the lunar and the solar year was also prevalent in the times of the Rig Veda. At 1.25.8, the Rig Veda speaks of twelve and thirteen months in a year with clear mention of an intercalary month. The adjustments in the lunar and solar months were made and the calendar for the year was made by the astronomers after actually observing the movement of the sun and the moon from one zodiac to another. As a result of the adjustment due to Adhik-maas, even after thousands of years, we have Chaitra as the first month of Vasant Ritu. As against this, the English calendar counts only 365.25 days and marks the end of the year. This leads to the precision of equinoxes as discussed earlier. That is the reason why, some of our festivals like Diwali, Holi etc. come late in certain years while in other years they fall relatively early as per the English calendar.

Rahu and Ketu

Rahu and Ketu are the two imaginary planets and they actually refer to two points where every month, the moon cuts the path of the sun. These two planets are associated with solar and lunar eclipses. Let us understand this concept with the help of the principles of very simple astronomy.

We know that the sun appears to move around the earth in a path called ecliptic. The moon also moves in the same path but oscillates in a sine curve around the ecliptic in a range of about 5° on both sides in a regular manner, meaning thereby that it goes up to 5° on either side of ecliptic and

then comes back to the center. We may draw a simple analogy, if the sun is assumed to be moving in the middle lane of a highway (ecliptic), the moon moves from the left lane to the right lane thereby cutting the middle lane and then again moves from the right lane to the left lane thereby again cutting the middle lane. Thus, it cuts the middle lane (ecliptic and path of the sun) twice in a lunar month. These two points are called nodes and are on the opposite sides of each other, separated by 6 zodiacs or 180° . This process repeats itself every lunar month and because of the motion of the moon and the sun, the points of intersection keep changing every time in a predetermined manner. Astronomers and perhaps, astrologers called these two nodes -Rahu and Ketu. These are the points where the sun and the moon could overlap or remain directly opposite to each other in their paths and thus, only at these places the eclipses could occur. Since these two points were also located on the ecliptic, their position in the sky was also fixed in reference to the 12 zodiacs only. The constellations, in which the moon crossed the path of the sun, were (are) called the places of location of Rahu and Ketu.

Solstices and Equinoxes

Our ancient astronomers have mentioned these concepts in contemporary literature and these references can serve a useful purpose in the dating of such literature. We know that during the summers, in India, the days become long and the nights are short; while during the winters, the phenomenon reverses. Somewhere around March and September, the days and nights are equal. These periodic and cyclical changes take place because of the apparent movement of the sun from the north in the summers to southward direction and from the south in winters to the northward direction. In our traditional terminology, the movement of the sun from the north to southward direction is called Dakshinayan while the reverse movement is called Uttarayan. The southern declination when the sun appears to move between Cancer and Sagittarius is called Dakshinayan. The northern declination of the sun, when it appears to move between the signs of Capricorn and Gemini is called Uttarayan. The sun reaches the extreme north at around 21st June in its cyclical movement and it reaches the extreme south at around 22nd December. These two points are fixed and are called Summer Solstice and Winter Solstice respectively. Solstice means

'stand-still' and at these two points, the sun appears to remain motionless for a few days. Hence, the names - solstice. Similarly, the two equinoxes, vernal and autumnal fall on 21st March and 23rd September respectively and they mark the days when the days and nights are equal. Astronomically, equinoxes occur at the intersection of the celestial equator and the ecliptic.

Suggestions to the Readers for Examining the Slides

Every finding of this book is based on the sky view generated by using the software and is presented to the readers in the printed form as slides. Every slide shows or illustrates the actual planetary configuration present in the sky at a given point of time. Some of the slides appear very complex and baffling because they show plenty of details. However, all those details are not relevant for understanding the points being made in the book. No attempt has been made to hide or conceal the unwanted details in the slides because that would have sacrificed the objectivity of the output of the software and hence, the findings of the book.

Under the circumstances, the readers are advised to focus only on those planetary details that are relevant to the context and such details are described at length below every slide. A thumb rule in this regard is that the focus of attention in almost every slide would be located on the ecliptic - the path of the sun, which is shown by a yellow coloured line. It is recommended that the readers should look at the constellations and the planets located on this yellow line and get conversant with them because in almost every slide, the object under examination would invariably be located around this ecliptic line only. The rest of the details, howsoever complex they may appear in the slide, are irrelevant for the purpose of the astronomical dating.





CHAPTER III

A Brief Description of the Software used for the Dating

Today, MANY SOFTWARES are available under different names and brands, which are capable of generating the picture of the sky for any given point in time on the computer monitors. Some of these, which are generally available as freeware on different websites, are capable of going back only up to 4th millennium BC but these do not serve our purpose because the dates we are searching for go much beyond. A number of software are, though, capable of going back to any date in the history (99,999 BC) as well as the future. This study uses one such powerful software named Planetarium Gold for the dating of Valmiki Ramayana. The findings of the study have been reconfirmed by using at least two such softwares.

The software is easy to use and its essential feature is that for any given point of time, whether in the past or in the future, it generates on the desktop the position of the stars and planets. There is a provision to change and prescribe the exact location, in terms of latitude and longitude, of making the observation. Besides the visual display of the sky, the Planetarium Gold software used in this study, generates coloured printed slides wherein the date, time and place of observation is printed automatically. These prints cannot be altered, modified or tempered with manually by any means. One of the most interesting and helpful features of this software is the capability of showing the phase of the moon on any given date. This feature helps us to ascertain whether the day in question was a no-moon day, a full-moon day, or simply a day of increasing or

decreasing moon phase. Therefore, it becomes easy to find which lunar month was running on the date of observation and the lunar date.

The software provides the option of displaying the scientific names and the conventional names of all the stars and thus helps us in the identification of nakshatras. In this book, generally the scientific names have been used to refer to the nakshatras. These scientific names are mentioned as symbols - α , β , γ , δ , θ etc in the slides. This may be seen in slides 4 and 5. In the text, the reference to the stars or nakshatras are generally made as symbols followed by the name of the constellation in which they are located. In certain cases where the reference is too frequent and it is expected that the readers would be familiar with the star, the reference is made by using its common name. For instance, the Chitra nakshatra or the star Spica in Virgo constellation is the most frequently referred to star in the book. Therefore, after referring to it by the relevant symbol for a few times initially, the later part of the book mentions it as star Spica in Virgo constellation only. Needless to mention, the software displays the sky view after taking into consideration all the complicated cycles like precession of equinoxes and the shift in the position of Pole star due to precession of the earth's polar axis. The preciseness and accuracy of the software got demonstrated during the recent astronomical phenomenon like Mercury's transit of the sun in May 2003, the proximity of earth with Mars around 23 August 2003 etc. The software on the desktop showed precisely the same timings for these events as were observable in the sky.

The software enables one to ascertain the occurrence of solar as well as lunar eclipses very precisely. Not only the occurrence, even the time of the beginning of the eclipse, period of the eclipse, time of maximum overage and the time of the end of the eclipse can also be found out precisely. It was verified during the occurrence of the solar eclipse of 31 May 2003 and the lunar eclipse of 9 November 2003. Since the astronomical dating of the Ramayana includes, inter-alia, tallying the occurrence of three eclipses - two solar eclipses and one lunar eclipse - it may be stated that without these features it was, perhaps, impossible to prove the results of the study so conclusively.

Observation Vs Reverse -Calculation of the Planetary Positions

Some indologists have alleged that the Hindus backdated their so-claimed ancient literature by mentioning therein the reverse-calculated positions of different heavenly bodies. However, it is not so simple to reverse calculate the positions of planets or dates of occurrence of eclipses, 7,000 years in the past. In the same context, Professor N.S. Rajaram has commented that fabricating astronomical data going back thousands of years would need the knowledge of Newton's Law of Gravitation and the ability to solve differential equations. Since such advanced knowledge was not available in the Vedic period, the only possibility remains is that whatever planetary positions got recorded were based on actual observations. The fact that the ancient astronomers of this land had highly advanced and accurate information about the motion of planets and other celestial bodies, gets summed up by the comments of the French astronomer Jean Sylvain Bailly who stated that, "the motions of the stars calculated by the Hindus before some 4,500 years vary not even a single minute from the (modern) tables of Cassini and Meyer";.

Even after acquiring the abilities to solve differential equations and having advanced knowledge of mathematics, it was not possible to work back the planetary positions. If it were so, the accurate astronomical dating would have been already completed. As recent as the early 1990s, attempts were made to find out when the position of the planets described at the time of birth of Lord Ram existed in history, by doing manual calculations. The dates of the era of Lord Ram, arrived at on the basis of such manual calculations by various authors were verified by using this software. The sky view generated by the software for such dates showed clearly that the planets and other astronomical bodies were not located in the places where they were declared to be on the basis of manual computations. Given the complexity of the calculations, this was very much expected. One can, therefore, conclude that only with the advent of computer softwares like Planetarium Gold, has it become possible to recreate the exact view of the sky for any point in time in history and arrive at the most accurate and precise conclusions.





CHAPTER IV

The Relevance of the Valmiki Ramayana in Astronomical Dating

One CAN RAISE the question as to why and how the Valmiki Ramayana is so important from the point of view of dating history. This has been explained in the following pages.

The first and the most important point is that Maharishi Valmiki was a contemporary of Lord Ram. Valmiki's ashram was on the outskirts of the kingdom of Koshal. When Lord Ram proceeded to the forests, he went to Valmiki's ashram, met him and stayed there for some time. Later on, as per the epic, Sita gave birth to her children in the ashram of Valmiki. Maharishi Valmiki had seen Lord Ram in flesh and blood. This is not true for Goswami Tulsidas and that explains the difference in the content of the writings of these two authors. The description of Lord Ram in the Valmiki Ramayana is as real as could be of any prince with exceptional qualities-yet a human being. However, for Goswami Tulsidas, Lord Ram was god right from the beginning of the Ramcharitmanas. The reason lies in the fact that Tulsidas was born thousands of years after Lord Ram and by that time Lord Ram was already worshipped as a god. Contrary to this situation, in the times of Maharishi Valmiki, Lord Ram was adored as a valiant prince and therefore, his description in the Valmiki Ramayana is completely lifelike and realistic. Being a contemporary of Lord Ram, Maharishi Valmiki could leave an abundance of evidence in the shape of the description of planetary positions, as he was present to witness the same. However, such evidence is missing completely in the Ramcharitmanas for obvious reasons.

As we discussed in the previous chapter, the ancient astronomers, sages and seers could accurately identify all the planetary bodies and they had comprehensive knowledge of their cyclical motion. Their obsession and lore for the astronomical bodies ensured that the planetary positions got recorded at several places whenever they composed any literature. In the same fashion, Maharishi Valmiki has mentioned the position of stars and the planets, at various places in the Ramayana at the time of occurrence of certain specific events. For instance, the position of all the visible planets is given for the time of the birth of Lord Ram. Thus, if we have the means to find out when exactly the planets were in those positions, we can find out the date and time of birth of Lord Ram. Going forward, he describes that Lord Ram went to exile on his twenty-fifth birthday and simultaneously, tells the position of the planets in the sky. Thus, there is again a possibility of not only finding out the second date but also verifying the fact that after twenty-five years of the birth of Lord Ram, whether the planets were present in their described positions or not. Here lies the vital difference between Valmiki Ramayana and other ancient literature. What makes the Valmiki Ramayana immensely important from the point of view of astronomical dating is the fact that this book contains a series of planetary references in a sequential manner beginning with the date of birth of Lord Ram. If we fix the date for the birth of Lord Ram, then the planetary position for all the subsequent events should also tally. And if the entire sequence tallies, then there is no need for any further proof or evidence to support the authenticity of the recording, antiquity of the events as well as the accuracy of the findings.

Looking from the critics' point of view, even if the position of all the main planets is described once in a scripture, it can be easily argued that it does not prove anything because anybody can, at any time (even today), suggest / invent such a combination by placing nine planets against 12 zodiacs. After all, such a combination would have definitely existed in the sky at one point of time or the other. Actually, several European scholars have accused the authors of our ancient literature that they worked out the positions of the planets etc. to backdate the Vedas and other ancient scriptures. The obvious question is, how far is this possible? Inventing one set of planetary configurations of a past date is not difficult. But working out an entire sequence of such references dating back to more than 7,000 years

in the past is not possible manually. It is possible only when one uses computers and advanced software. Further, Valmiki has not confined himself only to mentioning the position of the planets. He has described a solar eclipse to mark the date of battle between Lord Ram and Khar. The software demonstrates that a solar eclipse actually took place at the time when exactly it should have occurred as per the author of Ramayana. A second solar eclipse has been mentioned after a period of about six months while describing the killing of Bali at the hands of Lord Ram and this also tallies when examined with the help of the software. Remarkably, even the third eclipse - a lunar eclipse this time, which is mentioned to have occurred about five months later, when Hanuman visited Lanka in search of Sita, also tallies flawlessly. These findings add immense credibility to the entire sequence of astronomical references given in the book because it can be stated without an iota of doubt that reverse calculations to suggest the occurrence of a solar eclipse, 7,000 years ago is not possible manually even by modern scientists.

Unlike Valmiki Ramayana, the other ancient literature, may it be the Rig Veda or Mahabharata, have very few, stray and vague astronomical references from which nothing concrete can be inferred. Such isolated references cannot be accepted as evidences since they would repeat very frequently. For instance, if somewhere it is stated that “at that point of time Saturn was located in Libra”, this does not prove anything because after an interval of every 30 years, Saturn comes back to Libra. Where does this take us? Even if the position of two or three planets is given, no concrete date can be assigned to the event because such positions of three planets will repeat frequently. However, when the positions of all the main planets, including the position of sun is provided, the description refers to a unique combination. Such a description acquires the characteristics of evidence.

Another related question can be as to why no attempts were made so far to work out the dates corresponding to the planetary positions stated in the Valmiki Ramayana? In this regard, it may be stated that some attempts have in fact been made, but these attempts were made by doing manual calculations. These calculations are highly complex because the planets do not complete their cycles in exact multiples of the year we have adopted. If

accurate and precise calculations are to be made, then the time taken by Jupiter to complete one revolution around the sun would have to be taken as 11.863013 years and not 12 years. Similarly, Saturn takes 29.4544832 years per revolution while for Mars, the time per revolution is 1.88089 years. Ascertaining the precise position of these planets in the historical past of 7,000 years, that too manually, is therefore almost impossible. Even if some estimates are made, they are far from being accurate and if they are inaccurate, they are of no value. Thus, the complexity of calculations is the main reason why no successful attempts were made thus far. However, the development and availability of highly precise astronomical software recently has made it possible to work out the precise time of various events recorded by Maharishi Valmiki in the Ramayana. By using such advanced software, scientists are able to track the movement of different astronomical bodies, predict the eclipses of the past and the future, place the different satellites into orbits of the earth etc. This book has also used an advanced planetarium software, to provide the precise dates for the birth and other important events that happened in the lifetime of Lord Ram as described by Maharishi Valmiki.





CHAPTER V

The Astronomical Dating of the Ramayana: the Date of Birth of Lord Ram

BEFORE WE PROCEED with the dating of the events, it is necessary to devise a system of making references to various shlokas of the Valmiki Ramayana. The Ramayana has seven chapters called Kaands. Every Kaand has several Sargas, which are serially numbered. Each Sarga has a number of shlokas, again serially numbered. Therefore, to refer to any particular shloka, the Kaand number, the Sarga number and the shloka number would need to be mentioned. Henceforth, in all discussions, the references to the shlokas will be made by mentioning the Kaand number/ Sarga number/shloka number respectively. For example, the 15th shloka in 32nd Sarga of Ayodhya Kaand (2nd Chapter) would be stated as 2/32/15.

To summarize, the names of the seven Kaands and the respective numbers are as given in Table 2.

Table 2

Number	Name of the Kaand (chapter)
1.	BAAL KAAND
2.	AYODHYA KAAND
3.	AARANYA KAAND
4.	KISHKINDHA KAAND
5.	SUNDAR KAAND
6.	YUUDH KAAND
7.	UTTAR KAAND

The Time of the Composition of the Ramayana by Maharishi Valmiki

At the beginning of the first chapter itself, at 1/4/1 it has been clearly recorded that “after the coronation of Lord Ram as the king of Ayodhya, Maharishi Valmiki composed these details running into 24,000 shlokas, divided into 500 sargas and six kaands”. Another important statement given at 1/3/39 is that “Valmiki composed the Uttar Kaand separately in which he mentioned the events likely to happen in future”.

These two averments are qualitative in nature and imply only one thing that whatever has been stated in the first six chapters of the Ramayana, had already taken place by the time the Ramayana was composed. It also leads to the conclusion that Maharishi Valmiki and Lord Ram, both were living personalities at the time of the composition of the Ramayana.

The fact that Maharishi Valmiki actually decided to write the seventh Kaand after a considerable gap of time from the completion of the first six chapters, becomes evident from a plain reading of the Ramayana. There is a clear discontinuity between the sixth chapter and the seventh chapter of the book. The Yuudh Kaand (sixth chapter) ends in a manner as if the book is concluded finally at that point, with the reinstatement of Lord Ram at

Ayodhya. However, a fresh beginning is made again when the Uttar Kaand (seventh chapter) begins. This discontinuity is not seen when we move from the first chapter to the second, second to the third and so on till the sixth.

The Date of Birth of Lord Ram

The first instance of the recording of the planetary positions in Valmiki Ramayana is obviously for the date of birth of Lord Ram. The birth of the child was long awaited by the entire kingdom of Koshal. Prince Ram and his three younger brothers were born in the royal family. Hence, it is all the more explicable that the time of birth of the princes was recorded meticulously. Finding the date of birth of Lord Ram truthfully and accurately is crucial because this date would serve as the basis for all the subsequent findings of dates.

It has been recorded at 1/18/8, 9 that Prince Ram was born on the Navmi tithi of Shukla Paksha of Chaitra month (ninth day of the increasing phase of the moon in the lunar month of Chaitra). At that time, the nakshatra was Punarvasu, and sun, Mars, Saturn, Jupiter and Venus were in Aries, Capricorn, Libra, Cancer and Pisces respectively. Lagna was Cancer and Jupiter and moon were shining together.

The above mentioned description of the planetary positions on the particular day of the lunar month of Chaitra make a very unique configuration. For the sake of clarity, these conditions can be summarised as follows:

- 1) Sun in Aries;
- 2) Saturn in Libra;
- 3) Jupiter in Cancer;
- 4) Venus in Pisces;
- 5) Mars in Capricorn;
- 6) Lunar month of Chaitra;

- 7) Ninth day after no moon (Navmi tithi, Shukla Paksh);
- 8) Moon near Punarvasu nakshatra (Pollux star in Gemini constellation) and
- 9) Cancer as Lagna or Cancer constellation rising in the east.
- 10) Jupiter above the horizon

Using the software, the date when the five main planets were seen in the stated positions would be ascertained first. Thereafter, the conditions regarding the date, phase of the moon and the lunar month would also be matched with the description given in the Ramayana. The methodology employed in this regard and the important intermediate dates have been discussed and illustrated in the following paragraphs. This will help in two ways - firstly, the reader will get the benefit of understanding and appreciating the process involved; secondly, and more importantly, it will demonstrate the uniqueness of the planetary configuration being explored.

The basic principle to follow in this regard would be to attempt in bringing the slowest moving planet in the described position first. The sun is the slowest moving planet in the sky as was described while discussing the precession cycle. After the sun, Saturn is the next slowest planet completing the cycle in about 30 years followed by Jupiter, which completes the round of ecliptic in about 12 years and so on.

The first observation is that the sun was in Aries. At present (AD 2004), the sun remains in Aries from about 19 April to 13 May. One thing that is certain about the birth of Lord Ram is that he was born before Christ. On this issue, even the European historians do not have any reservations. Therefore, it would be interesting to find the position of the sun around AD 1. The software shows that in AD 1, the sun remained in Aries from 23 March to 15 April. The shift in the position of the sun by about 24-25 days is illustrative of its movement due to precession since there is a gap of about 2,000 years between AD 2004 and AD 1. Therefore, keeping the dates between 23 March and 15 April, which would ensure that the sun remains within the boundaries of Aries, the positions of other planets would be examined in AD 1.

Maharishi Valmiki stated that Saturn was in Libra and Jupiter was in Cancer. The software shows that in the year AD 1, between 23 March and 15 April, Saturn was in Gemini. Since Saturn moves across one zodiac in about two and a half years, two years before AD 1 i.e., in 2 BC it was located in Taurus. Moving backwards in this manner, it is noted that in 18 BC the Saturn was located in Libra. Therefore, between 23 March and 15 April in the year 18 BC, two conditions prescribed by Maharishi Valmiki were fulfilled - the sun was in Aries and Saturn was in Libra.

However, the software shows that in 18 BC the Jupiter was located in Pisces. The description in the Ramayana states that not only was the sun in Aries and Saturn in Libra, Jupiter was also in Cancer. We now move backwards at an interval of about 30 years because every 30 years Saturn would be in Libra and then attempt to find the year in which Jupiter was seen in Cancer. Between 23 March and 15 April in the year 48 BC, Saturn remains in Libra but Jupiter is located in Leo. During the same period in 78 BC, Jupiter was seen in Aquarius. In the year 108 BC, Saturn appears to be located outside the Libra constellation because it completes one revolution in little less than 30 years, while we are following an interval of exactly 30 years. Hence, an adjustment of one year brings back Saturn in Libra in 107 BC. Having brought Saturn back in Libra, let us see where Jupiter is located.

The software shows that in 107 BC, Jupiter was located between Virgo and Leo. In 137 BC, Jupiter was seen in Aquarius. In 167 BC during the above dates, Jupiter was located in Virgo. In 196 BC, Jupiter was again in Pisces. Going backwards in this manner and making suitable adjustments to the dates and years to keep the sun in Aries and Saturn in Libra, the software shows that in the year 666 BC, for the first time, Jupiter was seen in Cancer while Saturn was in Libra. This is shown in slides 6 and 7. To sum up, in the year 666 BC, the sun, Saturn and Jupiter were located in the positions mentioned in the Ramayana on the date of birth of Lord Ram.

Moving one step further, having brought three main planets in the described positions, we would now attempt to bring the other two planets also in the positions mentioned in the Ramayana on the date of birth of Lord Ram. Maharishi Valmiki states that Venus was in Pisces and Mars was

in Capricorn. In 666 BC, the sun was in Aries, Saturn in Libra and Jupiter in Cancer, but the problem is that Venus was located in Gemini and Mars was seen in Sagittarius. As per the Ramayana, at the time of the birth of Lord Ram, Venus and Mars were in Pisces and Capricorn respectively. Thus, we need to move further backwards and examine the planetary configurations in such a manner that the sun, Saturn and Jupiter remain fixed in Aries, Libra and Cancer respectively. The position of the planets is now examined at an interval of about 60 years because Saturn and Jupiter will together repeat their positions after an interval of a little less than 60 years. For keeping the sun in Aries, the dates of 23 March to 15 April are maintained.

The first important date to be noted, while going backwards, is the year 1698 BC, when in addition to the sun, Saturn and Jupiter in Aries, Libra and Cancer respectively, Venus was also located in Pisces as described by Valmiki. In this year, though four planets are in position as described by Valmiki, Mars is not located in Capricorn, it is actually present in Pisces constellation along with Venus. This may be seen in slides 8 and 9. Hence, this planetary position does not match fully with the description given in the Ramayana. Therefore, moving further backwards at an interval of about 60 years, we attempt to find the desired planetary configuration.

At this juncture, however, there is a need to make adjustments to the dates of 23 March and 15 April that are being maintained. Between these dates, the sun was seen in Aries in the year AD 1. However, now the year being explored in the study is 1698 BC. During this gap of about 1,700 years, the sun would appear to have shifted backwards by about one zodiac due to the precession cycle. This is evident from slide 8 that the sun is almost at the edge of the boundary of the Aries constellation for the dates adopted so far. Hence, to bring it well within the boundaries of Aries, the dates are shifted backwards by another 20-25 days. For 1698 BC, the sun remains in Aries during the dates 26 February to 22 March only. Thus, for a further backward journey in time, the dates would be kept between 26 February and 22 March to ensure that the sun was within the boundaries of Aries while examining the position of other planets.

The software shows that the next date where four planets viz. the sun, Saturn, Jupiter and Venus were located in the places mentioned by Maharishi Valmiki was 1 March 2374 BC (although Saturn was slightly out of Libra towards Scorpio). However, Mars was definitely out of place in Leo. Slides 10 and 11 may be referred for confirmation that Mars is not located in the place described by Maharishi Valmiki, while the other four planets are. Therefore, this date cannot be the date of birth of Lord Ram and we need to move further backwards in time and explore other dates.

Around the year 4000 BC, further adjustments are made to the dates of 26 February to 22 March adopted for 1698 BC because due to the gap of more than 2,300 years, the sun has again shifted backwards by another zodiac. Thus, during this period it remains in Aries during 17 January to 11 February. Moving further backwards in time, the software displays that on 18 January 4497 BC, the sun was located almost on the boundary of Aries, Jupiter was just within the boundary of Cancer, Saturn was seen almost on the boundary of Libra, Venus on the boundary of Pisces and Mars, for the first time, was seen within the boundaries of Capricorn. This combination, thus, deserves deeper scrutiny. The positions of these planets may be seen in slides 12 and 13.

The planetary positions seen in the sky broadly match the details given in the Ramayana. However, there are further finer details to be tallied. These planetary positions should have occurred on the navmi tithi of Shukla Paksha of the Chaitra month. In other words, on the ninth day from the no moon day (amavasya) of Chaitra month in that year the above described planetary positions should hold good. To verify this, we must find the date when the lunar month of Chaitra ended that year.

The software shows that on 18 January 4497 BC, the moon is in the decreasing phase (Krishna Paksh). Just six days before this date i.e., on 12 January 4497 BC, a full moon night had occurred, which means that 18 January 4497 BC was the seventh tithi of Krishna Paksha. However, the birth of Lord Ram was on the navmi of Shukla Paksha i.e., increasing phase of moon Hence, with a view to establish the identity of the month of that year, we examine the position of the moon on the previous full moon day.

On 12 January of 4497 BC, a full moon occurred near the star Spica or Chitra Nakshatra as seen in slide 14, and thus it marked the end of the month of Chaitra. It implies that the navmi tithi of Shukla Paksh must have been on 5 January of that year (7th day before full moon). We now examine the position of the planets on this date as shown in slide 15. The slide shows that the sun is out of Aries and is in Pisces. Thus, the date when the planets are in desired positions (18 January 4497 BC), the tithi is not navmi of Shukla Paksh and when Navmi of Shukla Paksh occurred, (5 January 4497 BC), the planets were not in the desired positions. This planetary configuration, therefore, cannot be the one that is mentioned in the Ramayana for the date of birth of Lord Ram. Therefore, with the help of the software, we move further backwards and ascertain the combination when all the conditions were fulfilled together.

The software displays that after a gap of another 700 years or so, in the year 5114 BC, all the five visible planets, surprisingly and convincingly, move into the positions described by Maharishi Valmiki. Are we nearing the destination? Since the software displays that all the five planets were in the desired positions in early January of 5114 BC, it is again crucial to ascertain the precise date of navmi of Shukla Paksha of Chaitra month that year and examine the planetary position on that day.

The first step is to ascertain the end of Chaitra month. The software shows that a full moon occurred on 16 January 5114 BC and on this day, the moon was near the Chitra Nakshatra (star Spica) thus marking the end of Chaitra month. This is illustrated in slide 16.

If 16 January was a full moon day, the navmi of the increasing phase of moon would be on 10 January 5114 BC (7th day before full moon day). Let us generate the sky view for the 10 January 5114 BC at 12.30 p.m. with the help of the software and examine the results. Yes! All the five planets were in place! The sun was seen in Aries, Jupiter in Cancer along with the moon, Venus in Pisces, Saturn in Libra, Mars was well within Capricorn. The place of observation has been kept as Ayodhya by providing the corresponding latitude and longitude. Look at slides 17 and 18 to see the position of all five planets as seen from Ayodhya and note that all the five planets are in the described positions. Slide 19 provides the picture of all

the twelve zodiac constellations and the planets including, Saturn and Mars, which were below the horizon at 12.30 p.m. of that date. Both these planets are in Libra and Capricorn constellations respectively as recorded by Maharishi Valmiki. These three slides confirm that date of birth of Lord Ram was 10 January 5114 BC.

We should now understand the meaning implied by the mention of Punarvasu nakshatra at the time of birth of Lord Ram. Referring to the basic concepts of astronomy, it may be recalled that the moon moves from one nakshatra to another every day. The Punarvasu nakshatra means that on the day of the birth of Lord Ram, the moon, in its orbit of 27 nakshatras, was near Punarvasu (star Pollux in Gemini Constellation, 78 β Gem, Sr. No. 7 in table 1). Here it should be further clarified that the movement of the moon among the nakshatras is such that, very frequently we find it to be positioned equidistantly between two nakshatras. In such cases, there arises a confusion of associating the moon with one particular nakshatra. In such a scenario, to clarify the position, one can count the number of days from the previous no-moon day (amavasya) and in this manner find out the particular nakshatra. In the instant case, one can point out that though the moon is declared to be at Punarvasu nakshatra, it is seen within the boundaries of Cancer constellation and thus, appears to be near the Pushya nakshatra. To arrive at an unambiguous conclusion about the position of the moon on this day, slide 20 may be referred to. This slide shows that the amavasya of Chaitra month was on 3 January 5114 BC and on that day the moon was on Revati nakshatra. Now, one can easily count that if the moon was at Revati on 3 January, it would be on Ashwini on 4 January, Bharni on 5 January and so on. In this manner, Punarvasu being the seventh nakshatra from Revati, it can be confirmed that the moon would be near Punarvasu on 10 January 5114 BC (may refer to Table 1). It should also be noted that for the full-moon to occur on Chitra nakshatra, the no-moon always occurs at Revati, barring the year in which the Adhik-maas is to be counted. In other words, for the month of Chaitra, amavasya falls when the moon is on Revati nakshatra and this day, traditionally, means the end of the year as per our lunar calendar.

We would now attempt to ascertain the precise time of the birth of Lord Ram by examining the meaning of 'Cancer Lagna'. Let us understand the

meaning of 'Lagna' at the time of birth. We know that due to the rotation of the earth on its axis, all the 12 zodiac constellations appear rising in the east in the course of 24 hours. This implies that after every two hours, a new zodiac would be seen rising in the east. The zodiac that is seen rising at the time of the birth is recorded as the 'Lagna'. To be precise, it means that when Lord Ram was born, Cancer constellation was rising in the east. Let us now see as to when the Cancer constellation rose in the east on 10 January 5144 BC.

Slide 21 shows the view of sky at 12.10 p.m. of 10 January of 5114 BC as seen from Ayodhya. One may see along the ecliptic and note that Cancer constellation is rising. On that date, actually, Cancer started rising at about 11.30 a.m. and continued to rise till around 1.00 p.m. after which the next zodiac-Leo started rising and the Lagna changed. The Ramayana further states that at the time of birth of Lord Ram, Jupiter was shining along with the moon. In this slide, note that while the moon had already risen, Jupiter was still not above the horizon. Next, look at slide 17, again showing the position of the sky at 12.30 p.m. when within Cancer, Jupiter had also risen. Jupiter came above the horizon only at about 12.15 p.m. Thus, the twin conditions that Jupiter was above the horizon and Cancer was the Lagna, suggest that the time of the birth of Lord Ram was between 12.15 p.m. and 01.00 p.m. Under the circumstances, it can be easily concluded that Lord Ram's time of birth was about 12.30 p.m. Even today, this is traditionally the time (midday) when the birth of Lord Ram is celebrated on Ram-navmi. Thus, the findings tally exactly with the age old tradition.

What does the entire discussion show? It obviously means that every statement made by Maharishi Valmiki about the planetary positions for the time of birth of Lord Ram is substantiated and verifiable. The tallying of the planetary configuration on the particular date (navmi of Shukla Paksha) of the Chaitra month that year illustrates the uniqueness of the planetary combination. Again, it can be easily argued that such a description of the sky cannot be made by a mere reverse calculations or pure imagination. It clearly implies that these details have been recorded on the basis of actual observation of these planetary bodies in the sky on the date of birth of Lord Ram.

Therefore, in terms of our present day calendar, the date of the birth of Lord Ram was 10 January 5114 B C. This date is going to be the base for subsequent findings. It should also be noted that in present times, we have the month of Chaitra around 15 March to 15 April but in those days, it was falling during the period 15 December to 15 January. The gap of about three-four months is on account of the precession of the sun over the period of 7,000 years.

The Date and Time of the Birth of Bharat, Lakshman and Shatrughan

Maharishi Valmiki has mentioned the time of the birth of the other three sons of King Dashrath as well. At 1/18/13-15 it is stated that after the birth of Lord Ram, Bharat was born in Pushya nakshatra and Pisces lagna while Lakshman and Shatrughan, being twins, were born in Ashlesha nakshatra and Cancer lagna. Pisces lagna means that at that time, the Pisces constellation was rising in the east. Since Cancer was rising around 12.30 p.m., when Lord Ram was born, the Pisces constellation would rise about 16 hours later because it is away from Cancer by seven constellations.

As shown in slide 22, the Pisces constellation started to rise in the east at about 5.30 a.m. on 11 January 5114 BC and it rose up to about 7.30 a.m. Therefore, it can be concluded that Bharat was born at about 6 a.m. on the morning of 11 January 5114 BC. Note that by this time, the moon would be clearly at Pushya nakshatra because it is the next day after the birth of Lord Ram when the moon was at Punarvasu nakshatra.

For Lakshman and Shatrughan, the time of birth remains the same as was in the case of Lord Ram because they were also born in the Cancer lagna, rising between 11.30 a.m. and 1.00 p.m. but since the nakshatra was Ashlesha, two nakshatras after the Punarvasu (refer Table 1), their date of birth must have been 12 January 5114 BC.





CHAPTER VI

The Astronomical Dating of the Ramayana: The Date of Exile of Lord Ram

The VALMIKI RAMAYANA makes it clear that King Dashrath had planned Lord Ram's accession to the throne of Ayodhya on the occasion of the latter's twenty-fifth birthday and as we all know, eventually, Ram went into exile that day itself. Since Lord Ram was born on the navmi of Chaitra Shukla Paksh, he must have completed twenty-five years on the navmi of Chaitra Shukla Paksh and therefore he should have gone to exile, on that day only. How do we know that Lord Ram had completed twenty-five years of age on the date of exile?

At 3/47/1-11, when Ravan approaches Sita with the intention of abducting her in the absence of Lord Ram and Lakshman, he asks her who she was and why was she staying in that forest. Sita replies that she was the wife of Lord Ram and after her marriage she had stayed in Ayodhya for twelve years. She goes on to say that at the beginning of the thirteenth year, King Dashrath wanted to crown Prince Ram but at the insistence of Kekeyi (step-mother of Lord Ram), he was sent to exile. Here she mentions that, “At that time my husband was 25 years of age and 18 years were counted about my age.” This unambiguous statement shows that when exiled, Lord Ram was twenty-five years of age. It also gives the inference that at the time of marriage, Lord Ram was thirteen years old.

Part of the above statement was again confirmed at 5/33/16-17, when Hanuman meets Sita in Ashok-Vatika and asks her who she was and what

she was doing there. Sita again replies that she was the wife of Ram and she had lived at Ayodhya for twelve years after marriage. She further states that at the beginning of the thirteenth year Ram had to proceed to the forests.

Further corroboration comes from a discussion between Ravan and Maarech, when Ravan was planning the abduction of Sita. At 3/38/1 -6, Maarech advises Ravan to avoid any conflict with Lord Ram and tells him that he had himself fought with Prince Ram at the time when the latter had gone to protect Vishwamitra's yagya just before Ram's marriage. In this context, Maarech mentions that when Sage Vishwamitra requested King Dashrath to send Prince Ram with him, Dashrath responded by saying "Ram is just twelve years old and does not even properly know the intricacies of war". This statement makes it clear that when Ram went with Sage Vishwamitra to protect his yagya, he was twelve years old. Thereafter, Lord Ram fought and defeated Maarech, Subahu and Taarka along with their army. After Vishwamitra completed his yagya, they all proceeded to the kingdom of King Janak where Prince Ram married Sita. Thus, one can see the coherence among all the statements discussed above to show that at the age of twelve he went to protect the yagya of Sage Vishwamitra, got married at the age of thirteen and after staying for another twelve years at Ayodhya, at the age of twenty-five he had to proceed to the forests at the insistence of his step-mother, Kekeyi.

But how do we know that it was his birthday on that day? For this, a few more statements would need to be looked at. Firstly, it was the month of Chaitra. This is mentioned at 2/3/4 when King Dashrath decides to give the crown to Lord Ram and says, "This month of Chaitra is beautiful and holy. The woods and trees are full of flowers. At this time coronation of Ram is most appropriate". Since Prince Ram was born in the month of Chaitra and had already completed twenty-four years before, the obvious implication is that he was nearing his twenty-fifth birthday at that time.

In a very important statement at 2/4/21 -22, King Dashrath calls Lord Ram one day before the date of his accession to the throne and tells him, "Today the moon is on Punarvasu nakshatra and tomorrow it will meet the Pushya nakshatra. In that Pushya nakshatra, I am eager to hand over the kingdom to you". This announcement clarifies that the

coronation of Lord Ram was planned in Pushya nakshatra of Chaitra month. Again at 2/15/3 the morning when Lord Ram was to be crowned, has been described. It has been stated that everyone was anticipating that, ceremoniously and formally the king would hand over the crown. But they were not aware of what had happened the previous night when Kekeyi forced the king to send Lord Ram into exile. In this context, the description of the morning of that day mentions that the sun rose up in the sky and the Cancer lagna came. Then the moon of the date of birth of Lord Ram also became visible. The 'moon of the birth of Ram' implies the moon of navmi of the Shukla Paksh (ninth day after no moon day). Considering all the statements discussed above, it can be concluded that Dashrath had decided to give the kingdom to Lord Ram on his completing twenty-five years of age - which was during those times, the most eligible age for the assignment.

Coming to the astronomical dating process, we can conclude, without any hesitation, at least one thing that Lord Ram went to the forests on the Pushya nakshatra day of the lunar month of Chaitra. It implies that on that day, the moon was to be on star Pushya i.e. the star β in Cancer constellation.

The planetary positions in this regard have been described in a very peculiar manner at 2/4/18. While discussing Lord Ram's coronation one day before the due date, Dashrath tells him, "Astrologers have told me that my nakshatra has been surrounded by malefic planets like Sun, Mars and Rahu. Under such planetary configurations, the king either dies or falls in some deep conspiracy. Thus, before I face any such problem, you takeover the kingship". The important part of the statement is that the nakshatra of King Dashrath was surrounded by the sun, Mars and Rahu. It implies that the three planets, the sun, Mars and Rahu were together in one zodiac constellation and perhaps around one nakshatra in that constellation. While the position of the sun and Mars can be seen easily, as these are visible planets, the position of Rahu needs to be ascertained, since it is only an imaginary planet. Another point to note is that the nakshatra of Dashrath is not named anywhere in the Ramayana and it needs to be found.

We add twenty-five years to the date of birth of Ram and reach the year 5089 BC and note that on 12 January 5089 BC, a full-moon occurred at star Spica in Virgo constellation, indicating the end of Chaitra month that year as shown in slide 23. Thus, the navmi of Shukla Paksha comes on 5 January BC, eight days before the full-moon day and on this date the coronation of Lord Ram to the throne of Ayodhya was planned. It further implies that King Dashrath intimated Lord Ram, the intention of handing over the kingship, one day before 5 January or on 4 January 5089 BC. This can be seen in slide 24 that the moon was located near Punarvasu Nakshatra on 4 January 5089 BC. We try to locate the position of the sun and Mars in the sky around this time to see whether these two are together or not. Slide 25 shows that on this date, the two planets were together in Pisces constellation. We know that Rahu is considered to be located in the constellation where the moon cuts the path of the sun in that lunar month. To find Rahu's position, we note that on the previous no-moon day, which was on 28 December 5090 BC, the moon cuts the path of the sun in Pisces constellation, as shown in slide 26, which confirms that Rahu was located in Pisces constellation too. Thus, the three planets, viz. sun, Mars and Rahu were together in Pisces. Does this mean that Pisces constellation and Dashrath had some relation? Yes indeed! Astrologically, the name has a direct relation with the zodiac constellation. Unlike modern western astrology, in which zodiac signs are directly related to the date of birth of a person, in the traditional Indian astrological system, the first letter of the name of a person has a direct relation with the twelve zodiac signs. Under this system, the person whose name begins with the Hindi word always has Pisces as the zodiac sign. Nakshatra being a part of the zodiac constellation, and the presence of the mentioned planets in the Pisces constellation during that period, proves the accuracy of the statement of Maharishi Valmiki about the astral configuration.

Analysing further, it may be noted that within the Pisces constellation, astrologically, the names beginning with word are assigned to the Purva-Bhadrapad nakshatra, which is the most westerly of the three nakshatras of Pisces. If we refer to slide 27, which shows the location of stars within the constellations, it can be noted that the moon is crossing the path of the sun, exactly at the Purva-Bhadrapad nakshatra (star ω) in Pisces constellation on 28 December 5090 BC. In fact, on 13 December 5090 BC, the sun and

Mars were exactly at Purva-Bhadrapad (Slide 28) and over the next two-three weeks, they moved towards the east. This can be noted by comparing their positions in slides 28 with those in slide 26 and again in slide 25. This comparison shows that during that period, (about three weeks) all the three mentioned planets were together passing through the Pisces constellation from west to east (from the Purva-Bhadrapad to Revati nakshatra). If King Dashrath was informed that his nakshatra was surrounded by the sun, Mars and Rahu, it implies that he must have consulted the astrologers about two weeks before the eventual discussion with Lord Ram on 4 January 5089 BC and in view of their advice must have decided to give the kingship to the latter on 5 January 5089 BC. It can, therefore, be concluded that Lord Ram went to stay in forests for fourteen years on 5 January 5089 BC.

It would be unfair not to mention here that in Ayodhya Kaand at 2/20/45, Lord Ram's mother Kaushalya makes a statement about his age when he seeks her permission to go to the forest. Few authors have mentioned that Kaushalya here says that seventeen years have elapsed since your birth" meaning thereby that Lord Ram was seventeen when he went to exile. However, certain other authors have interpreted the statement as a reference to the time period elapsed since Lord Ram's Yagopaveet or initiation ceremony. It has, thus, been explained that the statement was in conformity with the statements made by Sita, that Lord Ram was twenty-five years of age at the time of beginning of exile because seventeen years had elapsed since his initiation ceremony and the age of yagopaveet is generally at the age of seven or eight.

We can verify the accuracy of the interpretation of Kaushalya's statement by looking at the position of the sun, Mars and Rahu after a period of seventeen years from the birth date of Lord Ram. If he went to exile at the age of seventeen, these three planets should be located in the same zodiac and preferably near one nakshatra. Adding seventeen years to the date of birth of Ram, 10 January 5114 BC, we reach the year 5097 BC. The full moon at Spica, indicating the end of Chaitra, occurs on 10 January 5097 BC. The moon is on Punarvasu on 3 January 5097 BC as shown in slide 29. Thus, this should be the day when the discussion between King Dashrath and Ram had taken place. When we look at the positions of the sun and Mars on this date, these are not even together. The sun is in Pisces

while Mars is in Capricorn. As regards the position of Rahu, we find that on 25 December 5098 BC, the moon cuts the ecliptic in Aquarius as confirmed in slide 30. Hence, all the three planets were in different constellations if the statement was to mean that Lord Ram was of seventeen years when he left for the forest. Therefore, the correct interpretation of Kaushalya's statement was that she was referring to Lord Ram's initiation ceremony.

Thus, it can be concluded that Lord Ram was exiled on his twenty-fifth birthday, when he was to become king, on the navmi of Shukla Paksha of Chaitra month, Pushya nakshatra day and as per our modern calendar that date was 5 January 5089 BC.

The Date of the Death of King Dashrath

Since we have determined the date when Lord Ram left for the forest, we can also work out the date of the death of King Dashrath. At 2/63/4, it has been stated that it was the sixth night since Lord Ram left Ayodhya. If the first night were on 5 January, the sixth night would be 10 January 5089 BC. In the middle of that night, it is stated that King Dashrath remembered the wrong deed he had committed by killing an innocent young boy (Shravan Kumar), who was the only son of his blind parents. After narrating the entire episode, King Dashrath died. Therefore, it may be concluded that King Dashrath expired in the night of 10 January or early morning of 11 January 5089 BC.





CHAPTER VII

The Calendar During the Times of the Ramayana -the Calendar of Today

The RIG VEDA, unmistakably the earliest of all ancient Indian literature mentions six ritus at 1.23.15 and 3.56.3. It mentions a year of 360 days at 1.164.11 and further mentions a year having 12 or sometimes 13 months (due to Adhik-maas) at 1.25.17. Similarly, it mentions a year beginning with the month of Chaitra at 3.56.37. Hence, all the concepts of the traditional year, as we know of them today are as old as the Rig Veda. The Ramayana, being another ancient scripture, also has some terminology to refer to these concepts and names. There are frequent mentions of ritus, names of months and samvatsar (year) in the Ramayana. The compilation of all such information at one place can easily lead to the construction of a complete chart of different seasons, months etc. and it would be of great help in analysing the other planetary configurations described by Maharishi Valmiki.

What Used to be the First Month of the Year?

While the Rig Veda clearly mentions that the year used to begin with Chaitra, we need to examine a few statements to derive this conclusion in the Ramayana. At 3/47/1-11, Sita tells Ravan that she had stayed in Ayodhya for twelve years after her marriage. Then, at the beginning of the thirteenth year, King Dashrath wanted to crown Prince Ram but at the insistence of Kekeyi, Ram was sent to exile. We have noted that Lord Ram

went to exile in Chaitra month. Thus, one can infer that Chaitra was the first month of the year. This also gets confirmed from the following analysis.

In a very significant statement, when Hanuman met Sita at Lanka, she tells him at 5/37/ 7-8 that, "My life would remain only till the end of the current samvatsar (year). Out of the one-year period that the cruel Ravan had granted me, this is the tenth month running and only two more months remain after this". Analyzing the statement, we can infer that the end of the year long waiting period was to finish with the end of the year. This implies that Sita must have been abducted almost at the end of the previous year (Samvatsar). Only in that condition, the end of the year long period would coincide with the end of year itself.

After meeting Sita, Hanuman went back, met Lord Ram and brought them back to Lanka. A battle was fought and won with the killing of Ravan. Immediately after that, at 6/124/7, after killing Ravan, Lord Ram states that he must rush to Ayodhya because his period of exile was almost over. Actually he travelled by the Pushpak Viman, to reach Ayodhya quickly. Since Lord Ram went to the forest in Chaitra for a period of full fourteen years, his exile was to be completed only in the month of Chaitra. This implies that he had killed Ravan only at the end of the last month of the year or just at the beginning of Chaitra. If that coincides with the year long captivity period of Sita and the end of the Samvatsar, it obviously means that the year used to begin with the month of Chaitra, as it begins today as per the traditional Indian calendar.

The Number, Names and Sequence of Ritus in a Year

In the traditional Indian calendar, the year of twelve months is divided into three main seasons of four months each. These three main seasons are - grisham, varsha and sheet signifying heat, rain and cold respectively. But out of the four months of the season, the main characteristics of heat, cold and rain are shown in the middle two month; the first and the last months of the four-month period show the features of both the incoming and the outgoing seasons. Thus, there are three more intervening seasons, (mixed ones) formed due to the effects of the incoming and outgoing main season. Due to this reason, there is a further division of the year into six ritus of two

months each. The names and sequence of these six ritus are -vasant, grisham, varsha sharad, hemant and shishir.

In the Valmiki Ramayana too, the number of ritus in a year is stated to be six. At 2/25/15, when Lord Ram takes permission from his mother Kaushalya to go to the forest, she states, "Let six ritus, Adhik Maas, Samvatsar (year)... protect you in the forest Even otherwise, the names of only six different ritus appear in the text, neither less nor more. With the help of the following unambiguous statements, one can reconstruct the inter-se sequence of the ritus in a year. The statements are:

- At 2/63/15-16, it is stated that varsha started after the end of grisham.
- At 4/27/38, it is stated that sharad would start after varsha.
- At 3/16/1, it is stated that hemant started after the end of sharad.
- At 4/1/44, it is stated that the end of shishir would lead to vasant.

If we try and reconstruct the sequence of ritus on the basis of the above clear statements, we get the following two sequences:

grisham → varsha → sharad → hemant and

shishir → vasant

If the number of ritus is only six, then the only possible sequence is

grisham → varsha → sharad → hemant → shishir → vasant

The above sequence of the six ritus is still being followed as per our traditional Hindu calendar. Hence, we must conclude that there is continuity of the names as well as the inter-se sequence of the six ritus from the times of the Ramayana to the present day.

Let us now examine the issue as to which month of the calendar corresponded to which ritu.

The Months of the Ritus -the Entire Calendar

Today, the sequence and name of months as per the Indian lunar calendar is as follows: Chaitra, Vaishakh, Jyestha, Ashad, Sawan, Bhadrapad, Ashwin, Kartik, Margshirsh, Paush, Maagh and Phalgun. Every month ends at the full-moon (Purnamasi) and begins with the first day after full moon (Krishna Paksh or decreasing phase of moon) except Chaitra. The first month of our traditional calendar, Chaitra begins with the Shukla Paksh (first day after no moon day of Chaitra). Thus, the New Year begins in the middle of Chaitra; the first half of Chaitra (decreasing phase of moon) is counted in the previous year. Since twelve months are distributed among six ritus, every ritu has a two month period.

At 4/1/11&36, Lord Ram describes the vasant ritu in the entire Sarga and also mentions that it was the month of Chaitra. Thus, Chaitra was associated with vasant ritu. At 4/1/44, Lord Ram further states that the sheet (cold) ritu had just ended. This means that Chaitra used to be the first month (out of the two months) of the vasant ritu.

At 4/26/11 Lord Ram states, "this month of sawan is the first month of varsha ritu. There are three such other months of Varsha and that is why the full rainy season is called a chau-masa". Thus, sawan was the first month of varsha ritu. It is also important to note that in southern India, the monsoon rains are so heavy and stretched that a clear-cut four-month period is earmarked for varsha ritu. As a result, the sharad ritu is not so pronounced and generally not even counted as a ritu in south India. There the entire period of four months is for varsha ritu called Chau-masa. As against this, in north India, winters are four months long while varsha remains for only two months. Accordingly in north India, the sharad ritu is more pronounced and both the ritus - hemant and shishir are experienced and identified. At the time of describing the Chau-masa or four month long rainy season, Lord Ram was living at Kishkindha, which corresponds to 'Hampi' in Bellary district of Karnataka state of south India.

At 4/28/55, Lord Ram says, "Bharat must have completed all the necessary work before the onset of rainy season on the purnamasi (end of month) of Ashad". This means that varsha started at the end of month of Ashad. Thus, the month of Ashad preceded the month of sawan and Ashad was also the last month of the grisham ritu. Further, at 4/26/54 after

describing varsha ritu begining with the Sawan month, Lord Ram mentions the advent of the month of Bhadrapad. Thus, Bhadrapad was the second month of varsha ritu.

At 3/16/11, while describing hemant ritu, Lord Ram tells Lakshman that in these cold nights of hemant ritu, no one can stay at a place without a shelter. He further states "till the full moon (Purnamasi) of Pushya nakshatra (meaning end of Paush month) comes, it would not be possible to spend nights outside without a shelter due to cold". This statement makes it is clear that the end of Paush month, coincided with the end of hemant ritu.

From the above statements, the following relations emerge between the ritus and months:

- Sawan was the first month of varsha ritu.
- Hemant ended with Paush month.
- Chaitra was the first month of the vasant ritu.
- Varsha started at the end of month of Ashad.
- Month of Ashad preceded the month of Sawan.
- Month of Ashad was the last month of grisham.
- Month of Bhadrapad followed the month of Sawan in varsha ritu.

Based on the above combinations, we draw a complete table of the months and the ritus as follows:

Table 3

The Year, Seasons, Ritus, Months and Corresponding English Months

YEAR	SEASONS (4-MONTHS)	RITUS (2-MONTHS)	LUNAR MONTHS	COINCIDES WITH (5000 BC)	COINCIDES WITH (AS OF TODAY)
O N E Y E A R	GRISHAM	Vasant	Chaitra	December - January	March - April
			Vaishakh	January - February	April - May
		Grisham	Jyestha	February - March	May - June
			Ashad	March - April	June - July
	VARSHA	Varsha	Sawan	April - May	July - August
			Bhadrapad	May - June	August - September
		Sharad	Aswin	June - July	September - October
			Kartik	July - August	October - November
	SHEET	Hemant	Margsirsh	August - September	November - December
			Paush	September - October	December - January
		Shishir	Magh	October - November	January - February
			Phalgun	November - December	February - March

We already know that in the era of Lord Ram (about 5000 BC) the lunar month of Chaitra coincided with December-January of the modern English calendar. Since we have been able to work out the sequence of the months of the traditional Indian calendar, it is possible to see which lunar month coincided with the month of English calendar during the period of about 5000 BC. The same sequence for the present century has also been drawn and displayed in the last two columns of the Table above.

This chart confirms that our traditional Indian calendar and every component thereof, had existed even during the times of the Ramayana in the same form as we know of it today. What does it mean? It means that from the times of the Ramayana, till today, the system of calendar, system of counting months, seasons, their inter-se sequence etc. nothing has changed. It is once again a proof of the continuation of our civilization from that period to the modern times.



CHAPTER VIII

The Astronomical Dating of the Ramayana: the Date of Battle between Lord Ram and Khar - Dushan

ANY ARGUMENT THAT the kind of planetary configurations recorded in the Ramayana could have been manufactured by someone to backdate the scripture, would be set to rest when we prove the accuracy of the astronomical event described in the Ramayana to mark the battle of Lord Ram with Khar and Dushan. Maharishi Valmiki has described the occurrence of a partial solar eclipse, visible at Panchwati, on the day when Lord Ram fought a battle with these two demons and their army. Solar eclipses, whether total or partial, do not occur every now and then. Further, it is just not possible to make manual computations to work out the date of occurrence of the solar eclipse, which took place thousands of years ago with an intention to backdate the Ramayana. No astronomer or mathematician of even modern times is capable of doing so unless he uses advance computer software.

The description at 3/23/1 is as follows, "When Khar moved with his army towards Ram for the battle, a red coloured evening appeared to be descending all over." Shloka 3 states that the sun was covered by a dark coloured disc that had red coloured corners. Shloka 9 mentions that although it was daytime, it started appearing to be the evening. Shloka 12 clearly mentions that the sun was caught by Rahu because of which the sun lost its glow and brightness. Various animals and birds started making noises as if it was dusk.

The above description is clearly that of a solar eclipse. It is possible to verify the occurrence of a solar eclipse with the help of the Planetarium Software but for that we need to find out when exactly the battle between Lord Ram and Khar took place. Even if we were able to fix a time period of three to four months, it would be enough to find out whether any such eclipse took place during that period of few months or not. To ascertain the time period when exactly the battle took place, we will again examine various statements made in the Ramayana.

For those who are not well versed with the Ramayana, the important events of this part of the story may be narrated to make the sequence clear. Lord Ram was living with Sita and Lakshman at Panchwati. Surpnakha, the sister of Khar, Dushan and Ravan went to Panchwati and saw Lord Ram and Lakshman. She proposed to them, but they refused. Lakshman, at the instance of Lord Ram cut her nose and ears. She went back to Khar-Dushan and instigated them to kill Lord Ram. Thereafter, the battle between Lord Ram and Khar took place resulting in the latter's death. After the death of Khar - Dushan, Surpnakha, went all the way to Lanka, met Ravan and asked him to take revenge. Ravan went to Panchwati and with the help of Maarech abducted Sita. Ram found her whereabouts with the help of Sugreev. He fought the battle with Ravan, killed him and rushed back to Ayodhya as his exile period was nearly over. From the above narration, it is clear that the battle between Lord Ram and Khar took place after Surpnakha went to Panchwati and before Ravan abducted Sita. We would, therefore, try to fix the dates for the occurrence of these two events and then try to find out whether a solar eclipse occurred during this period or not.

From the analysis of different statements in Chapter VI, we concluded that Ravan had abducted Sita almost at the end of the thirteenth year of exile. The reason being that Sita was given a one-year waiting time by Ravan. That one-year period ended with the end of the exile period of fourteen years. Therefore, it can be safely concluded that Sita remained in Ravan's captivity during the entire fourteenth year of exile or she was abducted at the end of the thirteenth year. Further, as the battle between Ram and Khar definitely took place before the abduction of Sita, it can be held that this battle was fought definitely before the end of the thirteenth year of exile. Moreover, since the solar eclipse had occurred when this

battle was fought, we can keep the end of the thirteenth year in exile as the outer limit for the occurrence of the solar eclipse.

The date of end of the thirteenth year of exile can be worked out by adding thirteen years to the year of exile (5089 BC) and finding out the full-moon night when the moon is near the Uttara Phalguni nakshatra or the star 5 p in Virgo constellation. That marks the end of the last month of the lunar year-Phalgun. Slide 31 would show that this occurred on 20 December 5077 BC. Since the first 14 days of the month of Chaitra (Krishna Paksh) are counted at the end of every year, to be accurate, we may conclude that the thirteenth year of exile ended on 3 January 5076 BC. By this date, Sita was abducted by Ravan. As the battle between Lord Ram and Khar was definitely fought before Sita's abduction, it is safe to conclude that the solar eclipse, which took place during the Lord Ram-Khar battle should have occurred before 3 January 5076 BC.

We would now try to find when Surpnakha went and met Lord Ram at Panchwati because the battle between Lord Ram and Khar had taken place only after her visit there. This date would set the other limit of the time period for the occurrence of the solar eclipse.

In Aaranya Kaand, Sarga 16 begins with the statement that at Panchwati, Lord Ram was living happily when sharad ritu ended and hemant ritu began. In the entire Sarga, Lord Ram describes the features of hemant ritu to Lakshman and Sita thereby indicating that even hemant ritu had also progressed. Then, in a very important statement, in shloka 14 Lord Ram states that "due to fog, the moonlight of full moon is not very bright." This implies the end of one month, out of the two months of hemant ritu. Till that time, at least, Surpnakha had not met them. Sarga 16 ends without clearly describing whether hemant ritu ended or not. After this, the episode of Surpnakha starts in Sarga 17 with her request to Lord Ram and Lakshman to marry her. Lakshman cutting her nose and ears is described in Sarga 18. Her instigation of Khar and Dushan to take revenge on Lord Ram and Lakshman has been mentioned in Sarga 19 to 22.

From the description in Sarga 16, it can be concluded that Surpnakha visited Panchwati after the end of the first month of the Hemant ritu in the

thirteenth year of exile and thus the battle between Lord Ram and the demon Khar must have taken place only after that point of time.

To ascertain when the first month of hemant had ended, we refer to Table 3. The first month of Hemant ritu is Margshirsh. The Margshirsh month ends on the full moon night when the moon is near Mrigshira Nakshatra or the star ρ in Taurus constellation (Refer Table 1).

To find out when this month ended in the thirteenth year of exile, we first add 12 years to the year of beginning of the exile (5089 BC) to reach 5077 BC. In this year, the full moon near the star ρ in Taurus constellation occurred on 22 September 5077 BC. Slide 32 shows this clearly. This implies that Surpnakha went and met Lord Ram at Panchwati only after 22 September 5077 BC and therefore, the Lord Ram-Khar battle must have been fought after this date.

Summarizing the above findings, we may say that the Lord Ram - Khar battle must have taken place only after 22 September 5077 BC and before 3 January 5076 BC. This period of less than three-months should have witnessed a solar eclipse, from Panchwati, if the recordings are correct in the Ramayana

Panchwati corresponds to the modern city of Nasik in Maharashtra state on the banks of river Godawari. The software is provided the inputs of the longitude and latitude for Nasik as 20° N and 73.5° E. It is simple knowledge that a solar eclipse can occur only on an amavasya (no-moon day) and thus, only once in a month. From the month of August onwards, the position of the sun and the moon on the no-moon days for every month is shown in slides 33, 34 and 35. What do we find in slide 35? A solar eclipse occurs on 7 October 5077 BC! It is a solar eclipse with the moon's disc substantially covering the sun. It should be noted that the eclipse occurs just at that time when Valmiki has mentioned it in the story - fifteen days later from 22 September 5077 BC, when the first month of the hemant ritu had ended. The gap of fifteen days is obvious because, the month end is the full moon day while the solar eclipse can occur only on a no-moon day, which comes fifteen days after the full-moon day.

The entire story does not end here. Further verification is also possible to confirm whether this solar eclipse was the one described in the Ramayana or not. Maharishi Valmiki has qualified the solar eclipse by two more statements. Firstly, while describing their battle, it is stated at 3/29/23, that Khar told Lord Ram, "I want to say many things (about my prowess) but I would not, because if in the meanwhile sunset occurs, there would be a break in the battle". (And I would not be able to kill you today). This makes one thing clear - the battle (and the solar eclipse) took place late in the day, perhaps, in the afternoon due to which Khar was apprehending the sunset. The solar eclipse of 7th October 5077 BC reached maximum coverage (for Panchwati at 15.10 hours and that should be the time when Khar started with his army towards Lord Ram, (shloka 1,3,9 and 12 stated earlier). Therefore, when Khar actually confronted Lord Ram, it should be late afternoon and that is why he says that if time is wasted in talking, sunset may occur.

Secondly, and more importantly, Valmiki has provided the graphic description of the sky on the day of solar eclipse at 3/25/5. The statement is, "In the middle of his army, Khar was looking like Mars in the middle of the Planets in the sky. It means that firstly, all the planets visible to the naked eye were in the sky and secondly, Mars was in the middle of them. A look at slide 36 would show that the planets, (which could be seen with the naked eye) were forming a chain at the time of eclipse. From east to west these were as follows: Saturn, Sun, Moon, Mars, Mercury, Venus, and Jupiter. And undoubtedly, Mars was in the middle of the chain! A conjunction of all the visible planets at a time is itself a unique event. This conjunction is qualified with the fact that Mars was in the middle of these seven planets. And the ultimate qualification is that the sun was eclipsed. These factual details, narrating the happening of such unique celestial events, cannot be conjectured by anyone. These are some of the rarest of the rare astronomical events, which got recorded because of their uniqueness.

These two statements further corroborate the fact that the solar eclipse described by Maharishi Valmiki was the one that occurred on 7 October 5077 BC.

To provide completeness to the process of inquiry, it may be stated that the eclipse of 7 October was the only solar eclipse to have taken place in the entire year of 5077 BC (the thirteenth year of exile) for Panchwati. This was verified with the help of software for all other amavasya (no moon) days of the year. The occurrence of only one solar eclipse in the entire year and that too at a time when it is described in the Ramayana, demonstrates that it is not a calculation, conjecture or coincidence, since we are exploring the 6th millennium BC. It obviously means that someone had actually observed this celestial event and the same was mentioned directly by Valmiki in his Ramayana. The further corroboration of the authenticity, accuracy and the antiquity of Valmiki Ramayana will be the confirmation of occurrence of two more eclipses - one solar and one lunar - in a span of just eleven months as discussed in the following two chapters.





CHAPTER IX

The Date of the Killing of Bali -Confirmation by a Second Solar Eclipse

WE HAVE SEEN that the battle between Lord Ram and Khar - Dushan took place on 7 October 5077 BC while Ravan abducted Sita towards the end of Phalgun of 5077 BC, which ended on 20 December 5077 BC. This shows that Ravan took about two months to plan and execute Sita's abduction.

It is clear that Lord Ram had not met Hanuman and Sugreev till the beginning of the fourteenth year of his exile because the beginning of Chaitra, (or the beginning of fourteenth year of exile), is described in Sarga 1 of the Kishkindha Kaand while the meeting of Lord Ram and Sugreev is detailed in Sarga 3 and 4 and there is continuity of the description between these sargas.

After meeting Sugreev, Lord Ram vowed to kill Bali (Sarga 8). Sugreev told him about the physical strength of Bali (Sarga 11). Thereafter, all of them went to Kishkindha, (modern 'Hampi' in Bellary district of Karnataka state) the capital of Bali's kingdom. Lord Ram sent Sugreev to fight with Bali once but in that conflict, Sugreev was badly hurt by the latter (Sarga 12). After recovering, Sugreev was sent again by Lord Ram for a second fight. Sugreev went and roared in front of Bali's fort (Sarga 14). At this point of time, Maharishi Valmiki refers to another solar eclipse by mentioning that, 'hearing the roar of Sugreev, Bali's face became red like the Sun caught by Rahu during the eclipse' (4/15/3) Sugreev had

gone to fight with Bali in the morning time because at 4/14/4 it has been mentioned that when Lord Ram told Sugreev to go and fight with Bali, "Sugreev was looking like the sun in the morning". Further, Ram assured him that "today, your enemy Bali will be killed by me positively' (4/14/10-12,14). Therefore, one can conclude that, perhaps, there was a solar eclipse during the morning, on the day when Bali was killed by Lord Ram.

The Ramayana describes that immediately after killing Bali and establishing Sugreev on the throne, the search for Sita's whereabouts could not begin because of the onset of the rainy season. Lord Ram says at 4/26/13 that "Varsha ritu has begun" and thus, they would have to wait for the rains to recede before starting search. Since varsha ritu comes after grisham ritu, we can easily conclude that Bali was killed very close to the end of grisham ritu.

To find out when grisham ritu did end, we refer to Table 3 and note that the last month of grisham was Ashad and it used to correspond with the months of March - April around the 5000 BC period. Moreover, a solar eclipse can occur only on the no-moon day, hence, if a solar eclipse had occurred, it should have been on the no-moon day of Ashad because immediately after the accession of Sugreev on the throne of Kishkindha, Lord Ram has described the advent of the varsha ritu beginning with the month of Sawan.

In the fourteenth year of exile, 5076 BC, the full moon day or purnamasi of Ashad falls on 16 April, when the moon is near the Purva-ashad nakshatra in Sagittarius constellation (refer Table 1 for details). This can be seen in slide 37. One can also count that if Chaitra month ends in January, A .had being third from Chaitra would end in April. The no-moon day thus comes to 3 April 5076 BC and we find the solar eclipse on that day! (Slide 38)

Once again, to complete the process of inquiry, it may be stated that in the entire year of 5076 BC, (fourteenth year of exile) this was the only solar eclipse to have occurred and it was clearly visible for Kishkindha's location. Actually, its coverage was a little less, when seen from southern India. For the northern part of our country, it was almost a total solar eclipse, which

reached its maximum coverage at about 9.30 a.m. For Kishkindha the eclipse started at around 7.30 a.m. and continued till about 11.30 a.m. Thus, the description in the Ramayana that it was in the morning time when Bali's face was looking like the eclipsed sun also matches with the actual time of occurrence of the eclipse.

We have so far seen that the description of the two eclipses, out of the series of three eclipses mentioned in the Valmiki Ramayana, matches with the details of the actual occurrence of these astral events. Before we go ahead with tallying the other planetary positions given by Maharishi Valmiki in the Ramayana we must note down that the important date found in this chapter is 3 April 5077 BC when Lord Ram killed Bali.





CHAPTER X

The Date of Hanuman meeting Sita at Lanka - Confirmation by a Lunar Eclipse

THERE IS A VERY graphic and detailed description of the events that happened in the fourteenth year of exile of Lord Ram. Hence, it is possible to make a detailed analysis and find out the date and time of happening of most of the events, which took place that year. The important fact from the astronomical dating point of view is that the mention of a lunar eclipse has been made repeatedly on the day Hanuman visited Lanka and met Sita. We know that a lunar eclipse occurs only on full-moon days. Therefore, we must ascertain when exactly Hanuman crossed over the sea and met Sita. Thereafter, it can be verified with the help of the software whether a lunar eclipse occurred that day or not.

First of all, we examine a statement made by Sita at 5/37/7-8, when Hanuman met her. She told him that, “My life would remain only till the end of the current samvatsar (year). Out of the 1-year period that the cruel Ravan had granted me, this is the tenth month running and only two more months remain after this”;

This statement has already been analysed in Chapter VII to mean that the one-year captivity period of Sita ended with the end of the year. If that was the case, then the other part of her statement implies that nine months of that year were over. The tenth month of the year was running and two more months remained. However, Hanuman had met her on the full-moon day, which itself means the end of the month. Therefore, one can argue that

either the ninth month (Margshirsh) had completed, the tenth month was running (being the first day of that month) and two more months remained. The other possible interpretation is that the tenth month of the year (Paush) ended and two more months (one day less than full two months) remained. One conclusion that can certainly be drawn is that Hanuman met Sita either on the full moon day of the month of Margshirsh or that of Paush. We would now attempt to find the answer and ascertain this date by analysing the description of the events mentioned in this chapter.

Immediately after crowning Sugreev as the king of Kishkindha, at 4/26/11-16, Lord Ram speaks to him beginning with “this month of Sawan is the first month of varsha ritu”. This shows that he was making the statement at a time when the month of Sawan had already begun. Lord Ram goes on to state that, “there are three other months of varsha and that is why the full rainy season is called a chau-masa (four-months). During this chau-masa period, almost every thing comes to a halt due to heavy rains. Thus, it is not the time to search for Sita.” Here he mentions to Sugreev that at the advent of the month of Kartik, you make efforts to kill Ravan (or make efforts to locate Sita).

If Sawan is taken as first month of varsha season, Kartik is the fourth month, hence, Lord Ram must have meant that the beginning of the fourth month of the rainy season would be the appropriate time to start the search for Sita (efforts to kill Ravan) because by then the rains would begin to recede. The advent of Kartik means the end of the month of Ashwin.

The search for Sita was to begin in the beginning of Kartik but it got delayed. This has been unambiguously mentioned at 4/30/78 when Lord Ram tells Lakshman that Sugreev has forgotten his promise after getting the throne and wife. The entire Sarga 31 33 describes how an angry Lakshman went to remind Sugreev of his promise. Very importantly, it has been clearly mentioned that, Hanuman, who was Sugreev's deputy and was staying with him at Kishkindha, had realized that the time for starting the search for Sita had come. Hence, within the time, he had sent messengers to call all the monkeys (vanars) to the capital city of Kishkindha. He orders at 4/29/33 that any monkey, who would not come within a period of fifteen

days, would be awarded a death penalty. It is then stated that Lakshman came to Kishkindha and told Sugreev that the time fixed for starting the search for Sita had already passed. Thereafter, Sugreev, having realized his mistake immediately came to Lord Ram and apologized. As directed by Hanuman, monkeys from all over also came at about the same time (4/39/11-44) After that, the search for Sita began and a period of one month was fixed for completing the search as mentioned at 4/45/4.

From the above description, one can infer that the search for Sita which was to begin at the end of Ashwin or beginning of Kartik, got delayed and most probably began about fifteen days later, by which time, monkeys from different parts came to Kishkindha. A period of one month was fixed to find Sita's whereabouts. Hence, if the search began in middle of Kartik, it must have been scheduled to end by the middle of Margshirsh But it is explicitly mentioned at 4/50/3 and again at 5/57/17 that while the vanars (monkey army) were searching for Sita in the caves of Vindhyaachal hills, the period of one month elapsed. Thus, it means that till the middle of the month of Margshirsh, the monkeys were not successful in finding Ravan's (Sita's) whereabouts. Later, the vanars reached the seashore where they learnt that Ravan had kept Sita across the sea at Lanka. They needed someone to cross the sea, which Hanuman finally achieved. But he crossed the sea on a full moon day, a fact that has been clearly stated at many places in the Sunder Kaand. A full-moon day means completion of the month, and if it was so, it has to be the completion of the month Margshirsh.

The other possibility is that, if the statements 4/50/3 and 5/57/17 are taken to mean that the search began after the end of Kartik and a period of one month got over before Hanuman found Sita on the full-moon day, then the day on which the lunar eclipse took place must be the full moon day of the month of Paush.

Hence, there is genuine confusion about the exact date when Hanuman visited Lanka and met Sita. Whatever may be the basis, the ambiguity remains as to whether it was the full moon day of the month of Margshirsh or Paush. The answer can be provided by the software if it confirms whether a lunar eclipse, if at all, took place on any of these two days or not.

If yes, then it will be the answer to the question when exactly Hanuman visited Lanka and met Sita.

A detailed description of the time when Hanuman had crossed the sea and reached Lanka are available in the Ramayana. At 5/2/47-48, it is mentioned that Hanuman reached the other end of the sea (Lanka) well before sunset. He waited for the sun to set. Interestingly, the reference to a lunar eclipse is made at 5/1/192 when, on his way to Lanka, he had a tiff with a demon called Sursa in the middle of the sea. Thus, the lunar eclipse must have begun in the afternoon or late evening well before he reached the other end. Subsequently, at 5/5/5 it is mentioned that when Hanuman entered the fort after sunset, "the sky was decorated with the full moon". Again at 5/5/7 it is mentioned, "The way king beautifies the throne, the lion beautifies the rock it sits on, the same way the full-moon was beautifying the sky with its light."

The references to the lunar eclipse come later when after searching Sita for sometime elsewhere, Hanuman sees her at Ashok-Vatika. It is mentioned at 5/19/14 that Hanuman found Sita looking like a full moon, which was eclipsed by Rahu. At 5/29/7, Sita's face has been described as a full moon, which had just been released from Rahu's captivity (cleared of eclipse). Again at 5/35/87, it is mentioned that Sita's face resembled the full moon released from the 'grip of Rahu'. Collating all these statements, one can infer that a lunar eclipse must have occurred, well before sunset and must have ended late evening. Such an eclipse must have occurred on the full moon day/night of lunar month of Margshirsh or the Paush that year

Table 3 shows that the month of Margshirsh ended in September in those times. Thus, the full moon day / night in September works out to 12 September 5076 BC. On this day, the position of the moon was observed at an interval of five minutes after 4 p.m. in the evening. Till 4.10 p.m., the moon was displayed as the full moon but suddenly at 4.15 the software shows it as the moon of amavasya (no-moon). This feature is displayed in slides 39 and 40. Certainly, this is not a full moon turned into amavasya in a gap of five minutes. Actually, it is the beginning of the lunar eclipse that we have been looking for!

When the full-moon is suddenly shown as no-moon, it actually shows that the moon was under the shadow of the earth and hence, beginning to eclipse. The gap in the ecliptic line displays the shadow region of the earth through which the moon was passing during eclipse. At the time of observations shown in slides 39 and 40, the moon was below horizon. The moon remained eclipsed even when it rose in the east at about 6.30 in the evening as shown in slide 41. A magnified view of the moon can be seen at the time of rising above the horizon in slide 42. The eclipsed moon, at 6.40 in the evening, is shown in slide 43. Slide 44 shows that at 6.56 p.m. the disc of the moon is shown eclipsed by the software while at 6.57 p.m., the full moon is again shown by the software (Slide 45) indicating that the eclipse finally got over exactly at this time. This confirms, in the most authentic manner, that Hanuman visited Lanka and met Sita on 12 September 5076 BC.

To be fair and objective in the approach, the position of the moon for the entire twenty-four hours of full moon day/night of Paush was observed through the software. No eclipse takes place that day. Moreover, in the entire year of 5076 BC, this was the only lunar eclipse to take place.

The verification of the occurrence of the third eclipse by the software, allying verbatim with its description in the Ramayana, again corroborates in the most scientific manner the antiquity of the event and the historical existence of Lord Ram





CHAPTER XI

The Astronomical Dating of the Ramayana - other Dates of the Fourteenth Year of Lord Ram's Exile

The Date of Hanuman's Return from Lanka

As CONFIRMED IN the last chapter, Hanuman met Sita on the full moon night of 12 September 5076 BC. After dawn, he decided to take on a fight with the soldiers of Lanka (5/41/12-21). In that battle, he killed one of the sons of Ravan and hundreds of other soldiers. This is described in the Sargas 42-47. Later on, Meghnath the eldest son of Ravan, was able to capture Hanuman after a fight. (5/48/1-50).

After being captured by Meghnath, Hanuman was taken to Ravan's court, where he had an argument with the king of Lanka. Ravan ordered that Hanuman's tail be put on fire. Hanuman, with his tail on fire, then put the better part of Lanka on flames. This is narrated in Sarga 54 of the Sunder Kaand.

Undoubtedly, all this happened during the day on 13 September 5076 BC. After putting Lanka on fire, Hanuman was concerned about Sita's survival from the fire that was ravaging all around (5/55/8-10). Hence, he decided to meet her again before returning from Lanka. The meeting between Hanuman and Sita has been described in Sarga 56 of Sunder Kaand. Since it was again the night, it can be easily inferred that he must have met Sita again on the night of 13 September 5076 BC.

Thereafter, he began his return journey, which has been described in Sarga 57. The positions of different nakshatras and constellations have been mentioned in a poetic manner in the form of similes. It has been stated at 5/57/1-3 that when Hanuman was crossing the sea, "The Moon was like lotus while Sun appeared like a crane (5/57/2). In the sky, Pushya and Shravan nakshatras were seen like swans. Punarvasu appeared like a big fish. Mars and the large planet (Jupiter), the elephant of Indra - Airavat, Island and the Swati Nakshatra were seen to be moving like a swan" (5/57/3). After this description, at 5/57/14 it is stated that Hanuman reached the middle of the sea and touched the Sunaabh Parvat (hill). In the above quoted description of the sky, the important points to note are:

- While the moon was shining, the sun appeared (in the sky or it rose from the east) like a crane. Thus, the sun and moon were seen together for some time in the sky.
- The Punarvasu nakshatra located in Gemini, Pushya nakshatra located in Cancer and the Shravan nakshatra located in Capricorn constellation could be seen.

Note that from Gemini to Capricorn, there are eight constellations. Eight constellations cannot shine at the same time in the sky. At a time, only six remain above the horizon. The significance of mentioning the eight constellations in the context lies in indicating the time taken by Hanuman to cross the sea. One constellation takes about two hours to rise (12 constellations rise in 24 hours). Thus, if eight constellations could be seen in the sky while Hanuman was in the sea, it simply means that two constellations rose while he was crossing the sea. This means that the description of the sky is not for a point of time but for a period of time. After describing the above sky, Valmiki says that Hanuman reached the hill named Sunaabh in the middle of the sea. It shows that Hanuman took about four to five hours to reach this hill in the middle of his path while crossing the sea. Perhaps, he had rested there on his way back.

The above description of the sky tallies with the sky view generated by the software for the morning of 14 September 5076 BC as illustrated in slides 46 and 47. Slide 46 shows the position of sky on 14 September 5076 BC at 6:30 a.m. After meeting Sita for the second time on the night of 13

September, Hanuman must have started his return journey on the morning of 14 September 5076 BC. The slide shows that the moon is just under the Gemini constellation and it is about to set while the sun is rising in the east. Thus, the sun and the moon are seen together. Also, it justifies the fact that while the moon was already in the night sky, the sun, rising in the east “appeared” on the scene like a crane Gemini constellation, which has Punarvasu nakshatra, is looking like a large fish. Mars and Jupiter are shining. Also seen shining in the sky are Pushya nakshatra in Cancer constellation, and Swati nakshatra in Virgo constellation. Airavat or the legendary elephant of Indra corresponds to the Scorpio constellation, which looks like the trunk of an elephant. It may be noted that all these objects have been stated to be present in the sky and these are seen in the slide as well. Those astronomical objects, which are not seen in slide 46, although mentioned in the description, are Shravan nakshatra (Capricorn constellation) and the island. Now look at slide 47 showing the view of sky at 10.00 a.m. Note that Capricorn has risen fully and the star at the eastern corner of this constellation is Shravan nakshatra. Note that it is the Capricorn constellation, which itself looks like an island if the sky is viewed as sea. Thus, slide 47 shows the time when all the objects stated in this context had risen.

Therefore, this should be the time by which Hanuman reached the hill in the middle of his path while crossing the sea.

Having found that Hanuman started his return journey from Lanka at about 6:30 on the morning of 14 September 5076 BC, let us examine the appropriateness of this time from the point of view of crossing the sea. Hanuman had to swim across the sea. It was not sensible to venture into the sea in the night. However, there was an urgency too. He, therefore, waited for the first ray of the light in the morning to cross the sea. This is precisely shown in slide 46 that when he began his journey, the sun had not even risen above the horizon. Further, it needs to be pointed out again that the presence of certain planets along with certain specific constellations or nakshatras at a specific point of time refer to very precise planetary configurations. Unless someone could see and describe them, it is very difficult to conjecture about their positions in the sky. And if it is guesswork, then it cannot be found to be correct to the smallest detail. One

can expect that at least a few positions would be wrongly stated, if someone was conjecturing about the position of ten planetary bodies, as they existed thousands of years ago.

Leaving aside the astronomical dating process for a while, we will now discuss some interesting aspects related to Hanuman crossing the sea. There is a great deal of curiosity about Hanuman's flying prowess and it is generally believed that he flew over the sea to Lanka. This is not correct. Actually, he swam through the sea. Valmiki Ramayana clearly states at 5/1/29 that when Hanuman was contemplating crossing the sea, the sages, mystics and other learned persons gathered around and they were heard saying "He, the son of wind (Pawan), wants to swim across this sea". At 5/1/67 it is mentioned that Hanuman was like a boat in the sea. At 5/1/75 it is stated, "watching the Hanuman swimming across the sea, snakes living in the sea thought that Garuda was flying in the sky". 5/1/83 says that "watching the Hanuman swimming in the sea, the sages and devas showered the flowers on him".

Besides these clear and unambiguous statements, there are many allegories, which indicate that he swam across the sea. It should also be noted that when Hanuman was crossing the sea he rested at several hillocks. It has been stated clearly that while going to Lanka and while returning from there, Hanuman rested on Mainaak Parvat (hill) and Sunaabh Parvat (hill) in the middle of the sea. 5/1/103 states that the Mainaak Parvat came above the sea level and showed his many peaks to Hanuman and offered him to rest on them during his journey. Hanuman even met human beings living in the middle of the sea. This may be seen at 5/1/111 where it is mentioned that the Mainaak Parvat stood up in the form of a human and advised Hanuman to rest on his peaks. He also offered Hanuman sweet fruits and other things to eat. Later, at 5/11/150 Hanuman met a demon called Sursa and killed her. It is therefore clear that the entire stretch of about twenty-five to thirty kilometers was not deeply submerged. It appears that there were a series of small rocks or even small islands in the entire stretch that facilitated his journey from one end to the other. This is clear even from an archaeological point of view. Even today, the entire Palk Strait, between India and Sri Lanka has an ancient underwater sandbank known as Adam's Bridge, which connects India to Sri Lanka. Some new

pictures taken by a NASA satellite show this "bridge" in all its glory. The thirty-kilometer long connection, which stretches across the Palk Strait, is actually a narrow and shallow ridge of sand and rocks connecting Mannar Island in Sri Lanka to Pamban Island in India. The Sri Lankan archaeology department has said the connection could be two million years old. This connection was always available for the inhabitants of Lanka as well. This was perhaps the only source of accessing Lanka from main land India. The important point to note is that today, most of the stretch of this sand bank is submerged by about four to five feet. As has been discussed in the later part of this book, about 7,000 years ago, the sea level was about nine feet below the present level. Thus, at the time when Hanuman crossed the sea, most of these shoals were about four to five feet above the water. This explains the small peaks of Mainaak Parvat, which showed up when Hanuman reached inside the sea. However, this is not to take credit away from Hanuman of crossing the sea by swimming because though there were a series of peaks, they were disjointed and far apart from one another. This is true even today and we should not forget that sedimentation, over a period of 7,000 years could result in the filling; up of many of them. Thus, realistically thinking Hanuman must have swum through the distances between the adjacent peaks while crossing the sea both the ways. Even that would have needed amazing physical strength since the distance is a gigantic thirty-kilometers. None other in the army possessed such strength and thus, only Hanuman could do it. Later on, for enabling the entire army of monkeys to cross the sea, they needed to cover the gaps by building a bridge. This further explains the building of a bridge in five days across the sea by Lord Ram. Given the technology available even today, no machine can build a bridge of thirty kilometers in five days. They could do it because the way was available albeit with gaps, which needed to be bridged.

We now return to the main task of dating the events with the help of astronomical evidences.

The Date when Hanuman met Lord Ram - the Date of Starting the Journey to Lanka

After crossing the sea, Hanuman met his companions on the shore who were waiting there and then they all rushed to meet Lord Ram. Hanuman told him that Sita would need to be freed within two months otherwise she would be killed (5/65/26). Ram decides to immediately start the journey to Lanka. In this context, Lord Ram says at 6/4/6 "today is Uttara-Phalguni Nakshatra and tomorrow the moon would meet the Hasta Nakshatra. O Sugreev! we must start our journey immediately along with the army of monkeys". At 6/4/3 Ram says, "today the sun has already risen to the middle of the sky and this is the famous Vijay Mahurt. In my opinion we must start out journey towards Lanka in this Vijay Mahurt to win the battle against Ravan."

Let us see when was the day of Uttara-Phalguni nakshatra (Star 5 p in Virgo). Slide 48 shows that the moon comes at Uttara-Phalguni nakshatra in the noon of 19 September 5076 BC. Therefore, Lord Ram started his march towards Lanka in the afternoon of that day. It also implies that after crossing the sea in the early morning of 14 September, Hanuman with his companions took about six days to reach Kishkindha (Hampi in the state of Karnataka) where Lord Ram was staying.

It has been described subsequently that on their way to Lanka, the army travelled past many villages and through many forests. It is also stated that the army was moving day and night without resting (6/4/70). During the course of their journey to Lanka, the sky has been again described. At 6/4/48-50, Lakshman tells Ram that he could see a number of good omens suggesting a favourable outcome of their journey. He states, "The Venus has gone on the back side, Saptarishi (Ursa Major constellation) are visible brightly and they are moving around the pole star. Trishanku - our ancestor (Trishanku refers to the constellation CRUX in the sky) is brightly visible in the front. The Moola Nakshatra, which protects the Raksach (demons) is being inflicted with Dhumra-Ketu (node)".

To examine the veracity of the above statement, we look at the position of the sky on the day when Ram started the journey to Lanka i.e., on 20 September. Slide 49 shows the position of the sky for Kishkindha at 5.30 a.m. in the morning. Watch Ursa Major (Saptarishi) shining brightly in northern sky. Also note the small constellation called Crux, visible in the

southern sky. Since the army was moving from north to south or rather southwest, Crux must have been seen just in the front. Venus is not visible in the sky since it was located in the constellation of Capricorn that was rising after sunrise. Now comes the last part of the statement - Moola nakshatra inflicted by the Dhumra - Ketu, which refers to the node.

To find the position of nodes (Rahu and Ketu) at that time, we look at slide 50 showing the position of the moon only five days later and cutting the ecliptic at Sagittarius constellation. Note that Moola nakshatra (42 0 OPH) is a part of the Sagittarius constellation. Thus, once again we find that all parts of the statement made by Lakshman, describing the stay are correct and verifiable.

It is stated at 6/4/70 that the army was moving towards Lanka day and night without resting on the way. Thereafter, the army reached the seashore (6/4/98) and started to think about crossing the sea. After the discussions, it was decided that Lord Ram should worship the sea god and request the sea to allow the army passage across it. It is stated at 6/21/10 that Lord Ram worshipped the sea for three nights. Since his worship did not yield any results, he became angry and started killing the animals living in the sea. Then, Lord Ram was pacified by his companions and other sages and one of the monkeys named 'Neel' assured Lord Ram that he could make a bridge over the sea to Lanka (6/22/46). At 6/22/68-72, it is mentioned that they made a 14-yojan bridge on the first day; 20 yojan on the second day; 21 yojan on the third day; 22 yojan on the fourth day and 23 yojan on the fifth day. Thus, the bridge was built in five days.

We have, from the above description, the information that Lord Ram worshipped the sea for three days and the bridge was built in five days. They had begun their journey on 19 September 5076 BC. We do not know the number of days they travelled for before reaching the sea-shore. After crossing over to the other side of the bridge, they proceeded towards the fort of Ravan located at the top of the hill called 'Suvel'. At 6/38/19, it is mentioned that when the army reached the fort, the full moon night occurred. If we see slide 51, the next full moon night occurs on 12 October 5076 BC marking end of Paush. Till this time, the battle between the armies of Ravan and Ram had not begun.

The Yuudh Kaand has devoted more than thirty Sargas to describe the battle fought between the two armies. There are specific names and deeds of more than twenty main commanders of Ravan's army who came out of the fort with their contingents comprising a large number of armed warriors, fought the battle valiantly and then died in the war. This included the mighty brother of Ravan called Kumbhkaran. The above details include two occasions when Lord Ram and Lakshman were also gravely injured and their survival was almost ruled out. On both the occasions, because of help from other people, the two brothers regained consciousness, recouped their strength and fought back.

The reason for mentioning all these details here is that the Valmiki Ramayana is silent about the number of days for which the entire battle was fought. It has been mentioned, though, that Lakshman killed Meghnath after fighting for three days and three nights continuously and Ram killed Ravan after fighting for seven days. It is also mentioned that after everyone else was killed, Ravan came out for battle on a no-moon day (amavasya). By this time, almost all the main twenty commanders and Kumbhkaran had already been killed. Only Meghnath and Ravan remained in the Lanka camp. Moreover, both Ram and Lakshman had already been injured gravely twice and recouped. Now if the war had begun after the full moon day of Paush, and if Ravan came out on amavasya of Maagh (the following month) it will imply that the battle detailed above was fought in just about ten days (subtracting three days for Meghnath's battle with Lakshman). Moreover, it would imply that the entire battle, from the beginning to Ravan's arrival on the scene, was fought in the Krishna Paksh or the period of waning moon. But that is contradictory to the description of war in 6/75/50-51 and 52. Here it is mentioned repeatedly that the brightness of the moon, stars and jewels of the ornaments wore by the warriors lit up the night. This is clear that the brightness has to be only of the moon. The stars and jewels appear to be referred as an allegory. Further, a full moon is also referred to at 6/76/15. Now, if there were moonlit nights during the war including a full moon night also, before Meghnath's battle (three days) and Ravan's battle (ten days), then it clearly implies that war took more than a month to end as the moon-lit nights refer to the Shukla Paksh of Maagh. Therefore, it implies that the war stretched through the month of Maagh and the beginning of Phalgun.

The next important fact stated thereafter at 6/92/15 is that Lakshman killed Meghnath after fighting continuously for three days and three nights. It is further stated that when Ravan was mourning the death of his most able son -Meghnath, that night his deputy said, "today is the chaturdashi (14th tithi) of the Krishna-Paksh (decreasing phase of moon). You may go to the battlefield yourself, on amavasya (no-moon day) to win the battle." This shows that Lakshman killed Meghnath on the 14th day of the decreasing phase of the moon i.e. one day before the no-moon day of that month. Slide 52 shows that the no-moon day falls on 24 November 5076 BC and this suggests that Meghnath was killed on 23 November. Since he was killed after a battle that continued for three days, the battle must have begun on 21 November 5076 BC.

Thus, Ravan went out to the battlefield on 24 November 5076 BC. Initially, Ravan fought the battle with the other commanders of Lord Ram's army. It is stated that none of them could contain and defeat Ravan. This perhaps continued for three days because at 6/110/37, it is mentioned that the fight between Lord Ram and Ravan continued for seven days. Ravan was killed on the dashmi - 10th day of the increasing phase of moon. Counting the tenth day from the no-moon day of 24 November 5076 BC, we reach 4 December 5076 BC. Therefore, one may conclude that Ravan was killed by Lord Ram on 4 December 5076 BC.

After the above description, the Valmiki Ramayana is silent about the days as well as dates. No description of astronomical bodies is available to ascertain any dates. However, we must note that when Ravan was killed it was the 10th tithi of Phalgun month (increasing phase), which means that only five to six days were left for the month to end. After Phalgun, we count the first fourteen days of the Chaitra month in the year nearing its completion. These fourteen days correspond to the decreasing phase of the moon (Krishna Paksh) and hence, our year ends with amavasya or no-moon day of Chaitra. Let us examine a few details mentioned in the Ramayana about the date of Lord Ram's return to Ayodhya.

At 6/122/21, the Ramayana states that after Ravan was killed, various sages told Lord Ram that the period of fourteen years was almost over. On his return journey, Lord Ram came on a Pushpak Viman, which should be a

fast moving chariot. Finally, at 6/127/1 it is stated, "after the completion of 14 years Ram reached the ashram of Sage Bharadwaj (located on the outskirts of Ayodhya) on the Panchami (refers to the fifth day from the no-moon day or fifth day from the full moon day)."

It remains a matter of conjecture, whether Lord Ram reached Sage Bharadwaj's ashram on the Panchami of Krishna Paksh (decreasing phase) or Shukla Paksh (increasing phase of moon) of Chaitra. However, the mention of the words "after the completion of 14 years" very strongly suggests that it has to be the Panchami of the Shukla Paksh of Chaitra month. If it was the Panchami of Krishna Paksh, the statement becomes incorrect because the year does not end by then. Assuming that Lord Ram reached Sage Bharadwaj's ashram on Shukla Panchami of Chaitra, the corresponding date comes to 29 December of the year 5076 BC.

There is a lot of curiosity about the use of the Pushpak Viman by Lord Ram on the return journey. It is generally believed that they flew back on it from Lanka to Ayodhya. In this context, to be realistic, we must consider a few things. Lord Ram and Lakshman must have been completely exhausted and fatigued of the continuous travel, stress and fighting the enemy for two months prior to Ravan's killing. Sita was also equally weak and fragile because she was barely eating during the period of her captivity at Lanka (one year). There were many vanars like Sugreev, Angad, Jamvant etc and Vibhishan who were equally drained, yet they wanted to come along with Lord Ram to see him on the throne of Ayodhya. In that tired and weary physical condition, they were faced with the task of travelling from Lanka to Ayodhya, a distance of more than 2,000 kilometers. Under the circumstances, it was very obvious that they decided to take a chariot for the journey rather than travelling by foot - the way they went during their onward journey over a period of several years. The chariot or vehicle they used was the Pushpak Viman. It must have been a special vehicle with reasonable comfort for the passengers. Further, Lord Ram did not return straight from the battlefield to Ayodhya. He visited most of the places, where he had gone to during his onward journey, met the sages in their ashrams etc. Since the details of Lord Ram's return journey were not of any great significance, Maharishi Valmiki has very succinctly described these in

only one Sarga. Nevertheless, the fact remains that Lord Ram had travelled all the way from Lanka to Ayodhya almost by the same route, which he had taken for the onward journey.

The date of Ravan's killing was 4 December and the date Lord Ram reached Bharadwaj's ashram was 29 December. During this period of about twenty-five days, they travelled about 2,000 kilometers. This implies that they travelled about eighty kilometers per day, which appears to be highly plausible and very reasonable. The account of the Pushpak Viman that it used to fly appears to be only a metaphor or imagination because Valmiki Ramayana gives a very detailed description of various accomplishments of society at that time. It cannot be construed, from such details that people had flying machines. There is a mention of machines operating the gates of the fort of Lanka, machines used to blow up mounds of earth or small hills but nothing has been mentioned of such an advancement as that of the existence of airplanes. It is not difficult to comprehend how large the factories and how advanced the technology is required to manufacture a flying machine. The description of society and accomplishments thereof in the Ramayana clearly rule out the possibilities of someone possessing such a flying machine. Moreover, there is a clear mention at 3/42/9 that when Ravan and Maarech planned the abduction of Sita, the chariot of Ravan was like an airplane. This statement clarifies sufficiently that the fast moving chariots were referred to as airplanes despite being only chariots. Thus, to the best of our understanding the Pushpak Viman was a fast moving chariot, which in those days used to appear to be as fast as flying. In referring to those chariots, the word 'flying' was often used as an allegory.

Hence, to conclude the discussion we may hold that the most reasonable date for Lord Ram's return to Sage Bharadwaj's ashram was 29 December 5076 BC. The description of events in the Valmiki Ramayana show that Lord Ram reached the ashram of Bhairat at Nandigram the very next day, 30 December 5076 BC and met the entire family and the populace of Ayodhya waiting there for him. This concludes the core subject of ascertaining the dates of the era of Lord Ram. Moreover, it is clearly mentioned at 1/4/1 that Sage Valmiki composed this epic after the accession of Lord Ram on the throne of Ayodhya. Therefore, it can be easily

concluded that the composition of the Ramayana began around the beginning of 5075 BC.





CHAPTER XII

Archaeological Evidence Available to Support the Findings

THE ODDS AGAINST finding archaeological evidence for the times of Lord Ram are very heavy. Modern historians classify the period of the sixth millennium BC as 'pre-historic' and define it as a period for which no archaeological or literary evidence is available. The paradox, therefore, is that we are expected to look for evidence for the same 'pre-historic' age.

The possibility of having available with us, some archaeological remains of that time, is highly remote. There are hardly any structures being built, even today, which can survive the onslaught of natural wear and tear for seven thousand years. A period of 7,000 years means a substantial rise in the sea level as well as the dwelling level of the population. Hence, one has to perform the excavation at the levels where civilization existed 7,000 years ago, to explore the possibility of finding some remains of that time. However, at a time when the historians of the world have already dismissed the case of Indian history by concluding that the Rig Veda was written in 1500 BC and the stories of Lord Ram, Lord Krishna and others given in the Purans as unreal and invented, who would be zealous enough to carry out the onerous task of unearthing such 'difficult to find' evidences.

Some efforts on this front have been made recently. The latest archaeological findings show that Vedic culture developed in the Sapta Sindhu region or the Indus - Saraswati region. In this region, more than 1500 sites have been discovered and all of them show remarkable affinity.

Some archaeologists, who have studied the pottery styles and cultural artifacts have held that Indus - Saraswati culture shows a continuity traceable to at least 8,000 BC. Carbon dating of the remains of horses found at the Harappan sites have showed their existence long before 2400 BC. Very significant archaeological discoveries are being made at Dwarka, where an entire city has been found submerged under the sea. Archaeological research is a continuous process to find out the historical truths, and the future may reveal many evidences in support of the antiquity of Ramayana. However, for the present, at least two evidences already available can be examined. The first is the Saraswati river, which has now dried up but at one point of time was the mightiest river known.

According to the Rig Veda, the oldest composition of the entire ancient Indian literature, the Saraswati river was so vast and large that at some places its width was almost seven kilometers. At 6.61.2, the Rig Veda describes Saraswati as the river with so much power that it breaks the rocks of mountains standing on its two sides. Out of the seven rivers mentioned repeatedly in Rig Veda, including Indus and Ganges, Saraswati has been declared as the one with maximum water flow.

It has now been scientifically proved with the help of satellite imagery, that before changing its course at least four times, the Saraswati river dried up completely by about 1900 BC. It is reasonable to conclude that it must have taken several thousand years for a seven-kilometer wide river to dry up into the sands of the desert. An intermediate date available on the basis of satellite data combined with field archaeological studies, suggests that the Saraswati had stopped being a perennial river long before 3000 BC.

A Flowing Saraswati River in Ramayana

The Saraswati river has been described in the Valmiki Ramayana twice and at both the places, it has been described as a flowing river during those times. The first reference comes at a time when Prince Bharat was retuning to Ayodhya from his maternal uncle's kingdom, after the exile of Lord Ram and death of King Dashrath. Bharat was staying at the kingdom of Kekeya at that time, which is located in northwest region of India. We all know that Ayodhya corresponds to modern Faizabad, in Uttar Pradesh. Thus, when

Bharat was on his way from Kekeya to Ayodhya, at 2/71/1-5 the description of the journey mentions, "Bharat started his journey in the easterly direction. Crossed the Sudama river first and then the Sutudri river, which flows towards west." Sutudri river is now called Satluj. Thereafter, he crossed mighty hills and then reached the Saraswati and Ganga rivers. Subsequently, he reached the kingdom of Matsya and from there travelled in a northerly direction, on his way to Ayodhya. The description clearly shows that Saraswati river was flowing and it was located in the east of modern Satluj. Historians and archeologists agree that before drying up, the Saraswati was actually flowing in that region.

The next mention of the Saraswati comes at a time when Sugreev was sending vanars (monkeys) in all directions in search of Sita. Here, at 4/40/20-22, Sugreev tells them, "You search for Sita on and around the banks of Bhagirathi, Saryu, Kaushiki, Kalindi, Yamuna, Saraswati, Sindhu, Sone, Mahi and Kaalmahi rivers".

The above description may not state much, but they make at least one thing clear, that river Saraswati was among those rivers that were existing or flowing at that time. The banks of the river would exist only if the river exists. In view of these two unambiguous references to the Saraswati river in the Valmiki Ramayana, we can at least conclude that the dating of the period of Lord Ram has to be prior to 3000 BC. In fact, owing to the long time span since the times of the Ramayana, many of the geographical sites described in the Valmiki Ramayana have undergone drastic changes. Yet, a large number of such sites are available and are being researched.

The Remains of the Bridge Built by Lord Ram

The second, and a very remarkable archaeological evidence available today, is the 'remains of the bridge' built by Lord Ram on his way to Lanka. On the Indian side of the Palk Strait, there is a historical cum religious place called 'Cheedu - Karai' where the remains of the bridge can be seen even today at a depth of about ten feet under the sea. Another feature of the 'remains of the bridge' is that today these are about two kilometers inside the sea. This shows that since the times of building the bridge, the sea has taken over about two kilometers of land.

It is common knowledge that ever since the end of the last Ice Age, about 18,000 years ago or around 16,000 BC, the polar caps and glaciers are melting continuously. This phenomenon, coupled with the gradual warming of the earth, is resulting in the continuous rise of the sea level. All over the world, scientists have documented the retreat of mountain glaciers, including the Rocky Mountains of North America, the Alps of Europe, and the Himalayas of Asia. The overall rise in the sea level is a combination of all these factors and it is termed as Eustatic Sea Level rise.

The rate of rise in the sea level is a matter of intense research, nonetheless the sea level rise (SLR) is a real and continuous phenomenon. Large cities, near seashores like New York, are under the threat of slowly getting submerged and plans are being made to counter the problem. Committees have been constituted in several countries like Australia to study the process in its entirety and suggest the possibilities of town planning for future.

The website of NASA, GCMD (Global Change Master Directory) Learning Center provides valuable and documented information, in great detail on the subject. It mentions that the sea level is indeed rising and the rate of rise is determined by finding the average of changes in sea level over the entire globe for the past century. The site explains that the Intergovernmental Panel on Climate Change (IPCC) has estimated that the global average sea level has risen about 10 to 25 cm in the last 100 years alone (or 1mm to 2.5 mm per year). Further, IPCC estimates that sea level rise will not only continue into the next century, but will continue at an accelerated rate.

Several estimates of the rate of global sea level rise determined from tide gauge records have been summarized by different research scholars. These estimates do not differ widely though the accuracy of data depends on the length of the period for which data is available etc.

With a view to have a clear answer to the question, a research scholar on this subject was consulted with the following question:

“What was the average rise of sea level over the past 7,000 years? To be precise, from 5,000 BC till the modern times, how much has the sea

level has gone up?"

The reply received was as follows:

"Estimates for global average eustatic sea level rise for the last 7000 years range from 0.2 to 0.4 mm / year. Recent rates of global SLR for the last 120 years range between 1 -2 mm / year, thus clearly an acceleration in the trend in the last century over that of the 5-6 millennia."

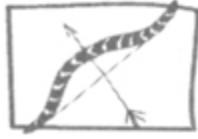
The reply makes one thing very clear, that there has been a remarkable increase in the rate of the rise of sea level over the period of time. As stated earlier, IPCC has estimated that the global average sea level has risen at a rate of 1 mm to 2.5 mm per year for the last 100 years while the reply above mentions a rate of 1-2 mm per year for last 120 years. However, when the average of the rate of rise in sea level is considered for 7,000 years, the rate is only about 0.2 mm to 0.4 mm per year, which shows that initially the rise in sea level was very slow and it gained momentum over a period of time. Today, it is rising as fast as 1.2 mm per year. In view of the above stated bias, if we work out the rise in sea level over a period of 7,100 years for the upper end of the estimated band, the rise in eustatic sea level comes to about 2,800 mms, which roughly corresponds to 9.34 feet. That is very nearly the depth at which the "remains of the bridge" are located today. If we discount the factor that over a long period of time even the 'remains of the bridge' must have also got eroded due to continuous erosion by waves of water, one may note that the 'remains of the bridge' are located at the same level where the sea level existed 7,000 years ago.

Analysing logically, a bridge is never built under the water. Evidently, the sea level is going up, albeit gradually. Over the centuries, slowly the bridge must have got submerged and was subjected to continuous erosion. The local populace though, must not have forgotten the location of the bridge. Over a span of seven thousand years today, the sea has stretched itself over the land by almost two kilometers and the remains have submerged by ten feet. Yet, the local people can locate the remains as has been done for ages.

This archaeological evidence actually offers a very precise and accurate computation for the dating of the Ramayana because the height of the sea

level above the 'remains of bridge' can be measured accurately and even if the conservative estimates for the rate of rise in sea level are adopted, the time of the construction of the bridge can be easily worked out. The rough computations, in any case, coincide remarkably well with the period of time found on the basis of the astronomical dating as detailed above. The mention of a flowing Saraswati river in the Ramayana puts the epic well before the dates assigned to it by most modern indologists. However, nothing more can be inferred from it in specific terms as far as precise dating is concerned. On the other hand, the rate of rise in eustatic sea level is a finding based on scientific research and if we divide the actual rise of sea level above the 'remains of the bridge' by the rate of rise of sea level, we can arrive at the number of years that have elapsed since the construction of the bridge. The computation in earlier paragraphs clearly show that the rise in the sea level over a period of about 7,100 years almost tallies with the depth of the remains of the bridge under present sea level. This confirmation in a very scientific manner, provides an independent interlocking evidence in support of the dating of the Ramayana carried out through the astronomical route in this book.





CHAPTER XIII

The Dating of the Ramayana -the Mahabharata War and the Beginning of Kali Yuga

IN THIS CHAPTER, we will examine the planetary configurations of a number of dates to ascertain which of them is the probable date of the beginning of the Kali Yuga. Thereafter, we will explore the possibility of finding a relationship between the dates of the era of Lord Ram arrived at in this book with the traditional Yuga system. However, before going into that discussion, let us first understand the views of modern historians about ancient Indian history including the dating of the Ramayana and learn why and how these views are changing recently.

Western Views about the Dating of the Ramayana

In the absence of any documented history of our country for the period prior to about 300 BC, the antiquity and dating of the ancient Indian civilization has remained open to speculation. The best evidence available with us, of the period of our 'glorious past' is the ancient sacred literature consisting of Vedas, epics, Upanishads etc. The dating of ancient Indian history is, therefore, almost equivalent to the dating of these scriptures. Indians claim this literature to be as old as the antiquity of any other civilization of the world. However, in the absence of any concrete evidence in this regard, modern historians were not ready to accept such claims.

The accepted version of the ancient history of our civilization is based on the Aryan Invasion Theory (AIT) authored by the Max Muller.

Propounded with the coloured vision of racial superiority, the AIT states that the original inhabitants of our country were the people of Harappan civilization or Indus Valley Civilization. Their script could not be deciphered so far, though recently it has been claimed to have been deciphered as later Vedic Sanskrit. The AIT states that in about 1500 BC, the northern part of India was invaded by a light-skinned race of nomadic people called Aryans, who descended from Central Asia and destroyed the inhabitants of Indus Valley Civilization. The barbaric Aryans then imposed their language and culture on the people of Harappan civilization. This supposedly marked the beginning of the early Vedic era during which Vedic literature began getting composed.

The Rig Veda, being the earliest among all the Vedic compositions, was accordingly assigned the date of about 1500 BC. The other Vedas and the rest of the Vedic literature is generally believed to have been composed in the later Vedic period of about 1200 BC onwards without assigning any specific dates to the individual compilations. Since the epics - the Ramayana and the Mahabharata cannot be classified strictly as Vedic literature, these are assigned vague time frames too. The one issue on which there is no dispute, however, is that the Ramayana is dated earlier than Mahabharata. According to modern indologists, the most 'reasonable' date of the Mahabharata war is around 950 BC. This was proposed by Pargiter mainly to conform to Max Muller 's date for the composition of the Rig Veda as 1500 BC.

While the Mahabharata is dated around 950 BC, the Ramayana has not been assigned any specific date under the AIT. In regard to this epic, the only belief that exists is that it is dated earlier than the Mahabharata. The logic extended is that while the Ramayana was a war between Aryans and non-Aryans, at a time when Aryans were trying to impose themselves over the non Aryans in this country, the Mahabharata was a war among Aryans who had already established their dominance over the entire country. In this background, it is not surprising that many historians believe the Ramayana and Mahabharata to be fictitious tales created by Brahmins.

Although lacking supporting scientific evidence, the Aryan Invasion Theory was accepted and promulgated, firstly, because it provided a

convenient explanation for the British subjugation of India. Secondly, it could reconcile the ancient Indian literature and history with the Biblical creation dated around 4000 BC, as the historians of that time could not conceive of any region outside of Europe that could have created a high civilization, centuries before their own. Lastly, the AIT theory created a further division between the people of India, in terms of north and south and helped in propagating the principle of divide and rule followed by the British.

Revising the Date of the Mahabharata - the Probable Dates

Numerous evidences from archaeology and literary sources are now available to prove that the AIT was just fiction without any sound basis. Contemporary researchers like Professor Subhash Kak, Dr. B.G. Siddharth, Dr. S.B. Roy, Shri B.B. Lal, and Dr. David Frawley have made rich contributions to scientifically disprove the AIT. The latest archaeological findings show that the Vedic culture developed in the Sapta - Sindhu region or the Indus -Saraswati region and the cultural tradition can be traced back to about 8000 BC without any break. The existence of the cultural continuity ever since that time dismisses the very basis of invasion by any tribe from outside the country. Further, the findings of Kenneth Kennedy of Cornell University suggest that there is no evidence of any demographic discontinuity during 4500 BC to 800 BC. Thus, one can conclude that there was no significant influx of people into this country during the mentioned period. This fact also gets corroborated from the reading of Vedic texts in which there is no memory of Aryans moving in this country from an outside or alien place. In addition to the above, there are certain planetary references in the Rig Veda that have been astronomically dated to at least 6500 BC. These findings prove the existence of history and culture of our civilization several millennia prior to the dates assigned to it under AIT.

As a result of an ever-growing number of deficiencies being found in the hypothetical Aryan Invasion Theory, there has been a general thinking of revising the dates assigned earlier to the ancient Indian scriptures. The preponderance of the evidence clearly suggest a far earlier date for the Mahabharata as compared to 950 BC assigned to it under the AIT. The revision of the date of the Mahabharata automatically affects the dating of

the Ramayana, because there is no doubt that the latter epic is dated much earlier than Mahabharata.

The probable dates for the Mahabharata, have been comprehensively discussed and analysed by Professor Subhash Kak in his book *The Astronomical Code of Rig Veda* (pp. 60-67). In this context, Prof. Kak has analysed in detail the three main Indian traditions about the date of Mahabharata war - Puranic evidence, Kali Yuga tradition by Aiyabhatta and the statement of Varahamihira to ascertain the probable date for the war. The findings and the conclusions of this book are being extensively quoted hereinafter and the probable dates of the Mahabharata examined therein are also discussed briefly.

The Purans contain detailed lists of the kings of the royal dynasties of ancient India. On the basis of these genealogies, it is possible to arrive at the probable date of the Mahabharata war. Hence, the first two dates for the Mahabharata war, considered by Prof. Kak on the basis of the Puranic evidence involves the detailed analysis and reconstruction of the genealogies of the royal dynasties mentioned in the Purans.

The first date examined by Prof. Kak for the war is 1424 BC, which is based on the description in Purans, mentioning a gap of 1,050 years between King Parikshit and the Mahapadam Nanda, the Nanda king who overthrew the Maurya empire and whose date of accession to the throne is a matter of documented history. However, this date comes in direct conflict with the latest archaeological findings in regard to Saraswati River. In the Mahabharata, the River Saraswati has been described as a major 'ocean going river' at that time. The archaeological evidence show that Saraswati river dried up as early as 1900 BC. Therefore, a dating of 1424 BC for the Mahabharata war is too late and thus, not acceptable.

The next probable date for the Mahabharata war is 1924 BC, which is again based on the Puranic genealogies. The year 1924 BC has been arrived at on the basis of the interpretation of some historians that the gap between King Parikshit and Nanda was 1,500 years. The count has also been 'obtained by adding up the durations of the reigns'. The only support for accepting this date for the Mahabharata war is that it coincides with the probable time of the drying up of Saraswati river (1900 BC) as per

the latest archaeological findings. It is believed that at this time a major natural catastrophe had occurred and perhaps the same has been poetically described in the epic. However, it may be seen, that this proposition has many variables and it does not pinpoint to any particular date. The archaeological finding of the drying up of Saraswati river, even if due to some natural catastrophe, around 1900 BC is an approximate date and it does not refer to any particular year. This date, therefore, has its own limitations.

The third probable date for the Mahabharata war is based on the statement of Varahamihira (AD 505) in his book, *Brihat Samhita*, that the commencement of the Saka era took place 2,526 years after the rule of King Yudhishtira. If this Saka era is considered to be the Salvahana era, which began in AD 78, then the year of Mahabharata war works out to be 2449 BC. Prof. Kak mentions that the date of 2449 BC may be correct only if "genealogies (mentioned in Purans) represent the chief kings". While concluding the discussion, he holds that the acceptance of this date would imply that "the genealogical lists are hopelessly incomplete, which is plausible if a great catastrophe, such as drying up of Saraswati, caused the tradition to be interrupted." Notwithstanding the doubts about the acceptance of this date for Mahabharata war, it remains a candidate among the other probable dates.

Some scholars are of the view that the Saka era referred to in Varahamihira's statement was the one started by an earlier Saka king and in their view the date actually works out to 3137 BC. This is the earliest date suggested for the Mahabharata war. The famous Indian astronomer Aryabhatta (AD 476) also worked out 3137 BC as the year for the Mahabharata war. Further, most historians hold that as per the famous Aihole inscription (AD 634) the date for the Mahabharata war is 3137 BC. In view of such a consensus, this date is considered the traditional date for the Mahabharata war. Given the considerable importance assigned to it, this date remains the most probable one.

Having discussed the different probable dates for the Mahabharata war, let us examine a few interesting details about the date of the war and the beginning of the Kali yuga and also the planetary positions for the beginning

of the Kali yuga. This offers us an opportunity to verify the correctness of the date of Mahabharata war and the date of beginning of the Kali yuga, with the help of the Planetarium software.

The Date of the Beginning of Kali Yuga - Astronomical Confirmation

The epic Mahabharata mentions that the Kali yuga began thirty-five years after the end of the Mahabharata war. Therefore, if we know the date(s) for the Mahabharata war we can deduce the dates for the beginning of the Kali yuga by adding thirty-five years to it. Further, it has been stated that the beginning of the Kali yuga was marked by the conjunction of all the seven planets (the sun and the moon included) visible to the unaided eye around the Revati nakshatra (star Zeta in Pisces constellation). The occurrence of a conjunction of main seven planets around the Revati nakshatra, can be verified with the help of the Planetarium software.

As discussed earlier, the dates of 1424 BC, 1924 BC, 2449 BC and 3137 BC are the four candidates for the probable dates for the Mahabharata war. Since the Kali yuga began thirty-five years after the war, the respective years for this event would be 1389 BC, 1889 BC, 2414 BC and 3102 BC. In regard to the year 3102 BC, Aryabhatta had worked out the date of the beginning of the Kali yuga also. This date is generally considered to be 17/18 February 3102 BC . We would see if there was a conjunction of all the seven visible planets around the star Revati (Zeta in Pisces constellation) in any of these years.

Before examining the planetary confirmations, it would be helpful to understand a simple astronomical concept at this juncture. If the conjunction is close enough and involves both the sun and the moon, then it has to be a no-moon day (amavasya) because these two remain together only on an amavasya day. The sun would be near Revati Nakshatra, located in Pisces constellation only once in a year. During the period under examination, the sun remains in Pisces constellation during the month of February, tallying in this regard with the generally accepted date of beginning of Kali yuga as 17/18 February 3102 BC.

Slide 53 shows the position of the sky on 9 February 1389 BC, which is the day when the sun and the moon are nearest possible to Revati nakshatra. For the other dates, when the sun is actually on Revati, the moon is not nearby and thus, the only nearest possible conjunction is on this date. It may be seen that only five planets the sun, the moon, Mercury, Venus and Saturn are in Pisces. Mars is in Taurus, while Jupiter is in Leo. Anyone would vouch that this is not even a near conjunction of seven planets. Therefore, this date cannot be accepted as the date of beginning of the Kali yuga

Slide 54 shows the position of sky on 11 February 1889 BC, the no-moon day when the sun and the moon were nearest possible to Revati. It may be seen that this is, again, a conjunction of only five planets because Jupiter is in Taurus and Mars is in Gemini. Obviously, this cannot be classified as a conjunction of seven planets and hence does not merit consideration as a probable date for beginning of Kali yuga.

Slide 55 displays the position of sky on 20 February 2414 BC. Again, this is the date when the sun could be nearest to the Revati nakshatra on a no-moon day because for other days, when the sun comes very close to Revati, the moon is very far away from the sun. As may be seen, this is also a conjunction of only five planets. Mars is in Taurus and Saturn is in Gemini. Hence, this configuration is also not a conjunction of seven planets.

Slide 56 shows the position of the sky on 18 February 3102 BC, the date worked out by Aryabhatta for the beginning of the Kali yuga. As may be seen, while the sun is in the middle of the constellation of Aries, Saturn is in Aquarius, Mars is too far away in Cancer / Leo and moon is not in the vision at all. Obviously, it was not an amavasya (no-moon) day, otherwise the moon would have been near the sun. Actually, the no-moon day, that year was on 1 February as shown in slide 57 and on this date, six planets were located around Revati. However, Mars is clearly very far away in Cancer constellation and therefore, there will be a hesitation in accepting this as a conjunction of all the main seven planets. Hence, this date also cannot be accepted as the authentic date for the beginning of the Kali yuga.

Now look at the position of the planets one year earlier i.e. in the year 3101 BC. Slide 58 gives the sky view on 22 January 3101 BC. One can see

that all the planets, including the sun and the moon are located around the Revati nakshatra. Saturn, the farthest among the seven planets from Revati is still located in the boundary of the Pisces constellation. Out of all the dates examined so far, this appears to be the best candidate for a seven-planet conjunction at Revati.

The no-moon (amavasya) day of 22 January 3101 BC has another very important significance. The full moon-day after 22 January occurs on 6 February 3101 BC and on this day, the full moon occurs at the Chitra Nakshatra as shown in slide 59. This implies that the month ended on 6 February 3101 BC was the month of Chaitra and therefore the no-moon day of 22 January 3101 BC was the amavasya of Chaitra, which represents the last day of the traditional Indian lunar year. We may recollect that in our calendar, the lunar year ends on the amavasya of Chaitra and the New Year begins with the first day of Shukla Paksh of Chaitra. Hence, the conjunction of the sun and the moon on Revati, along with the other five main planets, at the time of beginning of the Kali yuga (which is the same as the end of the Dwapar yuga), had a logical meaning. It meant that the end of the yuga was co-terminus with the end of the year and therefore, it corroborates the belief that the Yuga system was a system of counting long time periods in terms of years.

It may also be noted that in 3101 BC, the month of Chaitra is ending in February while in the times of Lord Ram (5114 BC) it used to end in January because over a span of about 2,000 years, the sun shifted in its position by about 25-30 days.

To be objective in our approach and fair to the findings, the position of planets one year before and after the years 1389 BC, 1889 BC, 2414 BC and 3102 BC were also examined for a possible conjunction. Beyond one year, the examination of the stellar position would not serve any useful purpose because Jupiter, which moves one zodiac per year, would move too far away from Revati to result in any meaningful conjunction. The examination of the position of planets for one year each on either side, confirmed that for none of these years a conjunction was observed around the Revati nakshatra, except for 3101 BC. Does it mean that this was the date of the beginning of Kali yuga? Instead of answering this in a

categorical 'yes' or 'no', we can conclude that if the choice of the dates is limited to the ones examined in this chapter, then the odds in favour of acceptance of this date as the date of beginning of the Kali yuga far outweigh the odds against it. We would now examine how the dating of the era of Lord Ram arrived at in this book fits within the Yuga system.

The Astronomical Dating of the Ramayana and the Puranic Concept of Yuga

Though the word yuga is found in Valmiki Ramayana (1/11/11), its details and description are available only in the Purans. Without venturing into the discussion about certain obvious inconsistencies in the Puranic description of the aspects of the yuga cycle vis-a-vis other ancient scriptures, it can be stated that the Yuga system comprises a cycle of four yugas with following details. Sat yuga of 4,000 years with 400 years long sandhya-period on either side; the Treta yuga of 3,000 years with sandhya a period of 300 years each on both sides; Dwapar yuga of 2,000 years with sandhya-period of 200 years on each side and Kaliyuga is of 1,000 years with sandhya, a period of 100 years on either side. Thus, the span of Sat yuga was 4,800 years (4,000 +400+400), Treta of 3,600 years, Dwapar of 2,400 years and Kali yuga of 1,200 years. One thing is absolutely clear that the Yuga system was again a method of time keeping when the time period spanned thousands of years. The four yugas are, thus, completed in 12,000 years and the cycle repeats itself.

In his book *Ancient Indian Historical Tradition*, Pargiter (1922) having examined the Puranic description of yugas, has conclude that the Sat yuga ended with the destruction of the Haihayas: the Treta yuga began approximately with Sagar (an ancestor of Lord Ram) and Treta ended with Lord Ram's destruction of Raksachs (demons); the Dwapar yuga ended with Bharat a war. The conclusion that 'Treta-Yuga ended with Lord Ram's destruction of demons is worth attempting verification and perhaps, draw some conclusions.

Before attempting the verification, we make an assumption for the subsequent discussion. The assumption is that in counting the length of yugas, the sandhya periods were within the overall yuga period. In

mathematical terms, instead of saying that the Sat yuga was of 4,000 years with two additional sandhya periods of 400 years on both side to make it 4,800 years long ($400 + 4000 + 400 = 4,800$), we assume that the sandhya periods were included in the main yuga period itself. This implies that the first 400 years of the yuga were of sandhya period, then 3,200 years constituted the main yuga and last 400 years were again in sandhya. In this manner, the entire yuga was of only 4,000 years with two inclusive sandhya periods of 400 years on both sides ($400 + 3,200 + 400 = 4,000$). Counting in this manner, the length of Treta yuga remains 3,000 years, Dwapar yuga of 2,000 years, and Kali yuga remains of 1,000 years, and the entire yuga cycle completes itself in 10,000 years and repeats again.

Assuming that the Dwapar yuga was of 2,000 years, if we adopt the year 3101 BC as a year for the beginning of Kali yuga, then it implies that the Treta yuga ended with the end of the year of 5101 BC. At this time, Lord Ram was just under thirteen years, because the year ends with amavasya of Chaitra while his birthday falls on navmi tithi of Shukla Paksh of Chaitra. Hence, there was a gap of just nine days. That shows that at the end of Treta yuga, Lord Ram was just nine days short of thirteen years. Was there anything special done by Lord Ram in that year of his life? Let us examine only two statements in this context and recall the sequence of events relating to this period of his life.

The sequence of events is as follows: Sage Vishwamitra wanted to perform a yagya but many rakshasas (demons) were creating hindrances. Thus, he came to King Dashrath and sought the help of Lord Ram and Lakshman in the completion of his yagya. Reluctantly, King Dashrath allowed the two brothers to go with the sage. Sage Vishwamitra, who himself was a great warrior, firstly taught the art of fighting and the use of weapons to both the brothers and then Lord Ram, while protecting the yagya, killed the rakshasas around the ashram of Sage Vishwamitra. After his yagya was over, Vishwamitra travelled with Lord Ram and Lakshman to the kingdom of King Janak (located at the foothills of modern Nepal) where Lord Ram broke the renowned bow of Shiva and fulfilled the condition for marrying Sita. Thereafter, the marriage of Lord Ram with Sita and the marriage of the other three brothers of Lord Ram were solemnized.

The first two statements to be examined is that when Sage Vishwamitra requisitioned the services of Lord Ram, King Dashrath, reluctant to send Lord Ram said, "Ram is just a 12 year old lad and does not even know the intricacies of war, how will he be helpful to you'" (3/38/1-6). This statement makes it clear that Lord Ram, at the time of going with Sage Vishwamitra had completed only twelve years of age.

The second statement to be examined is what Sita told Ravan, at Panchwati before her abduction. At 3/47/1-11 she states that, "I had stayed in Ayodhya after my marriage for twelve years and at the beginning of the thirteenth year, Lord Ram at the insistence of his stepmother, Kekeyi, had to leave for forests. At that time Lord Ram's age was 25 years ..." Thus, by reducing twelve years from the age of twenty-five years we can conclude that at the time of his marriage, Lord Ram was thirteen years old.

Therefore, we draw the inference that at the time when Lord Ram's association with Sage Vishwamitra began, he was more than twelve years and when his association with Sage Vishwamitra ended (with his marriage with Sita) he was thirteen years. In the intervening period, which means in the thirteenth year of his life, he had his first encounter with the rakshasas, whom he killed.

This description matches with the statement of Pargiter (quoted earlier) that "Treta ended with Lord Ram's destruction of raksachs". The phrase 'destruction of raksachs' does not refer to any specific point of time in the life of Lord Ram. However, we know that Lord Ram encountered rakshasas only during the thirteenth year of his age or during eleventh to fourteenth year of his exile when his age was between thirty-six and thirty-nine years. We may, therefore, say that the destruction of rakshasas by Lord Ram actually began with the his protection of the yagya of Vishwamitra and ended with the death of Ravan, Therefore, it becomes highly plausible that the Treta yuga ended in 5101 BC when Lord Ram had already begun his pursuit to remove the rakshasas from the face of the earth.

This proposition has another significant meaning. If the Treta yuga ended in 5101 BC, the Sat yuga must have ended in 8101 BC and it should have begun in 12100 BC to be precise. It simply means that even Lord

Ram's era is a matter of relatively recent history for Indian civilization. There should be a great history spanning at least 7,000 years before the birth of Lord Ram itself and that is evident from the reading of the Valmiki Ramayana because it gives the details of the historical past as it existed during the lifetime of Lord Ram. One such detail is the narration of the genealogy of Lord Ram's own family at 1/70/20-45 (repeated again at 2/110/3-35) in which he is himself mentioned at the generation number 39. The genealogies of many other royal families have also been described. For instance, 1/71/1-20 describes the genealogy of King Janak of Mithila. Another noteworthy point is the worship of the trinity Gods Brahma, Vishnu and Shiva during the time of Lord Ram and there is no mention of any details about the birth of any of these three Gods. About the Brahma, it has been mentioned at 1 / 70/20 that, "nothing is known about the birth of Brahma". This statement itself proves that there was a long historical past of our civilization prior to the time when Lord Ram took the birth and lived on the earth and the earliest part of it was almost obscure even during Lord Ram's times.

Therefore, we find that the date of 22 Januair 3101 BC as the date of the beginning of Kali yuga and the astronomical dating of the era of Lord Ram get interestingly linked up with the yuga system, provided we maintain the assumption that the yuga cycle completed in 10,000 years and not in 12,000 years. However, it is quite evident that these conclusions are based on assumptions and have their limitations. Further corroboration is required in terms of evidences — archeological, astronomical or otherwise to make a final judgement on the issue.

One may argue that it is justified to make such an assumption in regard to the yuga cycle described in the Purans. In this regard, one must point out that given the ambiguities and inconsistencies of this concept, it is not unjustified too. In this regard, it may be noted that the concept of yuga was not in prevalence during most of the period it covers. In simpler words, it means that in all probability this concept got the final shape during the Kali yuga (time of finalisation of Purans), although it described the time frame and the qualities of the society during the Sat yuga (Rig Veda), Treta yuga (the Ramayana) and the Dwapar yuga. The reason for holding this view is that although the Rig Veda employs the words 'during ancient times', 'long-

long ago' or 'in the past' etc to refer to certain events of the past, the word 'yuga' or 'Sat yuga' etc. has not been mentioned in it at all. The Ramayana mentions the word yuga at a couple of places but no description or detail of this concept is found in the entire text. This indicates that the concept of yuga may have been vaguely known in the times of Vedas and Ramayana but it developed at a very late date. This concept is found in the Purans and as the discussion in the next chapter would show, the Purans were written over a very long period of time, running into several centuries, and were finally concluded well after the beginning of the Kali yuga. The mention of the names of the kings of the Kali yuga as well as their deeds is a clear indication in this regard. Since the yuga system is an exclusive contribution of the Purans, it is easy to understand that this concept got the final shape as late as the second millennium BC, when as per the yuga cycle, the Kali yuga had already begun. It appears to be logical also because at that time our history had grown so old that to distinguish its different earlier phases, it was necessary to refer to Sat yuga, or Treta yuga or the Dwapar. In view of the above, it is not surprising that we find inconsistencies in this system, which need to be ironed out by making suitable adjustments or assumptions.

However, to maintain the objectivity of the analysis carried out in this book, no hasty conclusions are drawn. The concept of yuga is highly oblique and needs lots of explaining and clarifications. More research and analysis of our ancient literature is still needed, particularly in the case of Purans, to lift the veil of mystery that shrouds them.





CHAPTER XIV

Astronomical Dating -Explanation of Some Related Issues

THE METHOD OF using astronomical references as evidences to ascertain the dates of the events in history has not been used so far in a big way, although a number of attempts have been made. A number of questions are, therefore, expected to crop up requiring clarifications. Understandably, all the issues arising out of the discussions in the book cannot be anticipated, nevertheless the forthcoming discussion would, perhaps, clarify some of the apprehensions.

The Awareness of Astronomy in the Times of the Ramayana

The first and the most basic question is that how do we conclude that the people of the times of Lord Ram knew of the planets, stars, the astronomical cycles like month, years etc. and how can one believe that such knowledge was put to use in day to day life. The first point to note in this regard is that the reference to the astronomical bodies is not peculiar to the Ramayana alone. The entire ancient Vedic literature has abundance of such references, which shows the awareness of those people about the various planetary bodies. The fact that there was a widespread knowledge of astronomy becomes clear when we understand what the Vedangas were and who used to study these treatises. The important point to note is that one of the Vedangas was astronomy (jyotish shastra) and therefore, the knowledge of this Vedanga automatically implied a comprehensive understanding of the astronomical phenomenon.

For the benefit of those who are not very well versed with Vedic literature, it may be mentioned that there are six books called Vedangas - or the arms (limbs) of Vedas. The learning of these six sciences was necessary to learn and understand the main Vedas. These works are supplementary texts based on the Vedas and these six sciences were primarily learned to master the Vedas and perform the rituals described therein. The six Vedangas are:

<i>Shiksha</i>	- dealing with the method of pronunciation
<i>Vyakaran</i>	- grammar
<i>Chhandas</i>	- prosody, contents of the verse
<i>Nirukta</i>	- origin and development of words
<i>Jyotish</i>	- astronomy or astrology
<i>Kalpa or Kalpa Sutra</i>	- dealing with performance of rituals.

Clearly, the first four out of the six Vedangas described above, deal with the construction and pronunciation of language while the fifth and the sixth dealt with the specialised knowledge of the relevant subjects.

There is no doubt that the Vedas and Vedangas existed well before the birth of Lord Ram. At a number of places, the Ramayana contains clear statements about the knowledge of the Vedas and its six arms (Vedangas). Very importantly, there is mention of only three Vedas in Ramayana - the ones that are traditionally considered to be the most ancient. To be explicit, the Ramayana mentions the names of Rig Veda, Yajur Veda and Saam Veda. There is no mention of Atharva Veda as such. To illustrate this point, reference may be made to 4/3/27-29 where the meeting between Hanuman, Lord Ram and Lakshman has been described. Here Lord Ram explains to Lakshman the perfection of the language spoken by Hanuman. He states, "one, who has not been a student of Rig Veda, who does not remember Yajur Veda by heart and who doesn't have the knowledge of Saam Veda cannot speak such language. Certainly, he has heard the entire Vyakaran (one of the Vedangas) several times that is why he has not used even a single wrong word while speaking to us."

This above statement made by Lord Ram has many implications. Mention of only three Vedas (not naming the Atharva Veda) shows that in those times Atharva Veda was not known as a Veda. There is consensus on this issue that originally only three books were regarded as Veda and at a later date the Atharva Veda was given the status of a Veda. There is a clear mention of one of the Vedangas - the vyakaran or the grammar in the above statement.

Quoting from 4/3/28 onwards, we can note how the language used by Hanuman was analysed by Lord Ram in terms of Vedangas. Lord Ram says, "Certainly, he has heard the entire Vyakaran several times, that is why he has not used even a single wrong word while speaking to us (Nirukta)... He has stated all the things precisely (Chhanda). He neither spoke too fast nor too slowly (Shiksha). His statements are unambiguous (Chhanda)" It can, therefore, be seen from the above analysis that Lord Ram and even Hanuman were proficient in the knowledge of the Vedas and Vedangas. The fact that Kalpa, the fifth Vedanga was also known becomes clear as 'Kalpa Sutra' is explicitly mentioned in the Ramayana at 1/14/40 in the context of performance of Aswamedha Yagya.

As regards the prevalence of the knowledge of the Vedangas during the Ramayanic times we may note that at 1/14/21, it is mentioned that in the Aswamedha Yagya performed by King Dashrath, none of the performer (sage) was such who did not know the 'six limbs of Vedas'. This also shows that there is no confusion about the exact number of Vedangas known at that time. Therefore, there is no reason to hold that the sixth Vedanga, the jyotish, was not commonly known to the seers, sages and the learned people of that time. It may be clarified that although in modern common parlance jyotish implies predictive astrology, yet in ancient times, jyotish meant astronomy in all its aspects. Mathematics was also considered a part of jyotish Vedanga. Now if the learning of Vedas and Vedangas was an essential part of knowledge to be acquired by students, then can there be any scope for doubting that knowledge of astronomy was absent or rare those days? All the sages, seers and the erudite were the ones who imparted the knowledge. Most of those belonging to the upper strata including the ruling class were the seekers of such knowledge. Thus, undoubtedly we can

conclude that the understanding of astronomy was widespread and extensive even during those days.

It may be clarified that no claim is being made that the people of Ramayanic times knew of the precession of equinoxes or other such complicated and very slow movements of heavenly bodies. Nevertheless, all the planets, zodiacs, nakshatras and other prominent constellations were identified and the main cycles of these astronomical bodies were accurately known and understood. As may be seen, all the astronomical references examined for the purposes of dating in the earlier chapters were made in terms of very simple observational astronomy. The references could have been recorded by any one who was able to correctly identify the zodiacs, planets and understand their respective cycles of rotations. The mention of six ritus, samvatsar, dakshinayan and uttarayan etc. are a part of such simple basic and applied knowledge that people of those times had acquired. Such knowledge helped the people to make exact and accurate observations of the astronomical phenomenon. We had a scientifically designed calendar based on the movements of both the sun and the moon in the sky. These calendars were fine tuned by observing the precise movements of these bodies in the sky on regular basis. More importantly, what can be seen and inferred is that same set of principles are used even today for the purposes of our traditional calendar and the dates of our religious festivals are determined on the basis of that calendar. This remarkable continuity from the times of Lord Ram simply proves that our civilization has been in existence ever since such ancient historical times.

We would now deal with another issue, which has not been clarified, so far, in terms of history. This issue is regarding the place of the Ramayana in the chronology of the composition of ancient Indian literature particularly in relation to Rig Veda and Purans. We all know that the Mahabharata period was definitely later than the Ramayanic times.

Valmiki Ramayana Vs other Ancient Compositions

An analysis of the text of Valmiki Ramayana and the Rig Veda shows remarkable similarity of concepts. In the foregone discussion, we noted that the Ramayan i mentions only three Vedas namely, the Rig Veda, Yajur Veda

and the Saam Veda. Interestingly if we examine the text of the Rig Veda, we would note that at 9.97.34, it also mentions only three Vedas. The specific names of Rig Veda, Saam Veda and Yajur Veda are mentioned at 1.7.1 without any ambiguity. These names are repeated at many places elsewhere also

Regarding the concept of astronomy and seasons, the Rig Veda gives a very interesting description at 1.164.11-15. It states that the wheel of the sun with 12 spokes keeps moving eternally and never gets old and 360 days and nights remain in it. This circle of the Samvatsar (year) keeps revolving untiringly. Six ritus of two months each like Chaitra also keep rotating endlessly. Sometimes there is a seventh ritu of Adhik Maas (intercalary month) but it has only one month while the regular six ritus have a pair of months. At several places, for instance at 1.24.15, it mentions six ritus and at 1.25.17, it mentions 12 and 13 months. Again at 3.56.2, it mentions that a regular year has six ritus, and thereafter the names of these ritus are mentioned as - vasant, sharad, hemant, grisham, varsha and sheet. We should recall that the same concepts were prevalent during the times of Ramayana also. This analysis shows that the time of composition of the Ramayana was not very much later than the time of composition of the Rig Veda. However, it would also not be correct to hold that the Ramayana is as old as the Rig Veda. The reason being that despite the remarkable similarity between the terminology used in the Rig Veda and the Valmiki Ramayana, all such terms and concepts of the Rig Veda find a place in the Ramayana while it is not true otherwise. The story of the Ramayana shows far more advancement in every sphere of life, thus indicating a lapse of long period of time after the Rig Vedic period. Hence, a reasonable conclusion can be drawn that the era of the Ramayana can be dated only after the early Vedic period when the first three Vedas were already composed and perhaps, the Atharva Veda was known as Niti-Shastra only dealing with the rules of governance.

With a view to finding a link between the time of composition of the Rig Veda and the Ramayana, we would analyse the names of the kings mentioned in the former book. The Rig Veda mentions Surya-vansi (Solar dynasty) people, which is generally considered to be the dynasty of Lord Ram. It mentions many kings, and the prominent ones are -Mandhata

(1.112.13 and 8.39.8), Devashray and Devaraat, sons of Bharat (3.23.2), Turvash, Yadu and Yayati (4.30.17 and 6.45.1), Riturath (5.36.6), Bharat (6.16.4), Trasdasu and Puru -sons of Purukutsa (7.19.3), King Nahusha (7.95.2 and 8.6.24), King Pruthu (8.9.10), King Pruthushruva (8.46.21), Vivaswan (8.113.8) Yayati, the son of Nahusha (10.63.1) and finally Lord Ram at 10.93.14. The list is not exhaustive and many other kings are named about whom very little references are available in other ancient literatures. At 10.60.2-3, it has been mentioned that, "we went to the kingdom of King Asmati, who defeats the enemies and who fulfills the wishes of everyone. The people of the kingdom, ruled by 'Ikswaku', 'grow like people living in heaven'. The preponderance of the names associated with the dynasty of Lord Ram becomes fairly evident from the above.

The important point to note is that several of Lord Ram's ancestors have been mentioned along with his name. However, no king of this dynasty, who ruled later than Lord Ram finds a mention in this book. This very strongly suggests that at the time of composition of the tenth book of the Rig Veda, Lord Ram was the ruling king of the solar dynasty.

The tenth book (part) of the Rig Veda is generally considered to be the latest and it can be conclusively held that it was composed around 5000 BC, during the times of Lord Ram. This becomes absolutely clear from the hymn of the Rig Veda at 10.93.14 in which the sage who composed this hymn states that "I have recited these prayers in the presence of Duseem, Prithvaan, Ven and mighty Ram. All these kings are very prosperous". The meaning of this hymn unambiguously shows that at that time, King Ram was a living personality.

In view of the above analysis of the contents of the Rig Veda and the Ramayana, it is easy to conclude that the Ramayana and the last part (tenth book) of the Rig Veda were composed almost at the same time, that is, around 5000 BC. The earlier parts of the Rig Veda must have been already in existence at the time of birth of Lord Ram.

As regards the composition of the Purans vis-a-vis Ramayana, one thing is clear that the Ramayana was composed much before the Purans. There are few obvious reasons for it. All the Purans contain the genealogy of the main royal families and these genealogies are in two parts. The first part

generally describes the lineage up to the kings that ruled till the Mahabharata war. The second part of the genealogy lists the kings "which were supposed to happen in future" i.e., after the Mahabharata war. Basically, the Purans are supposed to be the compendium of the stories about the kings. Thus, the existence of Purans should be as old as the tradition of the kings. The oral transmission of the deeds of the kings must have been present from very early times of civilization and it must have been revised and enlarged every time when either a new king took over or a remarkable deed was done. This endless process, however, was perhaps brought to a close much after the Mahabharata war when the Purans took their final shape or say, the present shape. Taking a rational and logical view, it is obvious that even the kings mentioned under the category "which were supposed to happen in future" must have been a matter of past at the time of closing the lineages. This peculiar way of listing the kings was, perhaps, adopted to confer an element of sacredness to the Purans with the ability to forecast the future by the priests through the Purans. The genealogy of 'Ikshwakus' in the Purans mention the name of the king 'Brihadbala' as witness to the Mahabharata war and the second part of the list mentions different number of kings till the last king 'Sumitra' with whom the lineage was to come to an end in Kali yuga (Matsya Puran 273.53). This shows that in all likelihood, the updating of the Purans ended well after the beginning of Kali yuga. On the other hand, the genealogy of Lord Ram's dynasty in Valmiki Ramayana ends after mentioning the name of Lord Ram. This indicates clearly that at the time of composition of the Ramayana, Lord Ram was the last king and not even his own son, Kusha is listed in the genealogy. Hence, the Ramayana has to be a much older compilation than the Purans. Moreover, in terms of yuga cycle, the Ramayana occurred in Treta yuga while the Mahabharata occurred in the Dwapar yuga. Therefore, as the Purans mention the Mahabharata war and the Dwapar yuga, these treatises have to be later creations than the Ramayana.

The Dating of the Era of Lord Ram and the Puranic Genealogies

Having discussed the relative antiquity of the Ramayana versus the Rig Veda and the Purans, we would now take up the issue of reconciliation

between the astronomical dating of the Ramayana arrived at in this book with the Puranic genealogies of different royal dynasties.

Remarkable efforts have been made by many indologists to reconstruct the history of ancient India by taking into account the genealogies described in different Purans. Many of the Purans give the details of the kings of the royal dynasties of ancient India. All these genealogies begin with mythical Manu. He had many children and main among them was Ikshwaku who founded the 'Solar dynasty' by establishing his kingdom with the capital at Ayodhya. Lord Ram belonged to this dynasty. The offspring of Ila, a daughter of Manu, established the name of the lunar dynasty. The Purans give the names of different kings of each dynasty. The most complete list of the Ikshwaku's dynasty appear in Vishnu Puran in which ninety-three kings, including Lord Ram, have been named between mythical Manu and King Brihadbala, who was reigning during the time of Mahabharata war. For the same time period, names of only fifty-three kings of the Purus' dynasty have been given, which suggests that this list may be incomplete. As discussed earlier, there is considerable confusion in the different genealogies, available in different Purans. Sometimes the names are given in different sequences and often additions and deletions of certain names are found. Further, in most of the cases, the exact time for which the individual kings lived or ruled over their kingdoms, is not available.

Notwithstanding the variations, comprehensive and consistent lists of the kings have been drawn by arriving at logical reconciliation among different Puranic lists. However, the time periods to be assigned to these kings remain somewhat inaccurate and open to speculation. Dr. Subhash Kak has given the royal genealogies in his book *The Astronomical Code of the Rig Veda* at pp. 55-57. In the list of kings of Ayodhya, which begins with Manu, the name of Lord Ram appears at generation number 65. It is mentioned that the king listed at generation number 94, Brihadbala was the witness of Bharata war. Taking a very broad view, one can conclude from the above that there is no inconsistency in the relative position of Lord Ram in the dynasty list and the period assigned to Lord Ram vis-a-vis the Vedas and Mahabharata. It has been already discussed that there was a historical past, and perhaps a very long one, before the birth of Lord Ram. The existence of at least sixty-four kings before him in the dynasty of

Ikshwakus vindicates that claim. On the other side of the genealogy, there are about thirty decedents (number varies) till Brihadbala for the gap of time between the Ramayana and the Bharata war. However, it would not be feasible to work out and match year by year the genealogies available in the Purans with the outcomes of the astronomical dating. The incompleteness of the lists of kings apart, nothing exactly and accurately can be stated about the length of reign of each of these kings. It would be noteworthy that the Valmiki Ramayana also mentions the genealogy of Lord Ram's dynasty at two different places. Beginning with mythical Manu, in the dynastic list Lord Ram appears at generation number 39. If Ramayana mentions Lord Ram at the generation number 39, then the Puranic list of the same dynasty showing him at generation number 65 raises more questions than answering them. So far as the story of the life of Lord Ram is concerned, the Valmiki Ramayana is certainly more authentic than Purans. Moreover, the antiquity of the Ramayana over the Purans can hardly be debated. Under the circumstances, it has to be concluded that the lists of the kings of these dynasties, as mentioned in Purans, are not accurate. As such, nothing much can be proved or concluded only on the basis of the Puranic lists. The time period for which the individual kings ruled being open to speculation, different historians assume different 'average time of reign' for the kings to arrive at their conclusions. A very pertinent question that comes to the mind of the reader of the Ramayana is how the fourteen years, for which Bharat ruled Ayodhya, albeit passively, will be accounted for in these lists.

Therefore, once again, from the discussion above, only a general conclusion can be drawn that the genealogical lists of the Purans are broadly consistent with the dating of the Ramayana arrived at in this book. The incompleteness of the Puranic genealogies and non-availability of the information about the number of years for which each individual king lived or ruled the kingdom, does not permit any accurate computations of the time period to arrive at meaningful conclusions. Given these improbable factors about the Puranic genealogies, it would be unfair and arbitrary to arrive at any conclusions in specific terms.

Summing up

Whatever be the conclusions, this study demonstrates, undisputedly, the pre-eminence of astronomical dating over all other methods of dating history in terms of preciseness, exactitude and objectivity. If the planetary configuration is comprehensive and explicit, it will refer to one and only one date as seen in case of ascertaining the date of birth of Lord Ram. Given the vast resource of ancient literature we possess, the methods of dating employed in this book should offer large opportunities of dating more such scriptures.

The history of ancient India is shrouded in darkness. The veil of the darkness can be lifted only if we are able to date our sacred Vedic literature accurately and also convince modern historians of such dating. Not every book on this shelf offers an opportunity, such as provided by the Ramayana, to determine the precise dates of the era of Lord Ram. The efforts made, so far, to ascertain the precise dates for major epochs of the history of Indian civilization have produced certain results but the gaps among them were too large to show any notable coherence in such results. One of the main reasons, in this regard, was that the dating of the era of Lord Ram was not arrived at, in a convincing manner, till now.

Considering the findings of this book, along with the outcome of other research works, one can now observe a faint but firm timeline for the ancient history of our country. The archaeological evidence suggesting the dating of the Saraswati Sindhu region sites up to 8000 BC, astronomical dating of some of the verses of the Rig Veda to 6500 BC, the astronomical dating of Ramayana at around 5100 BC, date of composition of the last Mandal (book) of the Rig Veda around 5000 BC, strong evidences suggesting the traditional date of 3137 BC (or 3136 BC) as the date of Mahabharata, probable dating of composition of Purans well after the beginning of the Kali yuga (3100 BC) and the drying up of the Saraswati river around 1900 BC, makes a highly agreeable and logically acceptable timeline of our civilization. It is hoped that the astronomical dating arrived at in this study will serve as a pivotal date in the reconstruction of this timeline of ancient India as and when more evidences are unearthed and many more dates of the historical events are discovered in the due course.





Glossary

Adhik Maas. Means extra month and is also called the intercalary month. It is the thirteenth month in the Lunar calendar, which is necessary to adjust the gap between the solar and lunar year. An intercalary month occurs in 32 solar months and 16 days. This means that an intercalary month occurs every three years and this year has thirteen lunar months.

Aihole inscription: Aihole is a village in the Bijapur district of Karnataka state in India. An inscription found on a Jaina temple here of AD 634 is considered to be a source of information for the dates of Mahabharata War and the beginning of the Kali yuga.

Airavat: Name of the legendary elephant on whom the mythical King Indra used to ride.

Amavasya: No-moon night.

Aryabhatta: The famous Indian astronomer, believed to have been born in AD 476. His magnum opus *Aryabhattia* deals with the complicated concepts of mathematics and astronomical calculations, which are found to be accurate by European astronomers in the thirteenth century AD when his books were translated in their own languages.

Ashlesha: Name of one of the 27 stars identified as nakshatras. It is located in the Cancer constellation. Refer to serial number 9 in Table 1 in the book.

Ashok Vatika: Name of the garden in which demon king Ravan kept Sita, Lord Ram's abducted wife in Lanka.

Ashwin: Name of one of the months of the Lunar year. It derives its name from the star called Ashwini - one of the 27 stars identified as nakshatras.

Ashwini: Name of one of the 27 stars identified as nakshatras. It is located in the Aries constellation. Refer to Serial Number 1 in Table 1 in the book.

Aswamedha Yagya: Also called horse sacrifice performed by the kings of ancient India in order to achieve the desired objectives or get the blessings of the God.

Atharva Veda. One of the four Vedas. In the beginning, this compilation was not given the status of a Veda and it was known as *Niti-Shastra* dealing with polity. It is considered to be the most recent treatise of the four vedas.

Ayodhya: The capital city of the kingdom of Koshal where Lord Ram and his ancestors ruled.

Bharadwaj: A sage who lived in the times of Lord Ram. His *ashram*, place of dwelling, was located on the outskirts of the kingdom of Koshal.

Bharat: One of the three younger brothers of Lord Ram and the son of Kekeyi, stepmother of Lord Ram, who was instrumental in sending him to exile.

Bharata war: Also called the famous Mahabharata War, fought between Kauravas and Pandavas much after the birth of Lord Ram.

Bharni. Name of one of the 27 stars identified as nakshatras. It is located in the Aries constellation. Refer to Serial Number 2 in Table 1 in the book.

Brahmins: One of the four classes in which the society of India was divided. Brahmins were teachers and preachers of the knowledge particularly related to God.

Brihadbala: The king of the royal lineage of *Ikshwakus* in which Lord Ram was also born. The Purans mention that King Brihadbala was a witness to the Mahabharata War.

Brihaspati. The planet Jupiter.

Brihat Samhita. The book written by the great Indian astronomer Varahamihira, which deals with astronomy and mathematics.

Chaitra: One of the months of the Lunar year. It derives its name from the star called Chitra - one of the 27 stars identified as nakshatras.

Chaturdashi: The fourth day from the no-moon night or the full moon night. Therefore, it occurs twice in one lunar month.

Chau-masa: A period of four months. This name is usually associated with the season of rains, which in south India is four months long.

Chitra: One of the 27 stars identified as nakshatras. It is located in the Virgo constellation. Refer to Serial Number 14 in Table 1 in the book.

Dakshinayan The annual periodic movement of the sun from the north to the southward direction is called Dakshinayan.

Dashmi: The tenth day from the no-moon night or the full moon night. Therefore, it occurs twice in one lunar month.

Dashrath: King of Koshal and father of Lord Ram.

Dawapar Yuga: As per the puranic concept of Yuga cycle, Dwapar is one of the four yugas and it is third in the serial number of the cycle.

Dhumra-Ketu: Name given to one of the two imaginary planets called Rahu and Ketu. It is also used to refer to the comets in the Hindi language.

Diwali: One of the main festivals of India. It is celebrated to mark the end of the exile period of Lord Ram and his return to Ayodhya.

Dushan: One of the two deputies of King Ravan, the other one being Khar. These two were in charge of the area around Panchwati, where Lord Ram was staying with his wife Sita and brother Lakshman during their thirteenth year of exile.

Dwarka: The capital city of the kingdom of Lord Krishna. Dwarka was recently in the news, for it was discovered submerged under the sea and the

exploration work is being done at present.

Garuda: The Hindi name for an eagle.

Gautam Buddha: The founder of Buddhism.

Godawari: One of the largest rivers of south India. Lord Ram had stayed for a long time on the banks of Godawari river during his exile period.

Goswami Tulsidas. A of seventeenth century AD poet, who composed one of the most sacred books on Lord Ram, called *Ramcharitmanas*.

Grisham: The Hindi language name for the summer season.

Haihayas: An ancestor of Lord Ram.

Harappan civilization: Name given to the ancient civilization remains found in the Harappa and Mohenjodaro and these sites are now located in the North-West Frontier Province of Pakistan. It is claimed by historians that the Aryans invaded the natives living in these areas and established their own kingdom.

Hemant: One of the six seasons of two months each in which the year is divided. It corresponds to the cold winter period.

Holi: The festival of colours celebrated in India on the full moon day of the lunar month of Phalgun every year.

Ikshwakus: The royal dynasty in which Lord Ram was born.

Indra: A legendary king, who finds numerous mentions in the Rig Veda.

Jyestha: Refers to one of the months of the lunar calendar of India. It is also the name of one of the 27 stars identified as nakshatras. It is located in Scorpius constellation. Refer to Serial Number 18 in Table 1 in the book.

Jyotish: Refers to the science of astronomy as well as astrology.

Kaand: One of the seven parts (chapters) in which the Ramayana has been divided.

Kali Yuga: As per the puranic concept of Yuga cycle, Kali yuga is one of the four yugas and is the last yuga of the cycle.

Kartik: One of the months of the Lunar year. Refer to Table 1 in the book.

Kaushalya: Lord Ram's mother.

Kekeyi: Stepmother of Lord Ram, on whose insistence he was exiled for fourteen years.

Ketu: One of the two imaginary planets - Rahu and Ketu. These are also called nodes and are associated with the occurrence of the eclipses.

Khar: One of the two deputies of King Ravan, the other one being Dushan. These two were in charge of the area around Panchwati, where Lord Ram was staying with his wife Sita and brother Lakshman during their thirteenth year of exile.

King Janak: Father of Sita, wife of Lord Ram. King Janak ruled over the kingdom of Mithila, located in the foothills of the Himalayas.

King Yudhishtira. The eldest of the Pandavas, who fought the famous battle of Mahabharata with the Kauravas.

Kishkindha: The place that corresponds to the modern 'Hampi' in the Bellary district of Karnataka state. The kingdom of Kishkindha was ruled by Bali, the estranged brother of Sugreev.

Koshal: The kingdom ruled by Lord Ram and his ancestors.

Krishan Paksha: The period of the decreasing phase of moon when it wanes from full moon to no moon.

Krishna: Or Lord Krishna is considered to be the next incarnation of Hindu God Vishnu after Lord Ram. Lord Krishna established the kingdom with the capital at Dwarka.

Kumbhkaran The great warrior brother of Ravan.

Lagna. Refers to the constellation seen to be rising in the east at any given point of time.

Lakshman: Lord Ram's younger brother, who accompanied him during his exile period.

Maarech: One of the demons who helped Ravan abduct Sita.

Mahabharata The epic, written by a sage named Ved Vyas, deals with the story of the Pandavas and Kauravas. The Mahabharata also refers to the historic war fought at a place called Kurukshetra between the Pandavas and Kauravas.

Mahapadam Nanda: The first ruler of Nanda dynasty, who established his kingdom on the ruins of the empire left by the great ruler, Ashoka of Mauryan dynasty.

Maharishi Valmiki: Sage and contemporary of Lord Ram who composed the Ramayana.

Mahavira: The founder of Jainism religion in India.

Mainaak Parvat: One of the hills, mentioned in the Ramayana to be located in the middle of the sea stretching from the Indian shore and Lanka. It is mentioned that on his way to Lanka, Hanuman rested for a while on this hill.

Mangal: The name of the planet Mars in Hindi.

Margshirsh: One of the months of the Lunar year. Refer to Table 1 in the book.

Meghnath: The eldest and the mightiest son of Ravan.

Mithila. The kingdom ruled by Sita's father, King Janak.

Nakshatras: Refer to the twenty-seven stars located at an approximate distance of 13° from one another along the ecliptic circle. These stars, thus, divide the ecliptic into twenty-seven divisions in such a manner that moon travels from one nakshatra to another every day. These are also called the lunar mansions since the moon stays in each of these divisions for one day.

Nasik: The modern place in Maharashtra state, which corresponds to Panchwati, the place where Lord Ram spent a part of his exile period.

Navmi: The ninth day from the no-moon night or the full moon night. Therefore, it occurs twice in one lunar month.

Neel: One of the deputies of Sugreev, who was well versed in the art of bridge construction.

Niti-Shastra: The treatise of polity. Later on, this was given the status of a Veda and came to be known as Atharva Veda.

Panchami: The fifth day from the no-moon night or the full moon night. Therefore, it occurs twice in one lunar month.

Panchwati: The place where Lord Ram stayed for a long time, particularly during the thirteenth year of his exile. It was from here that Ravan abducted Sita in the absence of Lord Ram and Lakshman.

Parikshit: The grandson of Arjun - one of the five Pandavas. Parikshit's father was Abhimanyu, Arjun's son.

Phalgun: One of the months of the Lunar year. Refer to Table 1 in the book.

Punarvasu: One of the 27 stars identified as nakshatras. It is located in the Gemini constellation. Refer to serial number 7 in Table 1 in the book.

Purans: Part of the ancient literature mainly dealing with the tales about the exploits of the famous kings of ancient India. One of the main concepts that they deal in is the Yuga cycle of counting the time period in years.

Purna-masi: The full moon night is called Purna-masi in Hindi language.

Purva-Bhadrapad. One of the 27 stars identified as nakshatras. It is located in the Pisces constellation. Refer to serial number 25 in Table 1 in the book.

Pushpak Viman: Name of one of the chariots possessed by the king of Lanka - Ravan. As per the story, this chariot was capable of flying.

Pushya: One of the 27 stars identified as nakshatras. It is located in the Cancer constellation. Refer to serial number 8 in Table 1 in the book.

Rahu: One of the two imaginary planets - Rahu and Ketu. These are also called nodes and are associated with the occurrence of the eclipses.

Rakshasas: Name assigned to the demons in ancient Indian literature.

Ramcharitmanas: The epic composed by Goswami Tulsidas in the seventeenth century AD.

Ravan: Ruler of Lanka, who abducted Sita. Lord Ram fought with the entire army of Ravan and finally succeeded in defeating him to release Sita from his captivity.

Revati: One of the 27 stars identified as nakshatras. It is located in the Pisces constellation. Refer to serial number 27 in Table 1 in the book.

Ritu / Ritus. Refer to the seasons - a period of two months. One year is divided into six ritus

Rohini: One of the 27 stars identified as nakshatras. It is located in the Taurus constellation. Refer to serial number 3 in Table 1 in the book.

Saam Veda One of the four Vedas and part of the most ancient sacred literatures of Hindus.

Sagar: One of the ancestors of Lord Ram.

Saka era: Reckoning of the year started by a king of Saka dynasty in India. It corresponds to AD 78 for the beginning of the era.

Salvahana era: It has the same meaning as Saka era that began in AD 78. It is believed that the Saka era was started by the Salvahana king. Therefore, this era is called Saka era as well as Salvahana era

Samhita: Refers to a compilation of text or verse.

Samvatsar: A year is called by this name in Sanskrit and Hindi.

Sandhya period: The period between the two yugas. It is the intervening period between the end of the outgoing yuga and the beginning of the incoming yuga.

Sapta - Sindhu Refers to the region drained by seven rivers.

Saptarishi: Name given to the constellation called Ursa Major.

Saraswati: The mightiest river mentioned in ancient Hindu literature particularly the Rig Veda. However, today it does not exist, since it dried up around 3000 BC.

Sat Yuga. As per the puranic concept of Yuga cycle, Sat yuga is one of the four yugas and it is the first yuga of the cycle.

Sawan. One of the months of the Lunar year. Refer to Table 1 in the book.

Shani: Name of the planet Saturn.

Sharad: One of the six seasons into which the year was traditionally divided in ancient India. It refers to fall or autumn.

Shatrughan: The youngest of the three brothers of Lord Ram.

Sheet: One of the three main seasons into which the year was traditionally divided in ancient India. It refers to the four month long winter season.

Shishir: One of the six seasons into which the year was traditionally divided in ancient India. It refers to the winters.

Shlokas: Means the hymns written in Sanskrit language and the smallest unit into which the epic Ramayana can be divided.

Shravan Kumar: Was an obedient son of his blind parents. He was killed accidentally by King Dashrath, the father of Lord Ram.

Shukra: Name of the planet Venus.

Sita: Wife of Lord Ram.

Subahu: One of the accomplices of Mareech, who was killed by Lord Ram while protecting the yagya of Sage Vishwamitra.

Sugreev: Name of the king of the monkeys who ruled over the kingdom of Kishkindha after the death of his elder brother Bali at the hands of Lord Ram. Sugreev deputed his large army of monkeys in search of Sita and this army fought the battle with the army of Ravan.

Sunaabh Parvat: One of the hills mentioned in Ramayana to be located in the middle of the sea stretching from the Indian shore and Lanka.

Surpnakha: Ravan's sister, who went to the dwelling place of Lord Ram at Panchwati and proposed to him. Later on, Lakshman cut her nose and ears and in a way, gave a challenge to her brother Ravan.

Sursa: One of the demons who is believed to be living in the middle of the sea. She was capable of increasing the size of her mouth infinitely. She challenged Hanuman when he was crossing the sea to go over to Lanka in search of Sita.

Surya-vansi: Name of the royal dynasty to which Lord Ram belonged.

Suvel: One of the hills, mentioned in Ramayana to be located in the middle of the sea stretching from the Indian shore and Lanka.

Taarka: The demon killed by Lord Ram, while he was protecting the yagya of Sage Vishwamitra.

Teiterya Samhita: One of the oldest sacred books of the Hindu literature.

Tithi: Refers to the date in the lunar month.

Treta Yuga: As per the puranic concept of Yuga cycle, Treta yuga is one of the four yugas and it is the second yuga of the cycle.

Trishanku: One of the ancestors of Lord Ram. This name is also assigned to one of the constellations called 'Crux' located in the southern sky.

Uttarayan: The annual periodic movement of the sun from the south to the northward direction is called Uttarayan.

Vaishakh: One of the months of the Lunar year. Refer to Table 1 in the book.

Vanars: In Sanskrit and Hindi languages, monkeys are called thus.

Varahamihira: One of the most famous ancient Indian astronomers who wrote his magnum opus *Brihat Samhita*.

Varsha: One of the three main seasons into which the year was traditionally divided in ancient India. It refers to the four month long rainy season.

Vasant Ritu: One of the six seasons into which the year was traditionally divided in ancient India. It refers to the spring.

Vedangas: Means the limbs of Vedas and these are six in number. The study of Vedangas was considered necessary to understand the text and contents of Vedas.

Vibhishan: The younger brother of Ravan, who joined Lord Ram in fighting against Ravan.

Vijay Mahurt Refers to :he auspicious time during which, it is believed, if someone embarks, he would win the war.

Vindhyachal: Mountain range south of the Himalayas in India.

Vishakha: One of the 27 stars identified as nakshatras. It is located in the Libra constellation Refer to serial number 16 in Table 1 in the book.

Vishwamitra: One of the most prominent sages of ancient India He was responsible for imparting warfare training to Lord Ram and his younger brother Lakshman. Before becoming a sage, Vishwamitra was a valiant king.

Vyakaran: Means grammar in Sanskrit and Hindi Language.

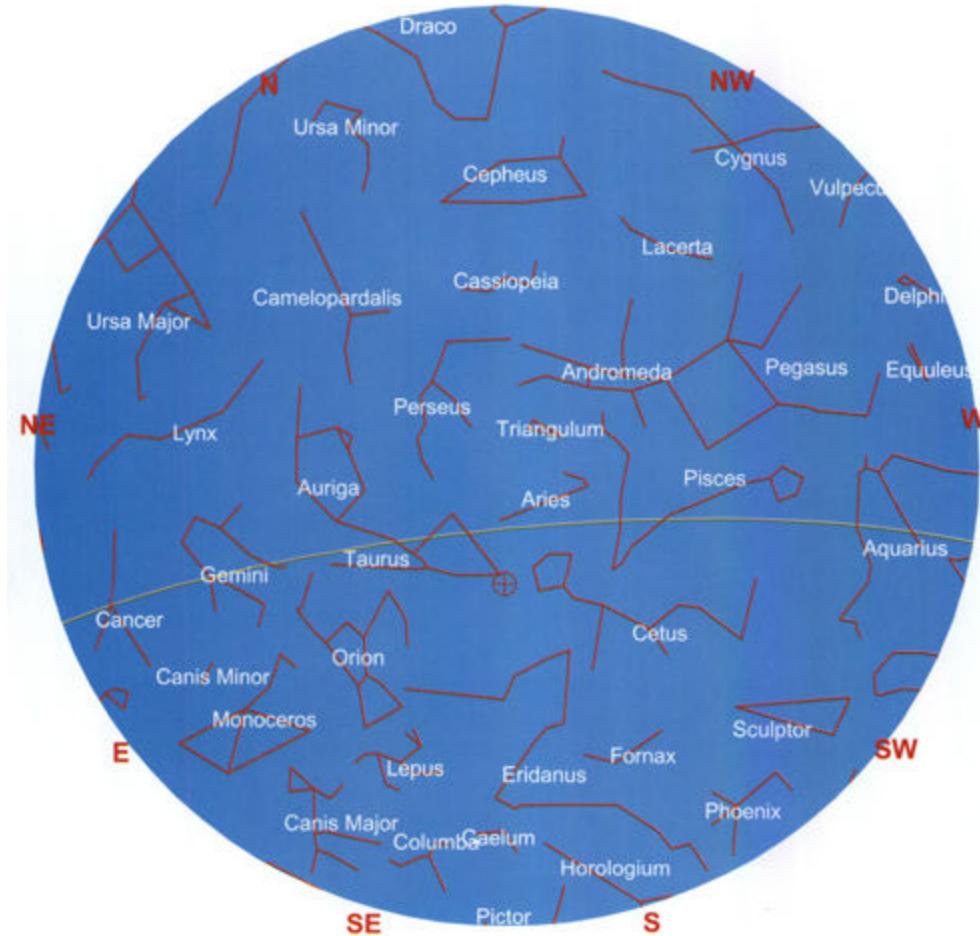
Yagopaveet. The initiation ceremony performed among Hindus at the age of about seven-eight.

Yagya: Means sacrifice performed in accordance with the procedure prescribed in Vedas to appease the Gods and receive their blessings.



SLIDE 1

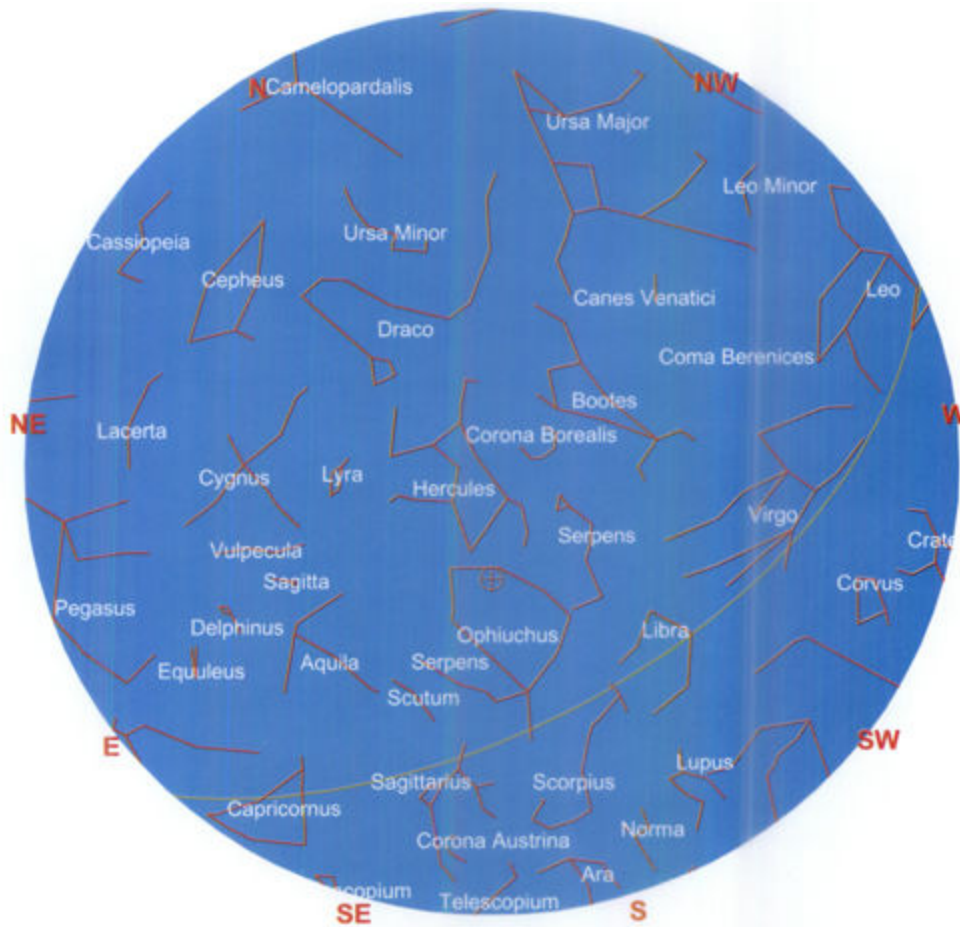
Six out of Twelve Constellations - Zodiac Signs



Out of the 88 constellations identified in the sky, this slide shows more than 40 such constellations (Red lines). However, important from an astronomical point of view are the six zodiac constellations viz. Cancer, Gemini, Taurus, Aries, Pisces and Aquarius, which are located in the path of the sun called ecliptic (Yellow Line). Every kink on the constellation line represents a star. In this manner, the image formed by the combination of stars gives the name to the constellation, which are named in white.

SLIDE 2

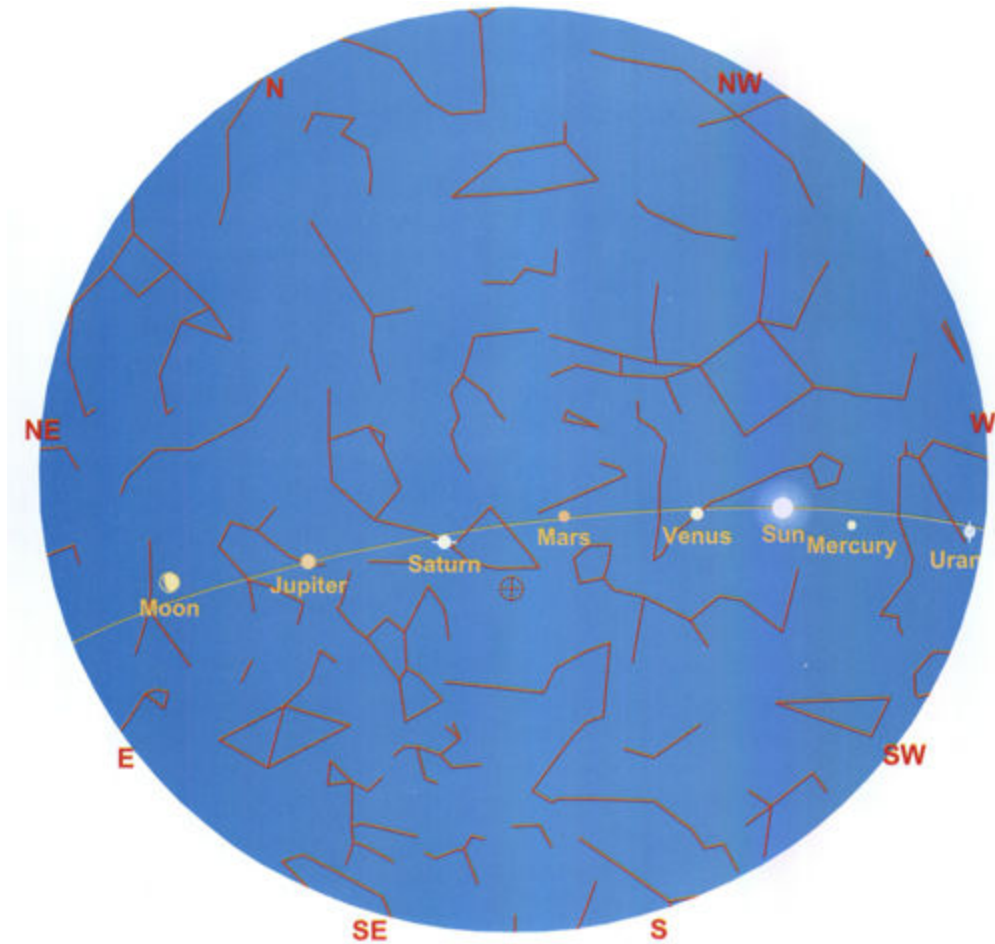
The Other Six of the Twelve Constellations - Zodiac Signs



This slide shows more than 40 constellations in dark red lines. Out of these, twelve constellations are identified as zodiacs. Six zodiac constellations were shown in slide 1 and in this slide the other six are shown viz. Capricornus, Sagittarius, Scorpius, Libra, Virgo and Leo. Note that these six constellations are also located on the path of the sun called ecliptic (Yellow Line). At any given time only six constellations are above horizon. Note that the software prints the date, time and location of the observation automatically.

SLIDE 3

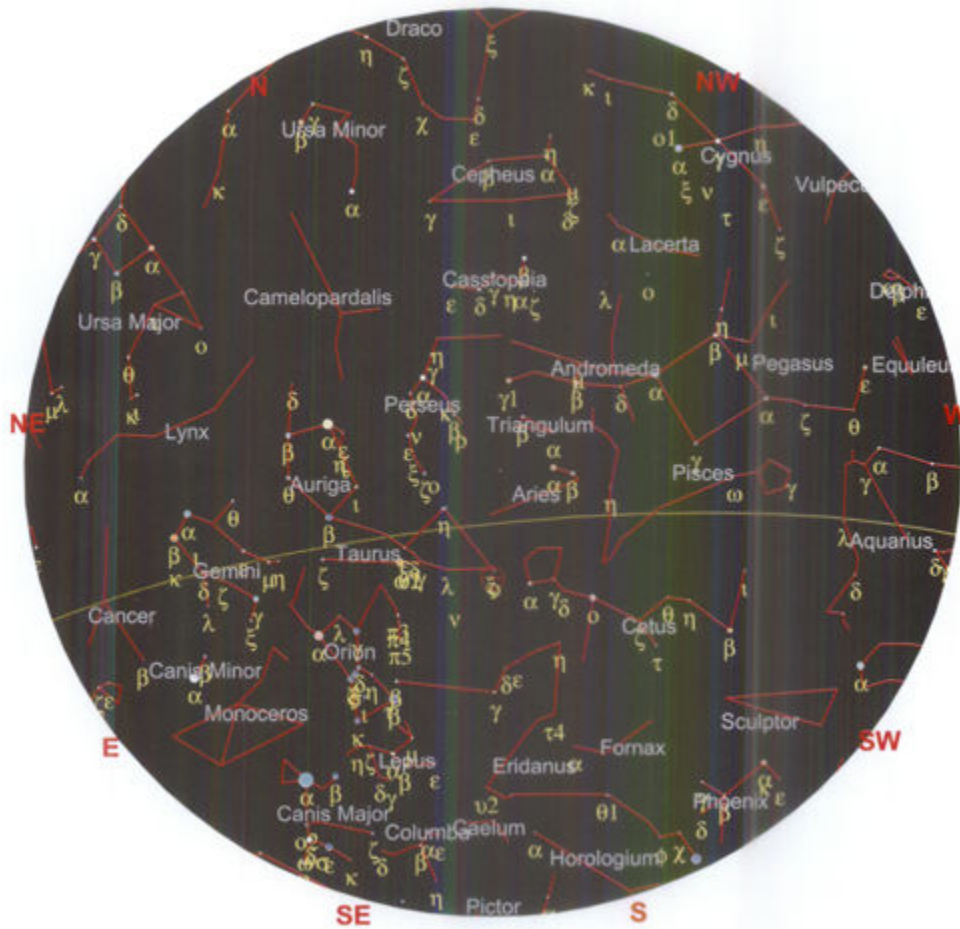
Main Planets of our Solar System and the Path of Their Revolution



This slide shows the following (from East to West) - Moon, Jupiter, Saturn, Mars, Venus, Sun and Mercury. These planets are named in yellow. Note that these are located on the yellow line known as ecliptic. Technically, they move in a small band around the ecliptic - up to 5 degrees on both sides. See that the moon is not exactly on ecliptic and so are Saturn and Mercury. We may say that the moon is in Cancer, Jupiter in Gemini, Saturn in Taurus, Mars in Aries, Venus and the sun in Pisces while Mercury is located in Aquarius.

SLIDE 4

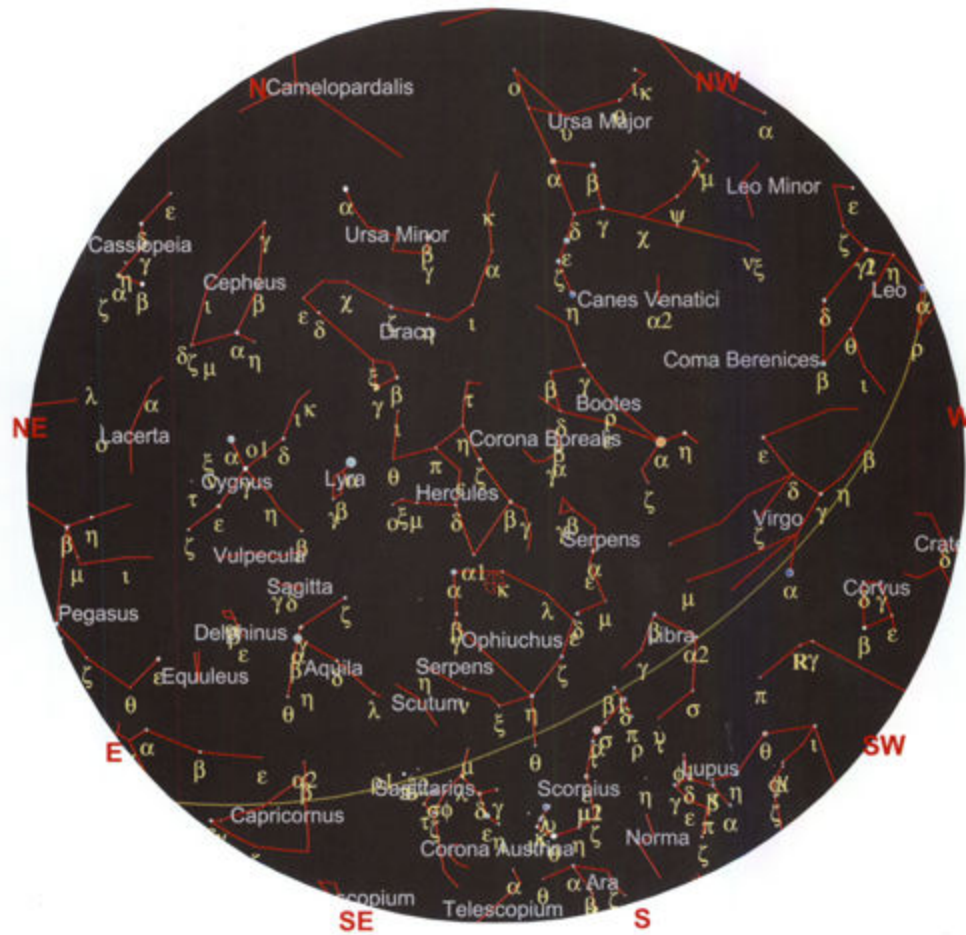
Nakshatras - The Stars in the Zodiac Constellations



Nakshatras are the 27 stars located almost equidistantly in the 12 Zodiacs. This slide shows the nakshatras located in Cancer, Gemini, Taurus, Aries, Pisces and Aquarius. Nakshatras were assigned specific names by the ancient astronomers and they have scientific names also. The star η in Taurus is Krittika nakshatra, star α in Taurus is Rohini, while star ρ in Gemini is Punarvasu (Sr. No. 3, 4 and 7 respectively in Table 1). In this manner, most of the listed nakshatras can be seen for the six Zodiac constellations shown above.

SLIDE 5

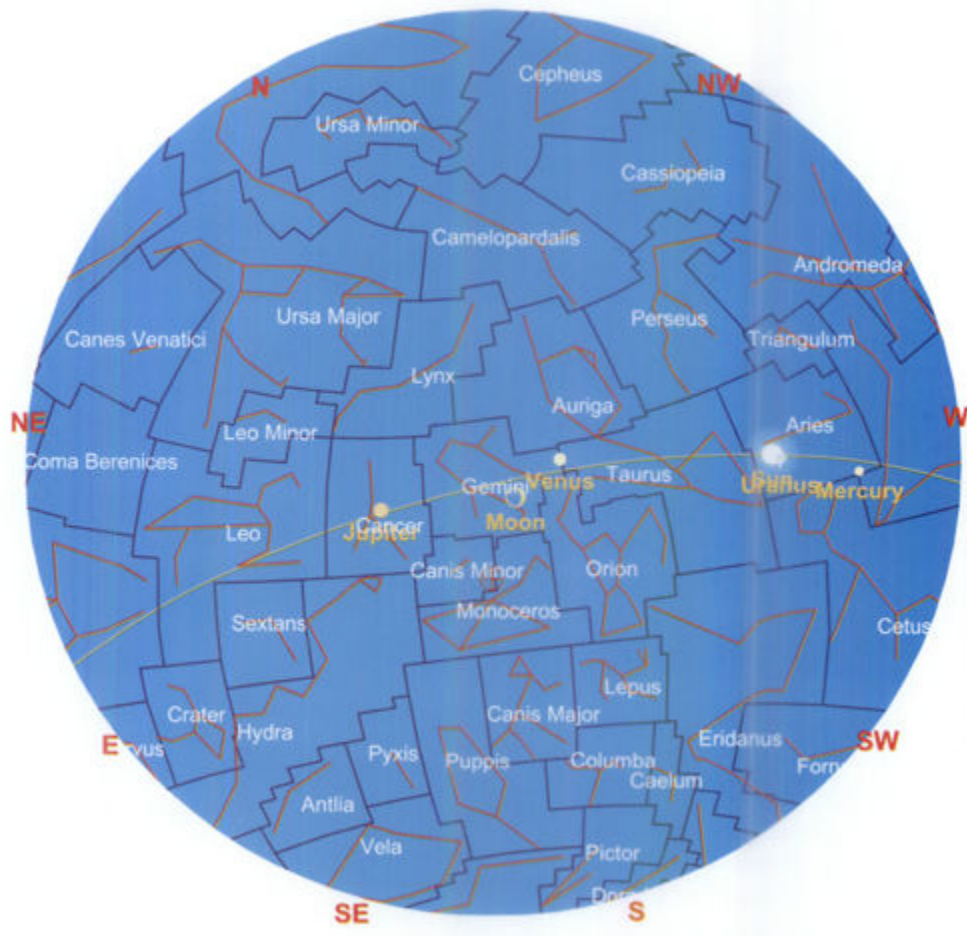
Nakshatras - The Stars in the Zodiac Constellations



This slide shows the nakshatras located in Capricornus, Sagittarius, Scorpius, Libra, Virgo and Leo constellations. For example, α in Leo is Magha, while α in Virgo is the famous Chitra nakshatra. Only once in a lunar year, the full moon occurs near the Chitra nakshatra and that marks the end of Chaitra month in that year. In this manner, the date of completion of all the twelve lunar months can be easily found and the name of the month can be easily identified for any year in the history. This is the significance of nakshatra in our study.

SLIDE 6

Examining the Planetary Positions - the Sun in Aries and Jupiter in Cancer



This slide shows the first date in BC era when the sun was in Aries, Jupiter in Cancer and Saturn was seen in Libra (shown in slide 7 for the same date). All the three planets could not be shown in one slide because these are spread over 7 zodiac constellations and at any given point of time, only six such zodiacs can be shown on one slide. Note that Venus is between Gemini and Taurus and not in Pisces, the place where Valmiki has stated it to be on the date of birth of Lord Ram. The constellation boundaries are drawn in blue.

SLIDE 7

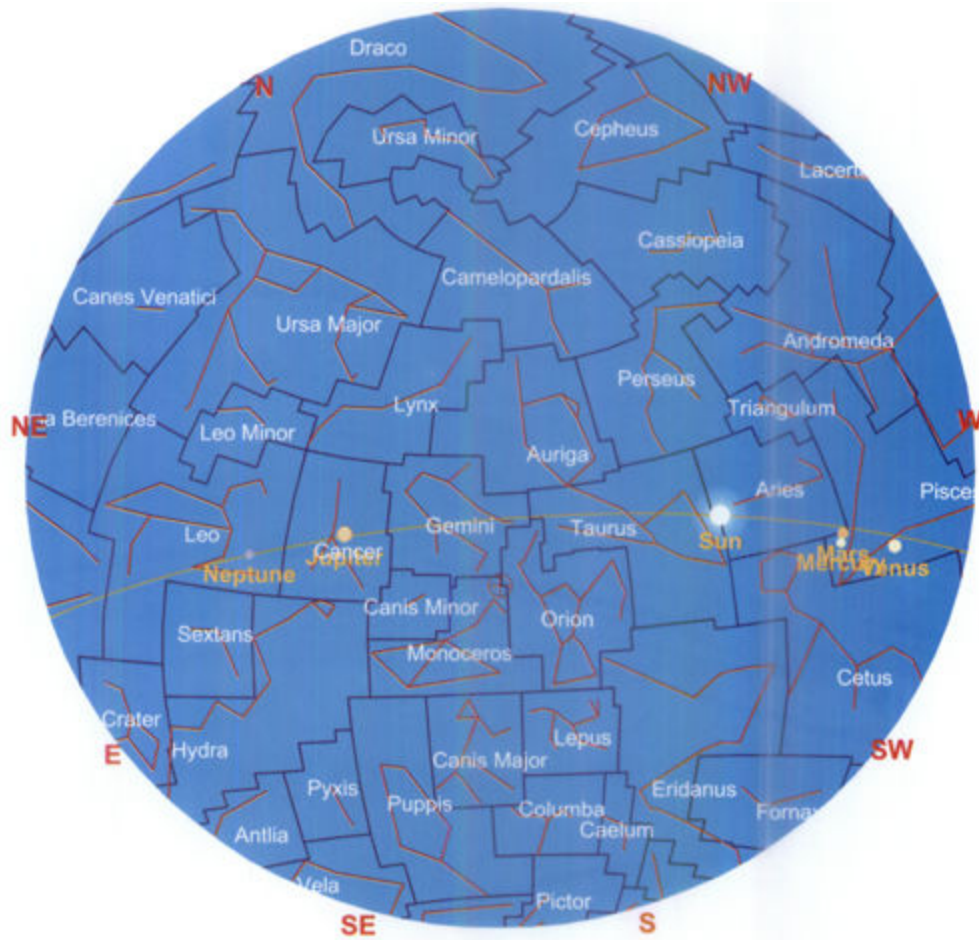
Planetary Positions - Saturn in Libra (while the Sun is in Aries, Jupiter in Cancer)



This slide is complementary to slide 6 and shows the date in BC era when for the first time Saturn comes to Libra while the sun was in Aries and Jupiter was in Cancer (last two planets shown in previous slide). Note that Mars is in Sagittarius and not in Capricorn, where it is stated to be on the date of birth of Lord Ram. Since Venus and Mars are not in the described positions, this cannot be the date of birth of Lord Ram. Note the date and time printed by the software. The longitude and latitude corresponds to Ayodhya.

SLIDE 8

Planetary Positions for Date of Birth of Lord Ram - 18 January 4497 BC



In our quest to find the date when the planetary configuration described in the Ramayana for the date of birth of Lord Ram was seen in the sky, we reach an important date of 23 March 1698 BC. The above slide and the following one show the sky view on this date. Note that the sun was in Aries, Jupiter in Cancer, Venus was seen in Pisces, while Saturn was in Libra (side 9). However, Mars was in Pisces and not in Capricorn where it is stated to be on that day. Note that the sun has moved to almost the boundary of Aries due to the precision cycle.

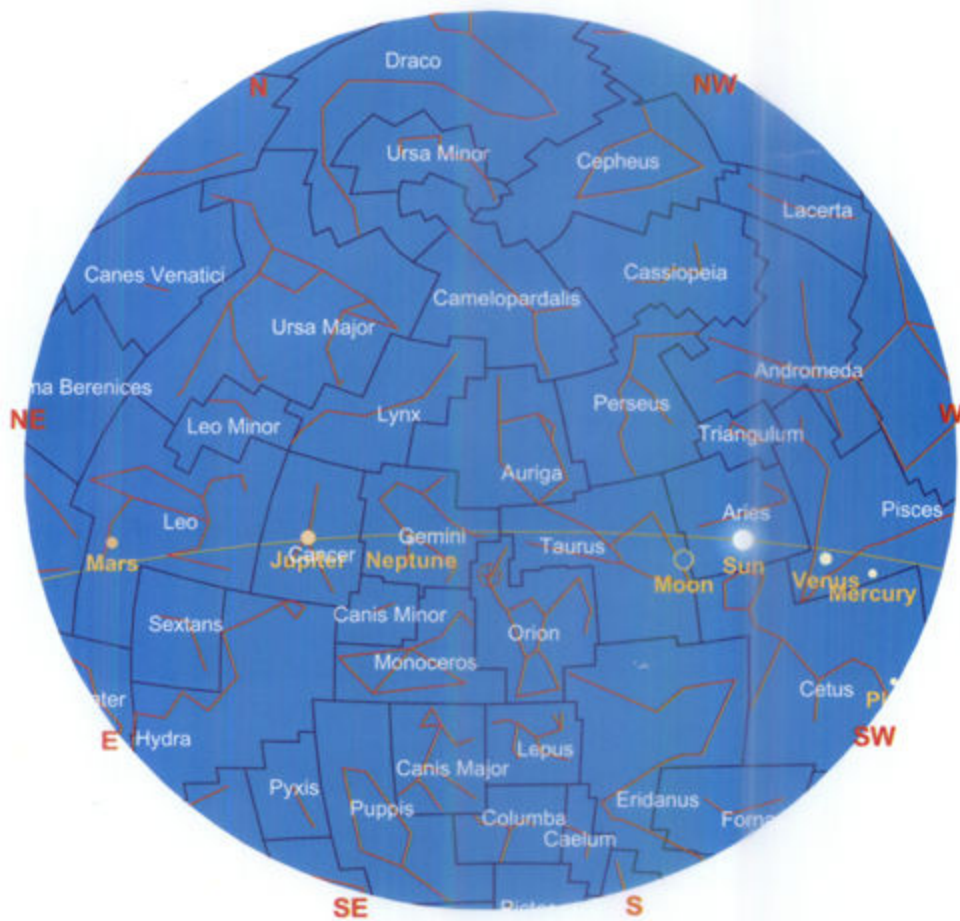
SLIDE 9

Planetary Positions for Date of Birth of Lord Ram - 18 January 4497 BC



This slide, along with the previous one, shows the sky view on 23 March 1698 BC. Saturn may be seen in Libra, while the previous slide showed that the sun was in Aries, Jupiter in Cancer and Venus was seen in Pisces. Since Mars was located in Pisces and not in Capricorn where it is stated to be on the date of birth of Lord Ram, this cannot be the one. Constellation boundaries drawn in blue would help in locating the planets within the zodiacs. Note the date, time and location printed automatically by the software.

SLIDE 10
Planetary Positions for Date of Birth of Lord Ram - 18 January
4497 BC



This slide along with slide 11 shows another set of planetary configuration close to the one described in Ramayana for the date of birth of Lord Ram. On 1 March 2374 BC, the sun was in Aries, Jupiter in Cancer, Venus was in Pisces and Saturn was seen in Libra (slide 11). But note that Mars is in Leo and not in Capricorn, the place where it is mentioned to be in the Ramayana on the birthdate of Lord Ram. One can see the no-moon day or amavasya day in the slide.

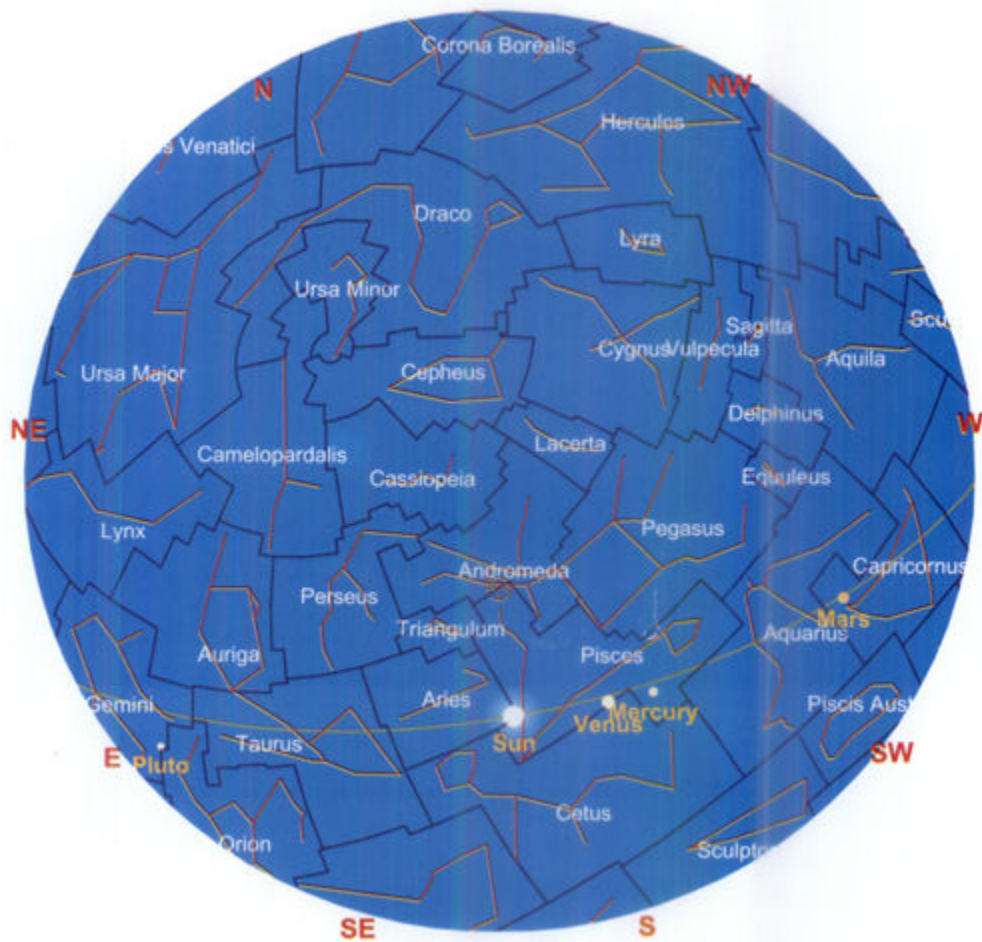
SLIDE 11
Planetary Positions for Date of Birth of Lord Ram - 18 January
4497 BC



This slide along with slide 10 shows the sky view on 1 March 2374 BC. Slide 10 shows the position of the sun, Jupiter and Venus. This slide shows that Saturn was in Libra. However, as Mars was seen in Leo and not in Capricorn, where it has been stated to be in the Ramayana, this cannot be the date of birth of Lord Ram. Therefore, we must move backwards in time to find out the date when the planetary configuration described in the Ramayana tallies with the sky view generated by the software.

SLIDE 12

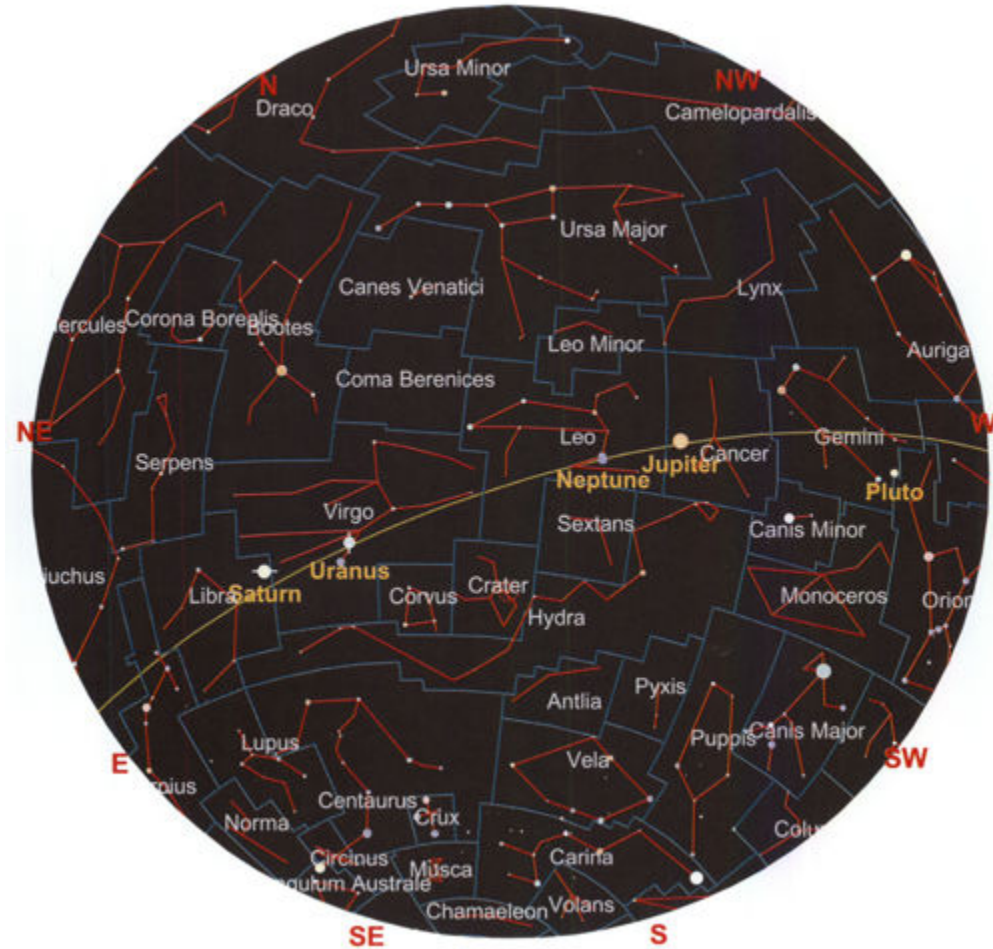
Planetary Positions for Date of Birth of Lord Ram - 18 January 4497 BC



Moving backwards by about 2,000 years as compared to the previous slide we come across the date when all the five main planets are seen in the places described in the Ramayana on the birthday of Lord Ram. This slide along with slide 13 shows the sky view on 18 January 4497 BC. The sun was in Aries, Venus in Pisces and Mars was seen in Capricorn. Slide 13 shows that Saturn was in Libra and Jupiter in Cancer on that day. Is this the date of birth of Lord Ram?

SLIDE 13

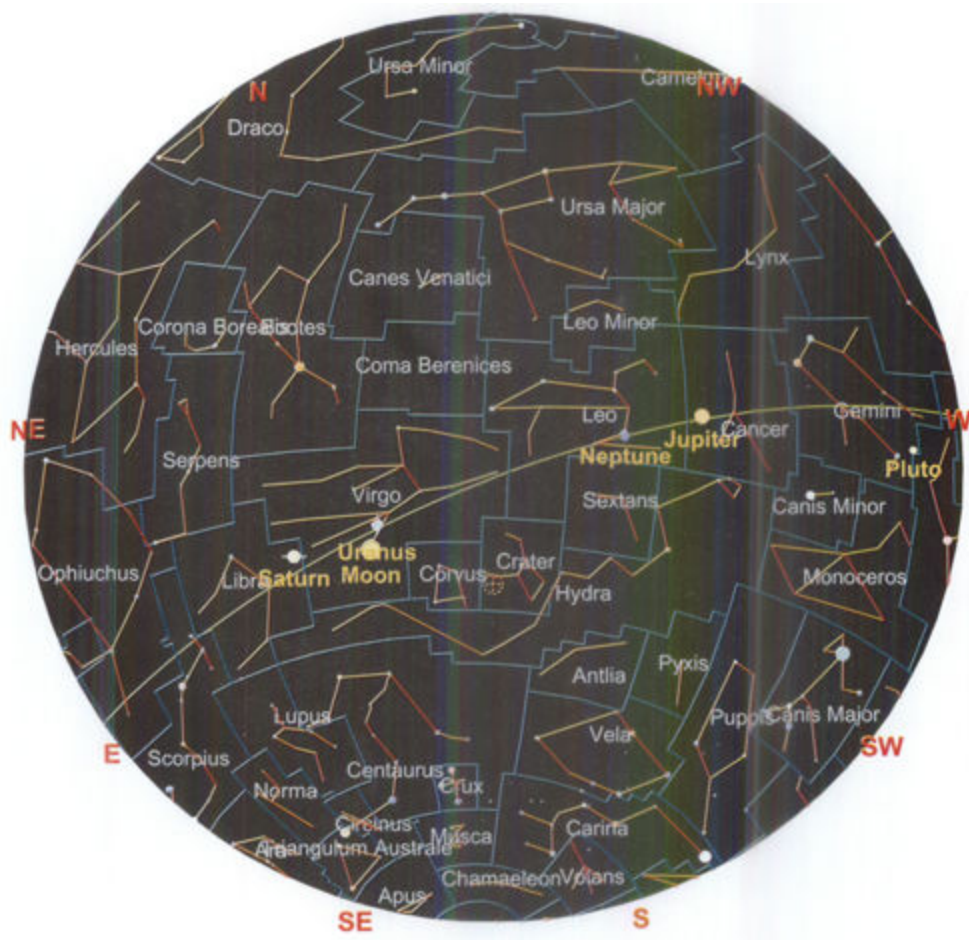
Planetary Positions for Date of Birth of Lord Ram - 18 January 4497 BC



This slide shows that on 18 January 4497 BC, Saturn was in Libra and Jupiter was in Cancer. Slide 12 showed that on this day the sun, Venus and Mars were in Aries, Pisces and Capricornus respectively. The question is whether this was the date of birth of Lord Ram. Remember that he was born on the navmi of shukla paksha of Chaitra. Thus, we should find the date of the navmi of the shukla paksha (9th day from no-moon day) in the Chaitra month that year, and then see whether these five planets remain in these positions on that day.

SLIDE 14

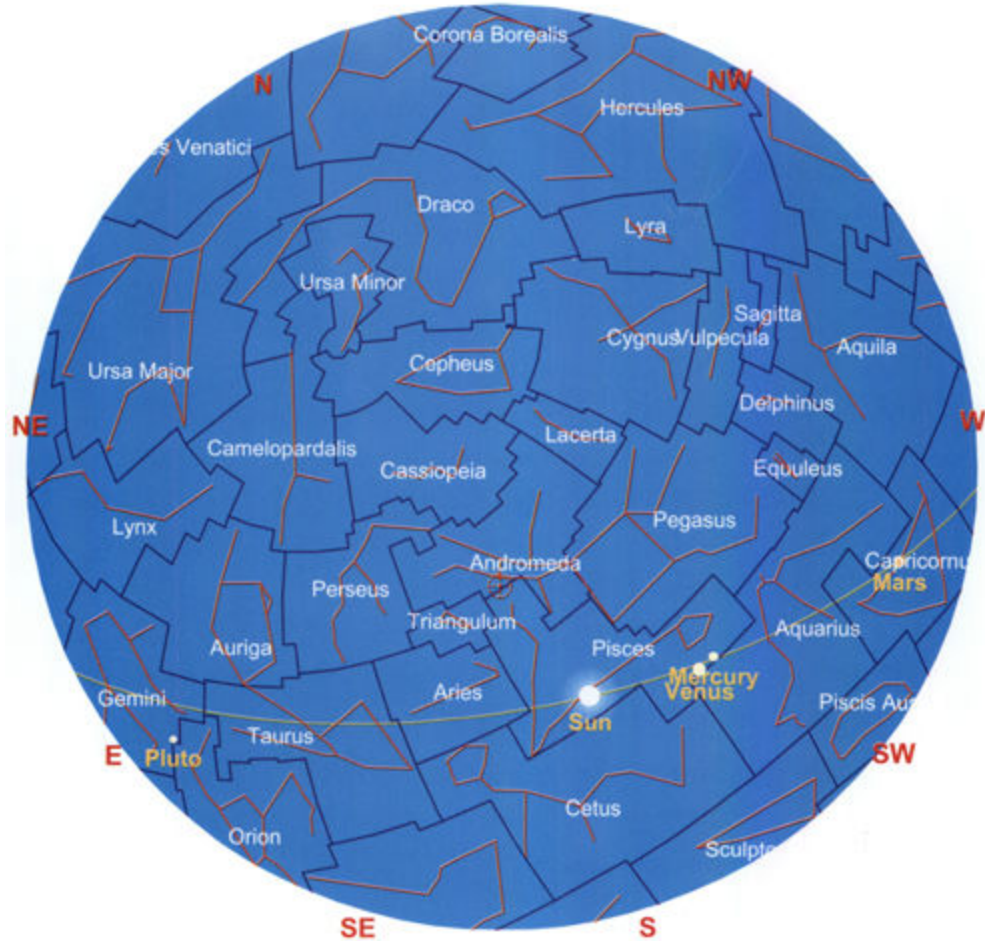
Ascertaining the End of the Chaitra Month - 12 January 4497 BC



This slide shows that the full moon of 12 January 4497 BC occurred near the star Spica or Chitra Nakshatra in the Virgo constellation. This marks the end of the month of Chaitra in the year 4497 BC. A full moon marks the culmination of the increasing phase of the moon, which spans over about 15 days beginning with the no-moon day. Thus, Navmi (9th day) of the shukla paksh (increasing phase of moon) should have been on 5 January that year. The following slide would depict the position of the planets on that day.

SLIDE 15

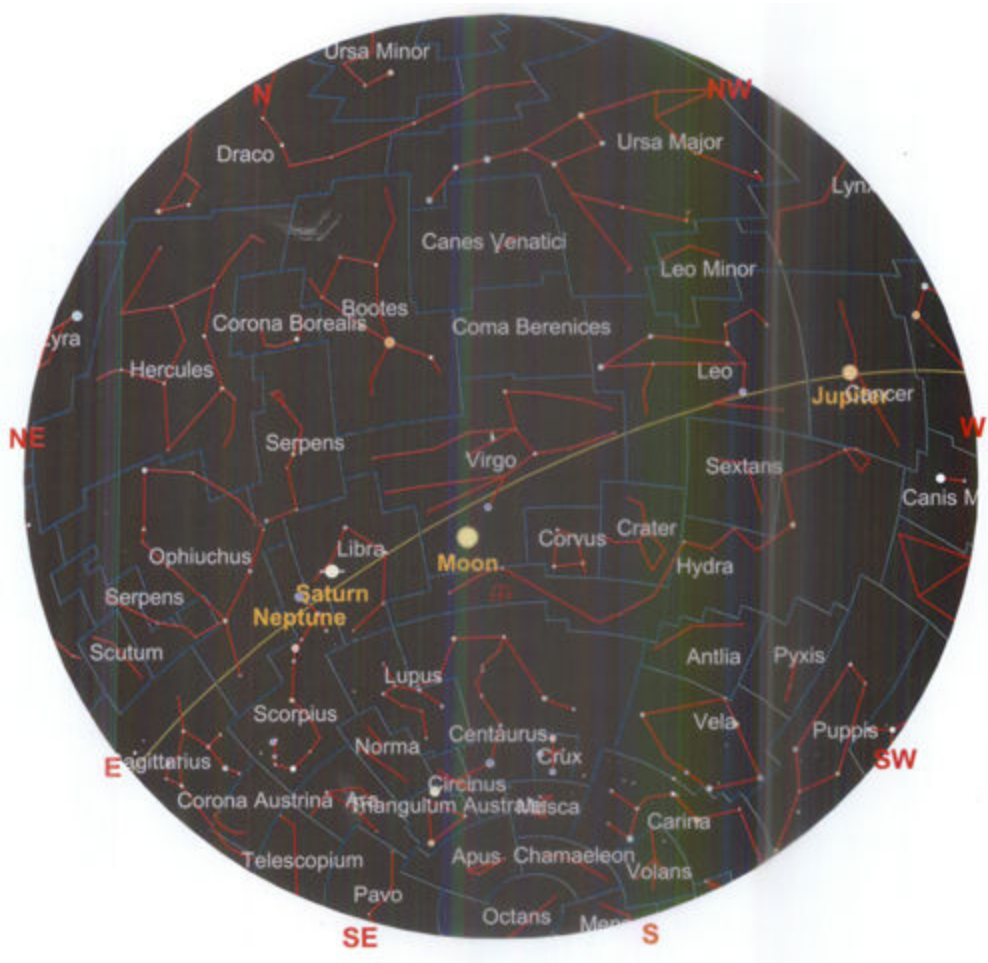
Planetary Positions on 10 January 5114 BC - Navmi of Chaitra Month



This slide shows the sky view on 5 January 4497 BC, which was the navmi of the Shukla Paksh of Chaitra month that year. The time of observation is mid-day and the latitude and longitude corresponds to Ayodhya, the birthplace of Lord Ram. We find that the sun has slipped out of Aries and actually it is seen in Pisces. Verification of the position of other planets would not serve any purpose since the sun itself is out of the place where it has been described on the birthday of Lord Ram. Thus, this cannot be the date of birth of Lord Ram.

SLIDE 16

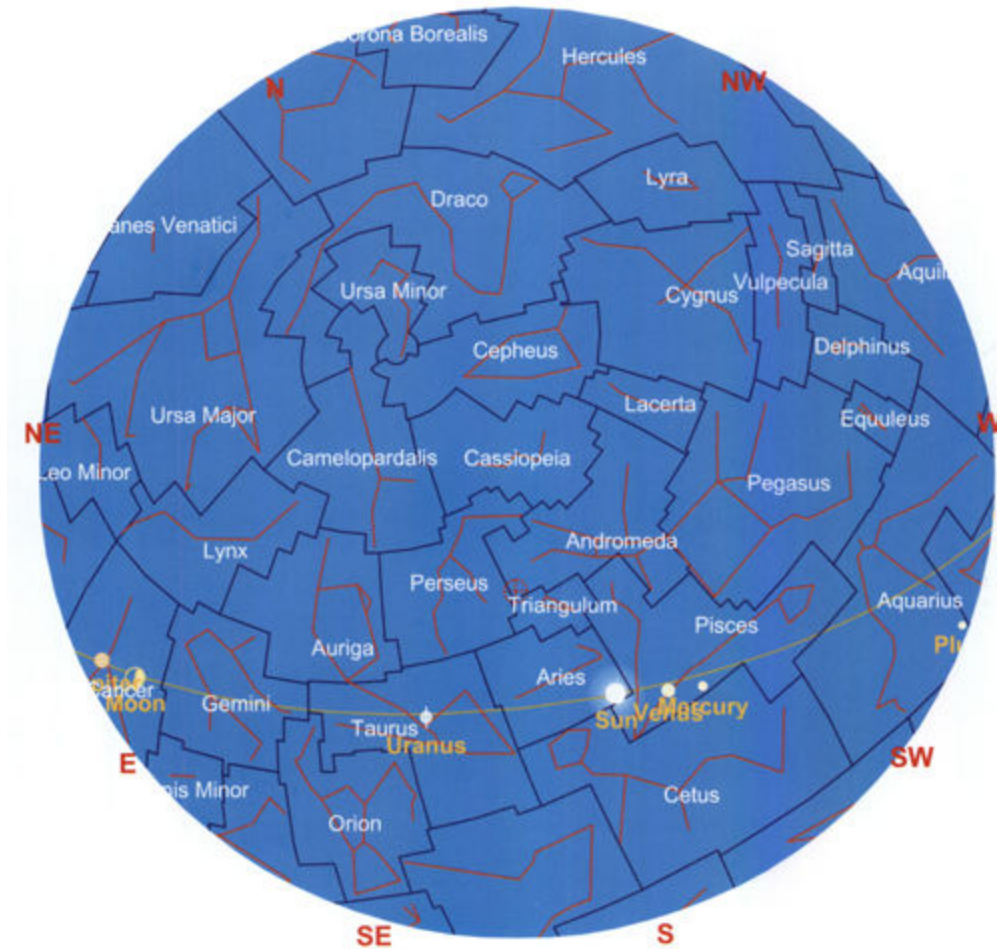
Ascertaining the Completion of Chaitra Month in the year 5114 BC



This slide shows the occurrence of a full-moon near the star Spica in Virgo constellation on 16 January 5114 BC, thereby implying the end of the Chaitra month that year. A full-moon marks the culmination of the increasing phase of the moon, which spans over about 15 days beginning with the no-moon day, the navmi of Shukla Paksh of Chaitra month should be seven days before this date i.e., on 10 January 5114 BC. The next two slides would show the position of the planets on that day.

SLIDE 17

Planetary Positions on 10 January 5114 BC - Navmi of Chaitra Month



Slides 17 and 18 together show the sky view on 10 January 5114 BC, which was the navmi of the Shukla Paksh of the month of Chaitra of that year. See that the sun is in Aries, Jupiter in Cancer, Venus in Pisces and the moon is near the star Pollux or Punarvasu nakshatra. The next slide confirms that Saturn was in Libra and Mars was in Capricornus. Note that the Cancer constellation is rising in the east at 12.30 p.m. Further, the latitude/longitude corresponds to Ayodhya, the birthplace of Lord Ram.

SLIDE 18

Planetary Positions on 10 January 5114 BC - Navmi of Chaitra Month



This slide is complementary to slide 17 and shows that as mentioned in the Ramayana, Saturn was in Libra and Mars was in Capricorn on 10 January 5114 BC. These two slides show that on the navmi of the Shukla Paksh of the Chaitra month of that year, all the five planets were located in the places mentioned in the Ramayana of the date of birth of Lord Ram. Thus, all the conditions described for the birth of Lord Ram are met on this date. The next slide will show the position of all five planets at one place at the time of the birth of Lord Ram.

SLIDE 19

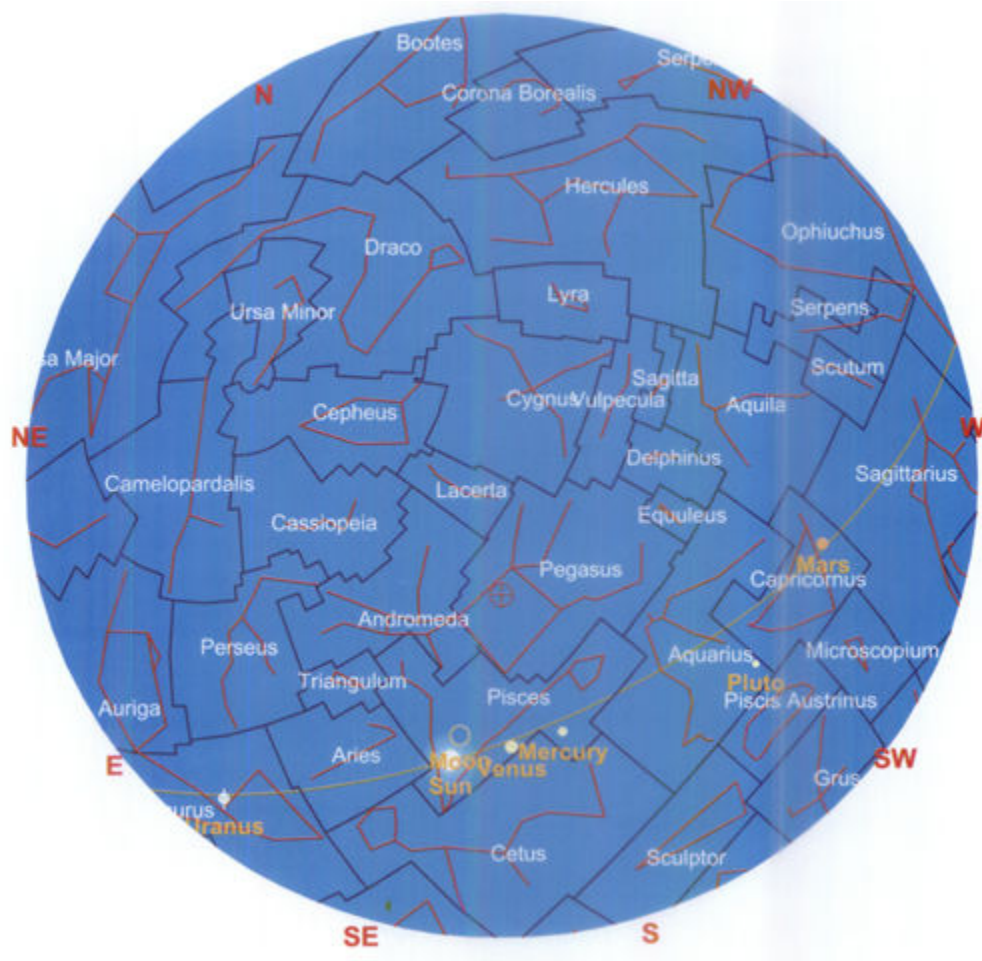
Date of Birth of Lord Ram - 10 January 5114 BC - Positions of all Planets



Do not get baffled while looking at the slide. Just watch all the twelve zodiac constellations around the circle - the ecliptic - and see the position of the planets. Note that the time of observation is 12.30 p.m. when Cancer constellation was rising in the east. See that Mars in Capricorn was just under the horizon while Saturn was well below the horizon at that time.

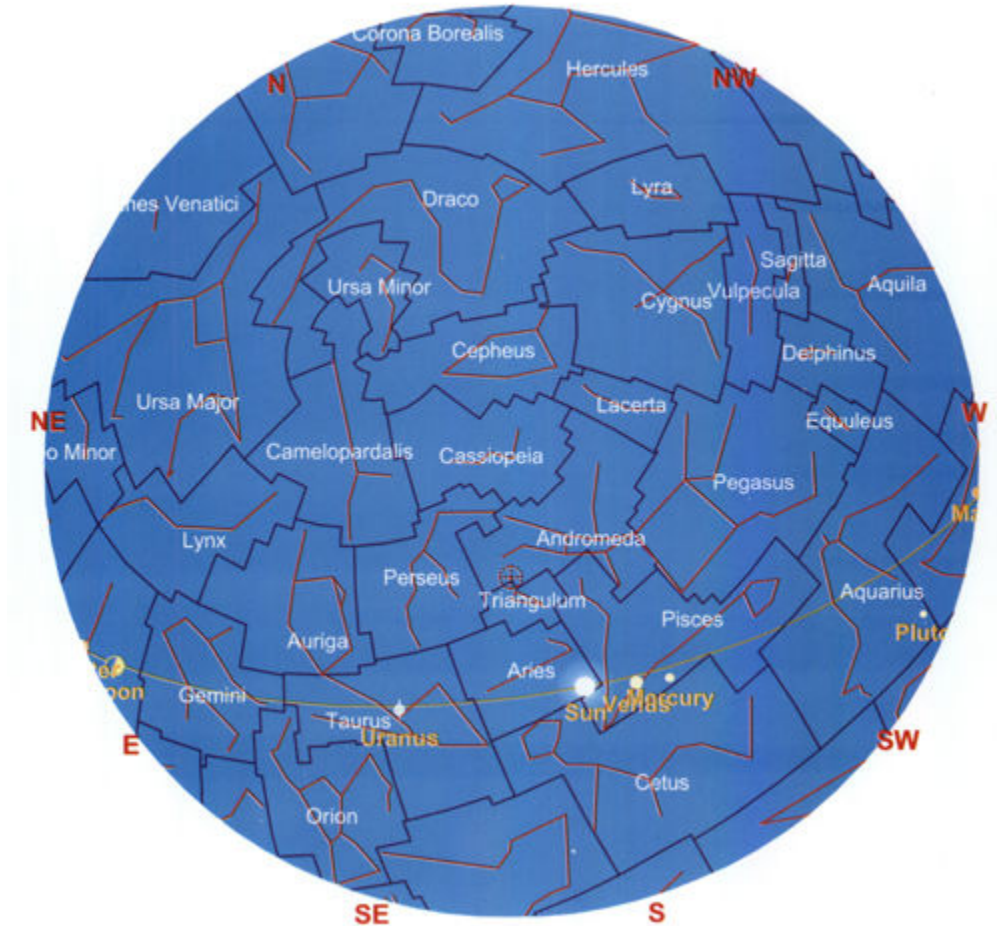
SLIDE 20

The Position of the Moon on Amavasya of 3 January 5114 BC in Chaitra Month



This slide shows that 3 January 5114 BC was the Amavasya or no moon day of Chaitra that year. Note that the moon is clearly located on the Revati nakshatra. If it were on Revati nakshatra on 3 January, it would reach Punarvasu nakshatra on 10 January 5114 BC, since there are six other nakshatras between these two and the moon moves by one nakshatra in a day. This clarifies the position of the moon on the date of birth of Lord Ram.

SLIDE 21
**The Time of the Birth of Lord Ram - Rising of Cancer Constellation
and Jupiter in East**



This slide helps in ascertaining the exact time of birth of Lord Ram. The Ramayana mentions that at the time of Lord Ram's birth, Jupiter was above horizon in Cancer zodiac. The sky view at 12.10 p.m. on the birthday of Lord Ram in this slide shows that Jupiter was not above the horizon till this time. Thus, Lord Ram should have been born after 12.10 p.m. Further, as Cancer rose only up to 1.00 p.m. that day, the time of the birth of Lord Ram cannot be beyond 1.00 p.m. Hence, we can conclude that Lord Ram was born between 12.10 p.m. and 1.00 p.m.

SLIDE 22

Time of the Birth of Prince Bharat - Rising of Pisces Constellation in the East



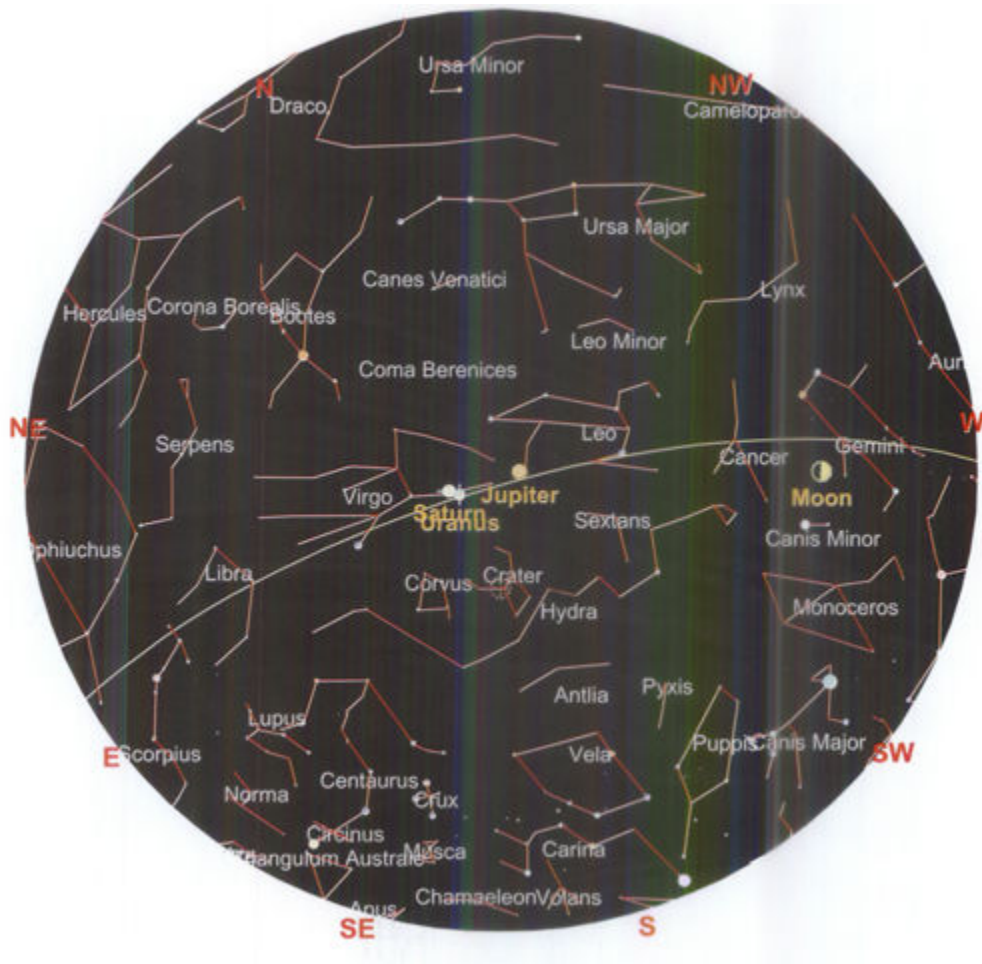
This slide shows the view of the sky at 5.30 a.m. on 11 January 514 BC, about sixteen hours after the birth of Lord Ram as seen from Ayodhya. The rising Pisces constellation implies the Pisces lagna in which Bharat was born. Pisces continued to rise up to about 7.30 a.m. that day. Hence, we can conclude that the time of the birth of Prince Bharat was around 6.30 a.m. on 11 January 514 BC.

SLIDE 23
Full Moon Day after Twenty-five Years of the Birth of Lord Ram -
12th January 5089 BC



This slide confirms that after twenty-five years of the birth of Lord Ram, the Chaitra month ended on 12 January 5089 BC. This gets illustrated from the observation that this was a full moon night and the moon was located near the Chitra nakshatra in Virgo constellation.

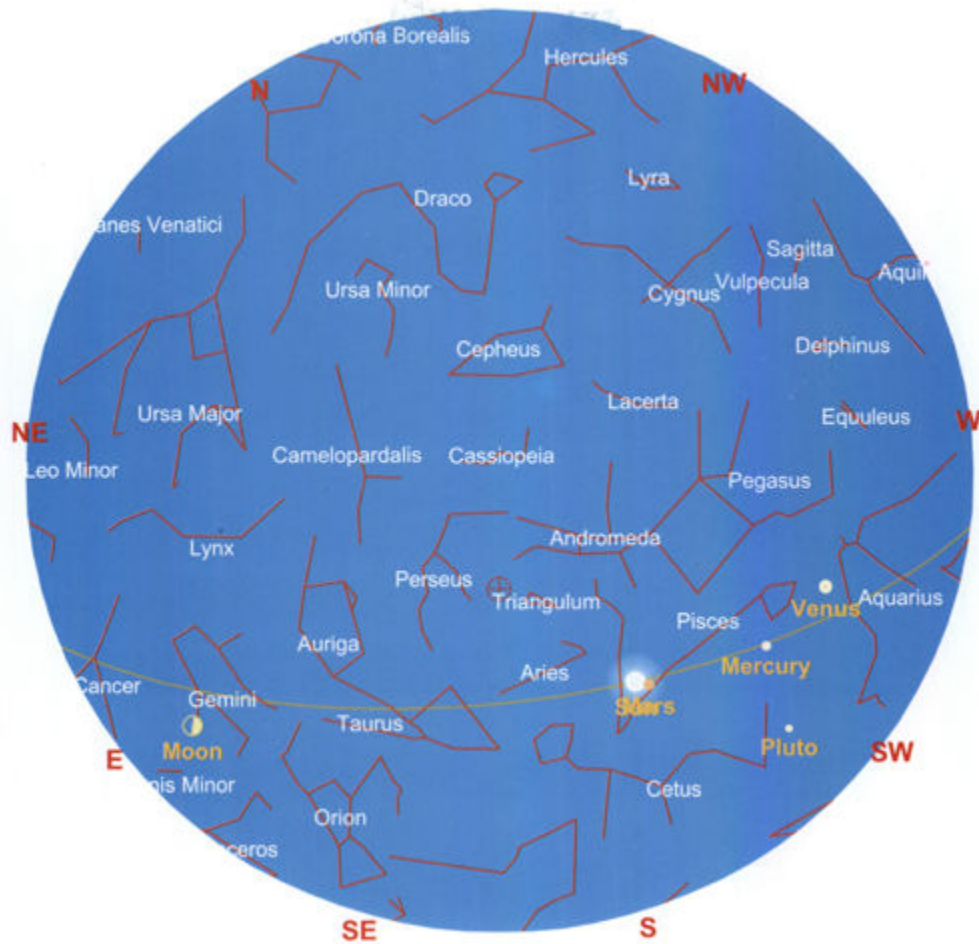
SLIDE 24
Position of the Moon - near Punarvasu Nakshatra on 4 January
5089 BC



King Dashrath told Lord Ram "today moon is at Punarvasu and tomorrow it will be at Pushya. Tomorrow you will be made king." This slide shows the date when this discussion took place between King Dashrath and Lord Ram i.e., 4 January 5089 BC. The moon was near the Punarvasu nakshatra in Gemini constellation as can be seen in the slide. It goes to the Pushya nakshatra next day - on 5 January, the day Lord Ram's coronation was planned.

SLIDE 25

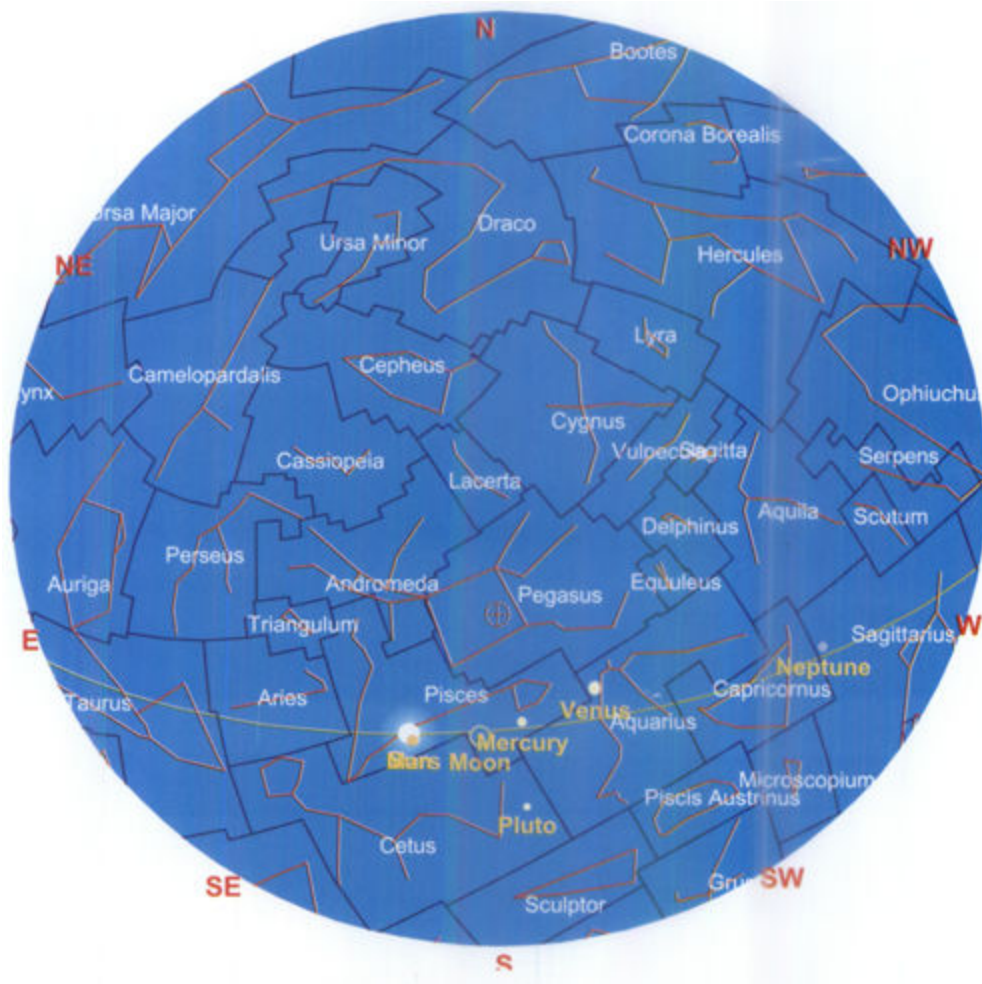
The Position of the Sun, Mars and Rahu on 3 January 5097 BC



King Dashrath mentioned to Lord Ram on 4 January 5089 BC that astrologers had told him that his own nakshatra had been surrounded by the malefic planets - sun, Mars and Rahu. This slide shows the position of the sun and Mars in the sky on 4 January 5089 BC. Note that both these planets were located together in Pisces constellation. The position of Rahu would be clear from the next slide.

SLIDE 26

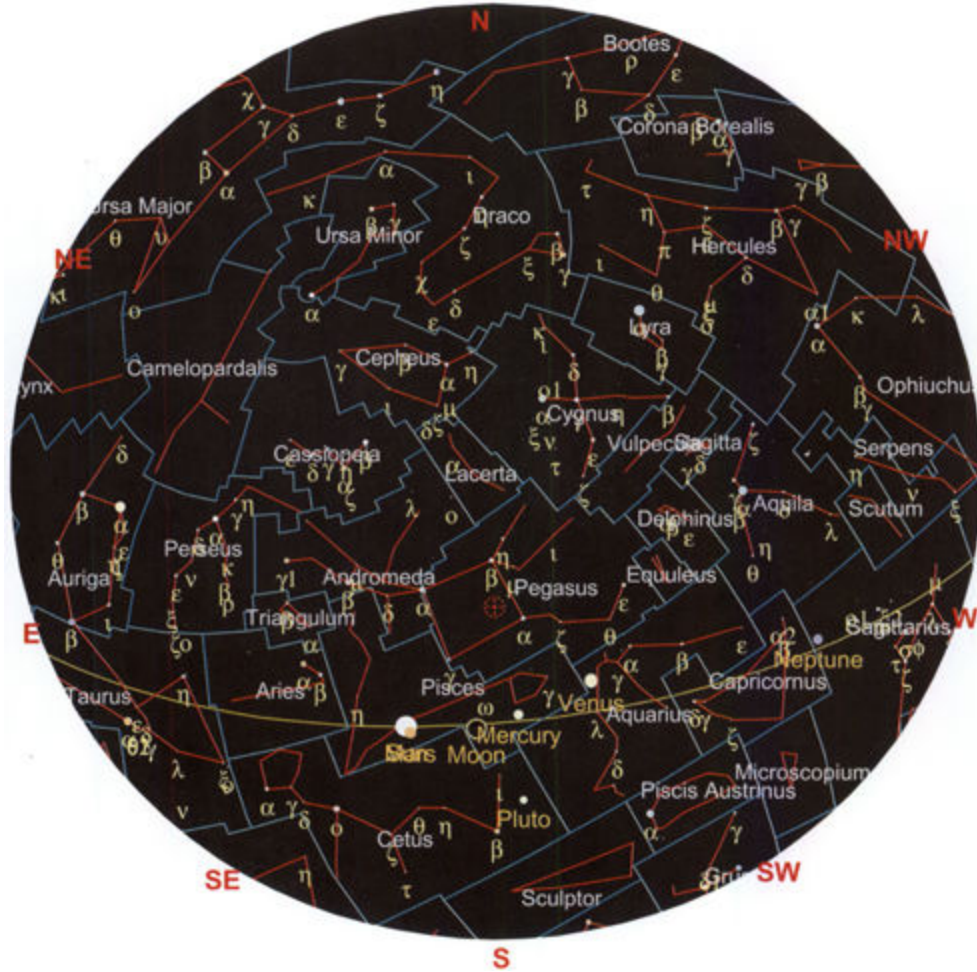
The Position of Rahu - the Moon Crossing Ecliptic on 28 December 5090 BC



The two imaginary planets - Rahu and Ketu are considered to be present in the constellations where the moon cuts the path of the sun (ecliptic) during every lunar month. This slide shows the moon crossing the ecliptic in the Pisces, while the sun and Mars are also located in the same zodiac. The names of the sun and Mars are overlapped and thus not very clearly legible. This confirms that all these three planets were located in Pisces at that time. Does this mean that Pisces constellation had some relationship with King Dashrath?

SLIDE 27

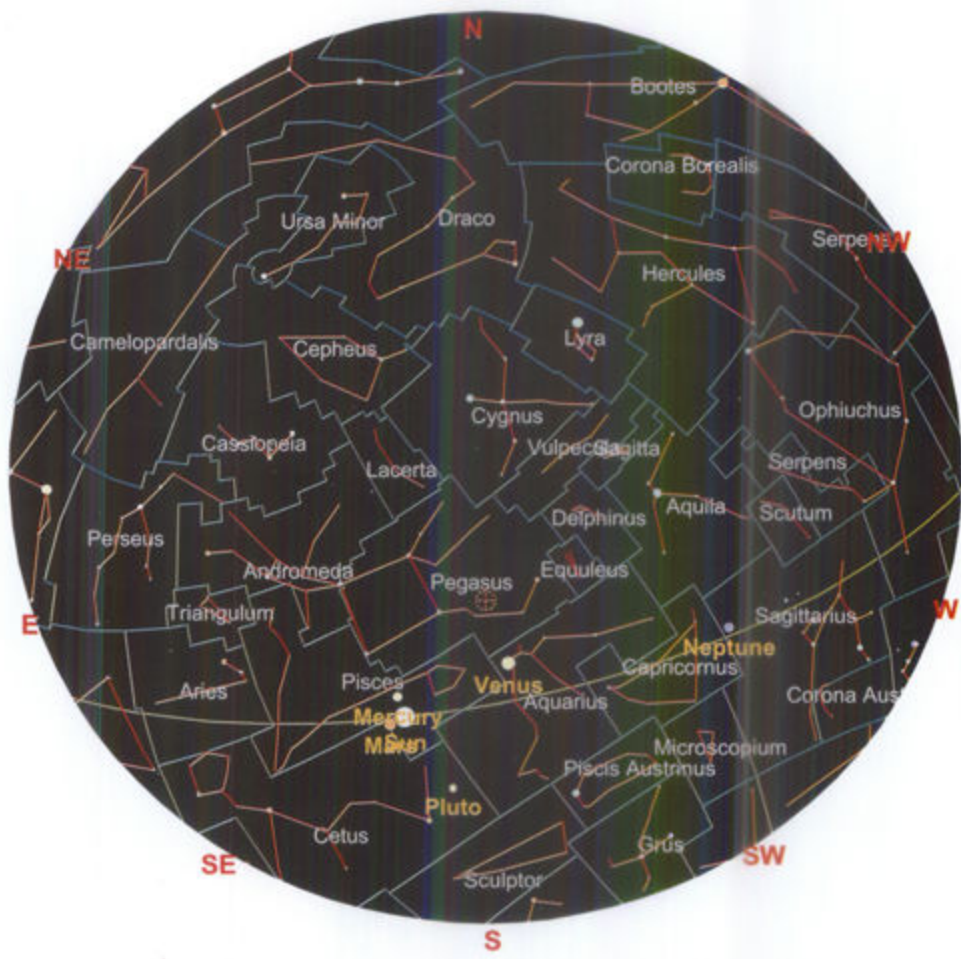
Ascertaining the Nakshatra of King Dashrath - the Position of Rahu within Pisces



This slide provides the names of different stars in the different constellations. However, important to us are the stars, which are identified as nakshatras in only the Pisces constellation. Note that on 28 December 5090 BC, the moon cuts the path of the sun exactly at star ω in Pisces constellation, which corresponds to the Purva-bhadrapad nakshatra. This nakshatra corresponds to those who have their names beginning with the Hindi word 'V' ('D') as per the traditional Indian astrological system.

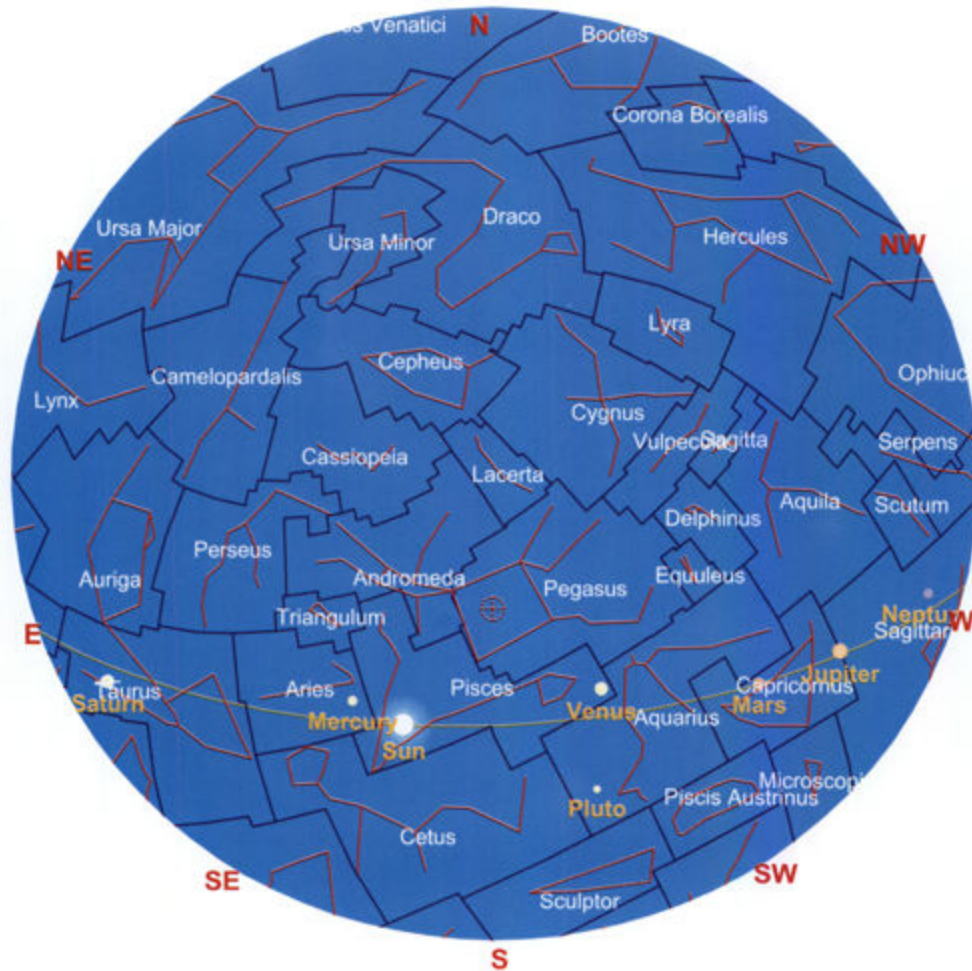
SLIDE 28

Ascertaining the Nakshatra of King Dashrath - the Position of the Sun, Mars and Rahu



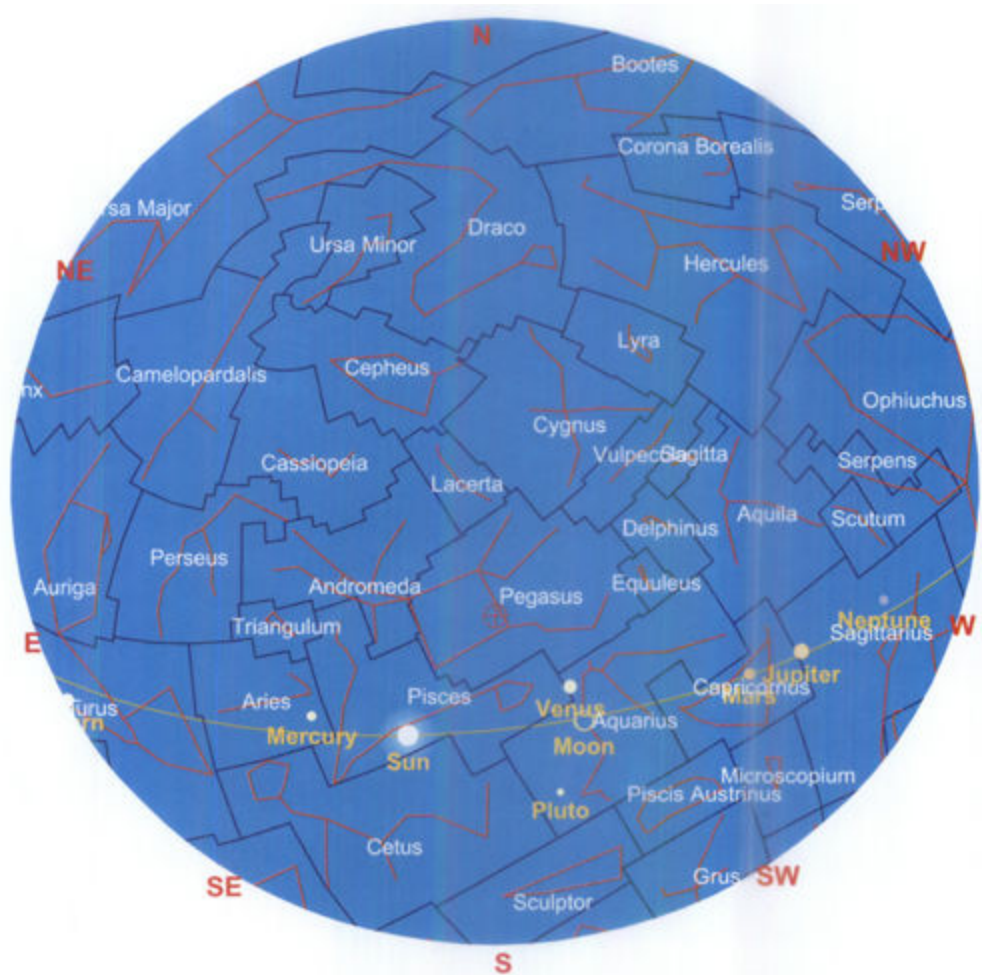
This slide shows that the sun and Mars were located almost at Purva bhadrapad nakshatra on 13 December 5090 BC, about three weeks before the discussion between King Dashrath and Lord Ram on 4 January 5089 BC. Comparison of the position of the sun and Mars in this slide with their respective positions in slides 26 and 25 would show that these two were moving together from west to east in the Pisces constellation during this period

SLIDE 29
The Position of the Sun, Mars and Rahu on 3 January 5097 BC



To check the accuracy of Kaushalya's statement made at 2/20/45, we need to examine the position of the sun, Mars and Rahu when Lord Ram was seventeen years old. These three planets should have been located in one zodiac constellation, around one nakshatra i.e., the nakshatra of King Dashrath. This slide shows the position of the sun and Mars on 3 January 5097 BC. The two planets are located in two different zodiacs. The sun is in Pisces while Mars is in Capricorn. The next slide clarifies the position of Rahu.

SLIDE 30
The Position of Rahu on 25 December 5098 BC - the Moon cuts
Ecliptic at Aquarius



This slide confirms that for the month of Chaitra ending in 5097 BC, Rahu was in Aquarius because on 25 December 5098 BC, the moon cuts the ecliptic in Aquarius constellation. Therefore, if it is assumed that Lord Ram went to exile at the age of seventeen, the planetary configuration mentioned in the Ramayana would not tally since at that time all the three planets were nowhere near to one another - the sun being in Pisces, Mars in Capricornus and Rahu in Aquarius.

SLIDE 31

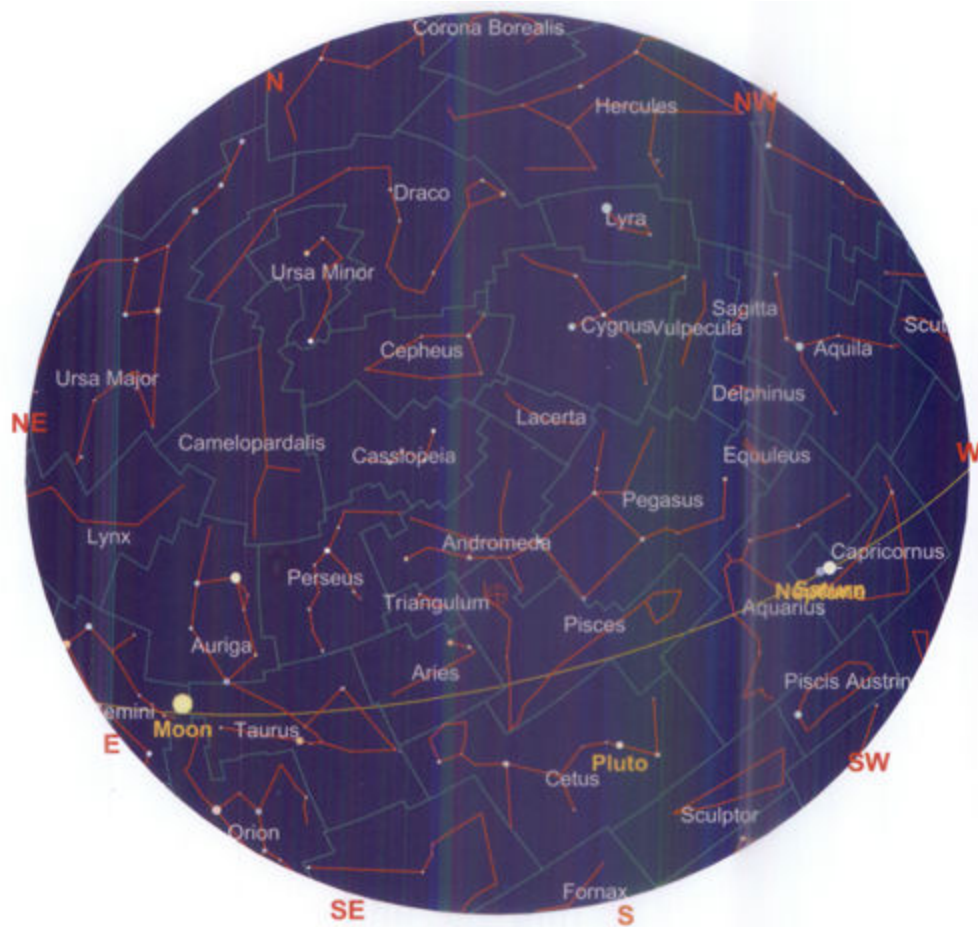
End of the Month of Phalgun - the Last Month of the Thirteenth Year of Exile



The twelve months of the lunar year end with the month of Phalgun when the full-moon occurs at the Uttara Phalguni nakshatra (star 5 P) of Virgo zodiac. The position of moon in this slide confirms that for the thirteenth year of exile of Ram, the Phalgun month ended on 20 December 5077 BC. As the first 14 days of the month of Chaitra (Krishna Paksh) are counted at the end of every year, we may conclude that the thirteenth year of exile ended on 3 January 5076 BC and the battle between Lord Ram and Khar must have been fought before this date.

SLIDE 32

The End of the month of Margshirsh - First month of Hemant in the Thirteenth Year of Exile



Surpnakha visited Panchwati after the end of the first month of the Hemant ritu i.e., Margshirsh. This slide shows that in 5077 BC, Margshirsh ended on 22 September since the full moon occurred near Mrigshira nakshatra (star β in Taurus). Thus, Surpnakha went to Panchwati only after 22 September 5077 BC and hence the Lord Ram-Khar battle must have been fought after this date and before 20 December 5077 BC, as concluded in the previous slide. We need to find whether a solar eclipse really occurred during this period or not.

SLIDE 33

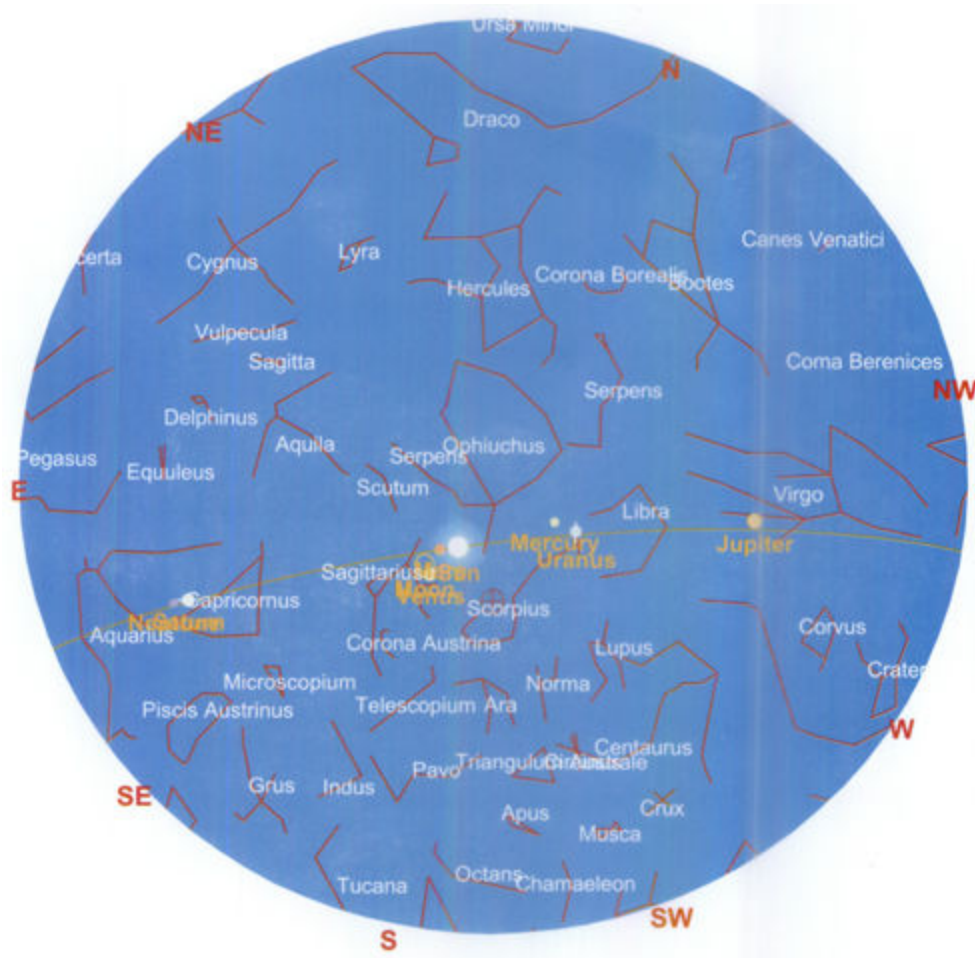
Looking for a Solar Eclipse on No-Moon day in the Month of August 5077 BC



This slide shows the position of the moon and sun during the daytime on the no-moon (amavasya) day in the month of August 5077 BC. The moon can be seen in the vicinity of the sun but it does not overlap the sun. No solar eclipse, therefore, occurs on this amavasya day. Note that the latitude and longitude correspond to Panchwati, which is now, the modern day city of Nasik in the state of Maharashtra.

SLIDE 34

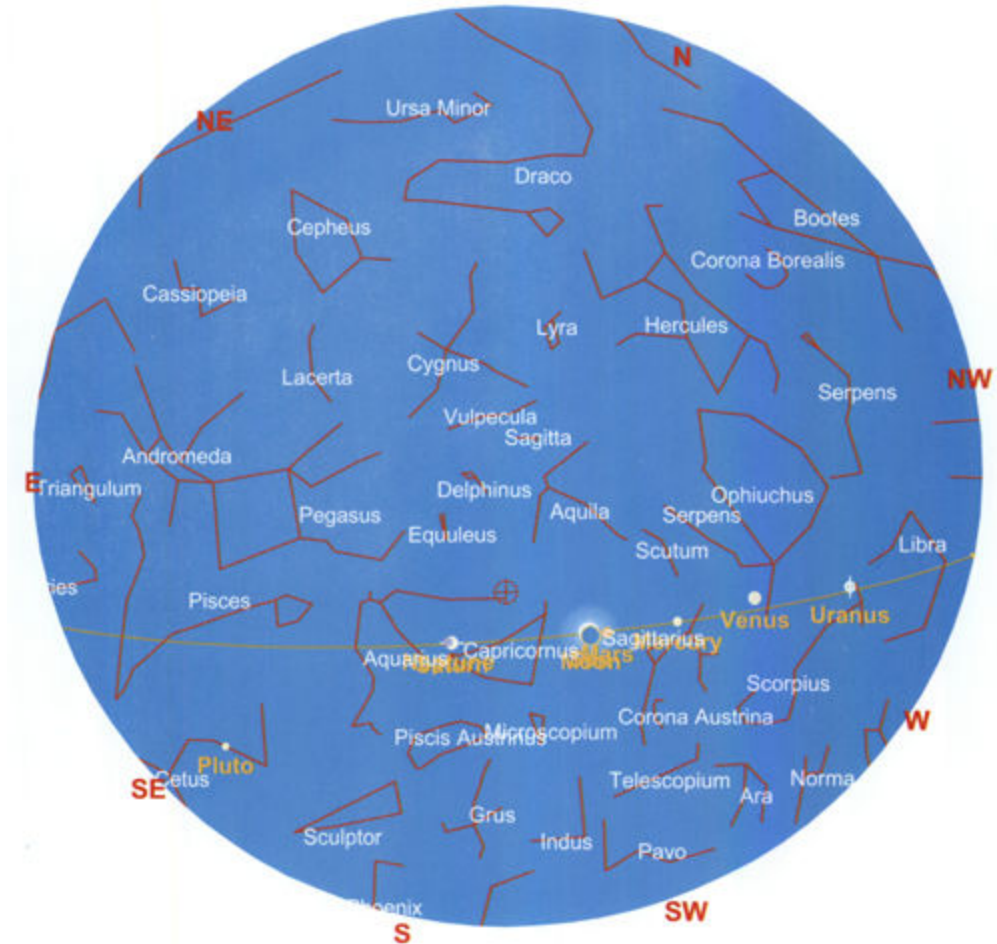
Looking for a Solar Eclipse on No-Moon day in the Month of September 5077 BC



This slide shows the position of the moon and the sun during daytime on the no-moon (amavasya) day in the month of September 5077 BC. The moon can be seen in the vicinity of the sun but it does not overlap the sun. No solar eclipse, therefore, occurs on this amavasya day as well. Note that the latitude and longitude correspond to Panchwati, which is now the modern day city of Nasik in the state of Maharashtra.

SLIDE 35

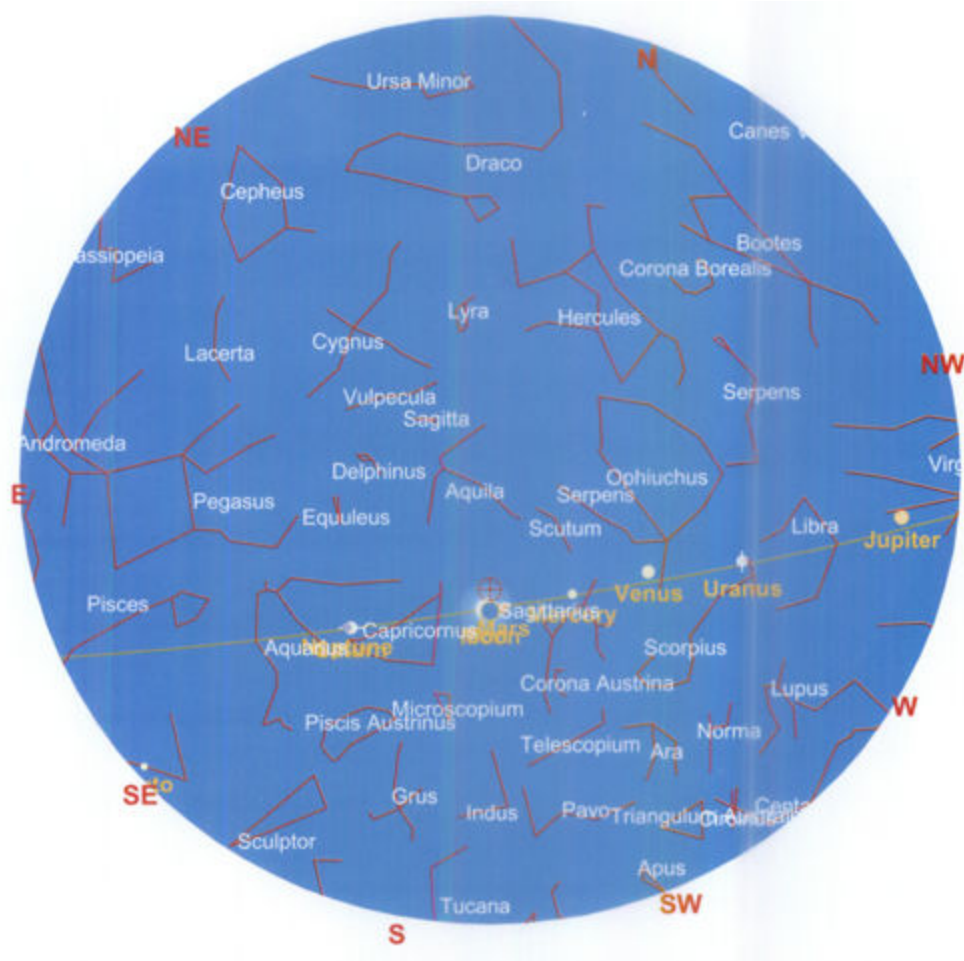
The Solar Eclipse we were Looking for! (Amavasya of 7 October 5077 BC)



This slide shows the Solar Eclipse we were looking for! Note that the disc of the moon is covering the disc of the sun partly, resulting in the solar eclipse. The time is 2.00 p.m. and the latitude/longitude are for Panchwati. The names - sun and moon - are also overlapping and thus not clearly readable. Interestingly, in the entire year 5077 BC, this was the only solar eclipse to have taken place for this location. This solar eclipse occurred on the day Lord Ram fought the battle with Khar. Further confirmation is available in next slide.

SLIDE 36

The Solar Eclipse of 7 October 5077 BC - the Position of the Planets in the Sky



See the sky view during the solar eclipse of 7 October 5077 BC. Valmiki states, "In the middle of his army, Khar, was looking like Mars in the middle of the Planets in the sky". The planets, which were visible to unaided eyes are: (East to West) Saturn, Sun, Moon, Mars, Mercury, Venus, Jupiter, and Mars was indeed in the middle of this chain! We have tallied a unique planetary configuration. Rarely, all the planets are in the day sky, rarer it is that Mars would be in the middle of them and finally, the occurrence of a solar eclipse at that time makes it absolutely unique.

SLIDE 37

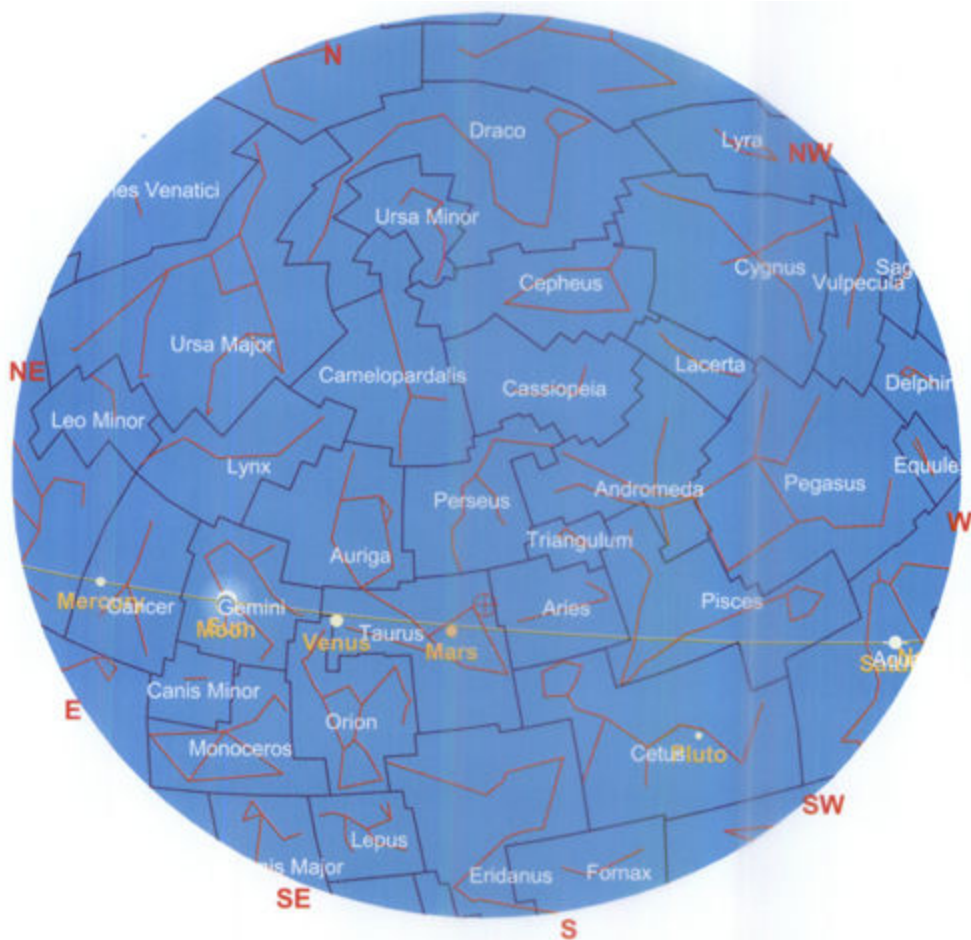
The Date of the Killing of Bali - Full-moon Day to Mark the End of Ashad in 5076 BC



This slide shows that on 16 April 5076 BC, the full moon occurred near the Purva-ashad nakshatra (star Nunki or 34 σ) in the Sagittarius constellation that marked the end of the month of Ashad that year. We know that a solar eclipse occurs only on a no-moon day. If the full-moon day of the Ashad was on 16 April 5076 BC, the no-moon day for that month must have been on 3 April. If at all a solar eclipse had occurred on the day Bali was killed by Lord Ram, it should have been on the no-moon day of Ashad. The next slide provides the confirmation.

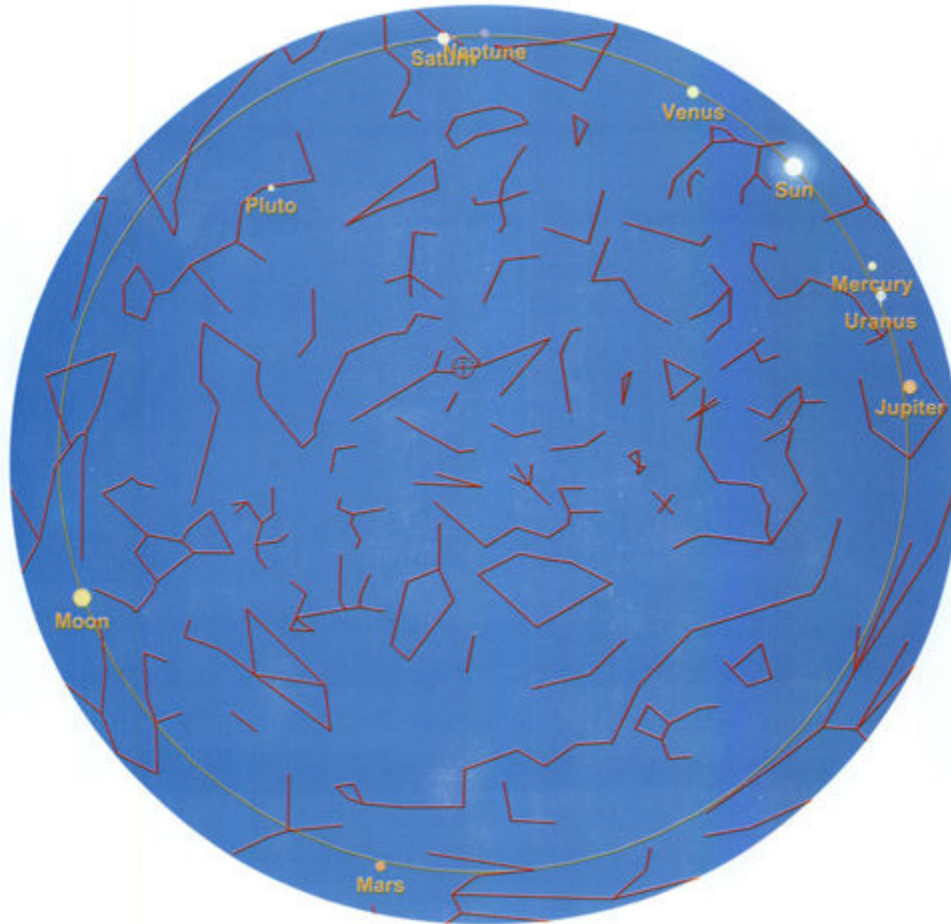
SLIDE 38

The Date of the Killing of Bali - the Solar Eclipse of 3 April 5076 BC



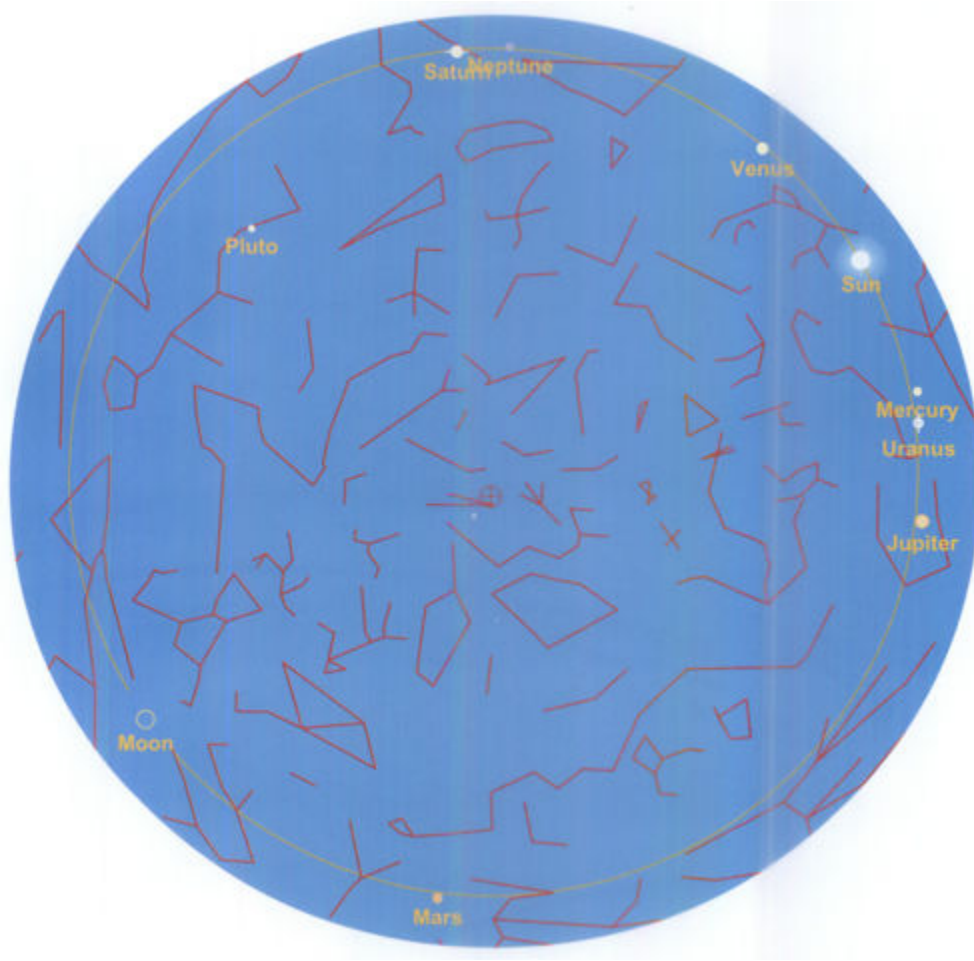
Maharishi Valmiki states that in the early morning of the day on which Bali got killed, Sugreev went and roared in front of Bali's fort. Hearing his roar, Bali's face became red resembling the sun that was caught by Rahu during the eclipse. (4/15/3). This slide shows that in fact, a solar eclipse had actually occurred on the no-moon day of Ashad month and that too in the morning hours as mentioned in the Ramayana.

SLIDE 39
The Date of Hanuman's Visit to Lanka - the Lunar Eclipse of 12
September 5076 BC



This slide, along with slide 40, will show how a lunar eclipse is shown by the software. Shown in this slide is the full moon that marked the end of the month of Margshirsh in the year 5076 BC. The time, as may be seen in the box at the left corner below is 4.14 p.m. The next slide shows the moon just one minute later i.e., at 4.15 p.m.

SLIDE 40
The Date of Hanuman's Visit to Lanka - the Lunar Eclipse of 12
September 5076 BC



This slide shows the moon after a gap of just one minute as compared to its position in slide 39. Note that the moon is now shown as if it is of a no-moon day (Amavasya). Actually, the moon is eclipsed and this time (4.15 p.m.) marks the beginning of the eclipse. Note the gap in the ecliptic indicating the shadow of earth through which the moon passes and gets eclipsed. Note that the coordinates correspond to Sri Lanka. At this time of evening, the moon was below the horizon.

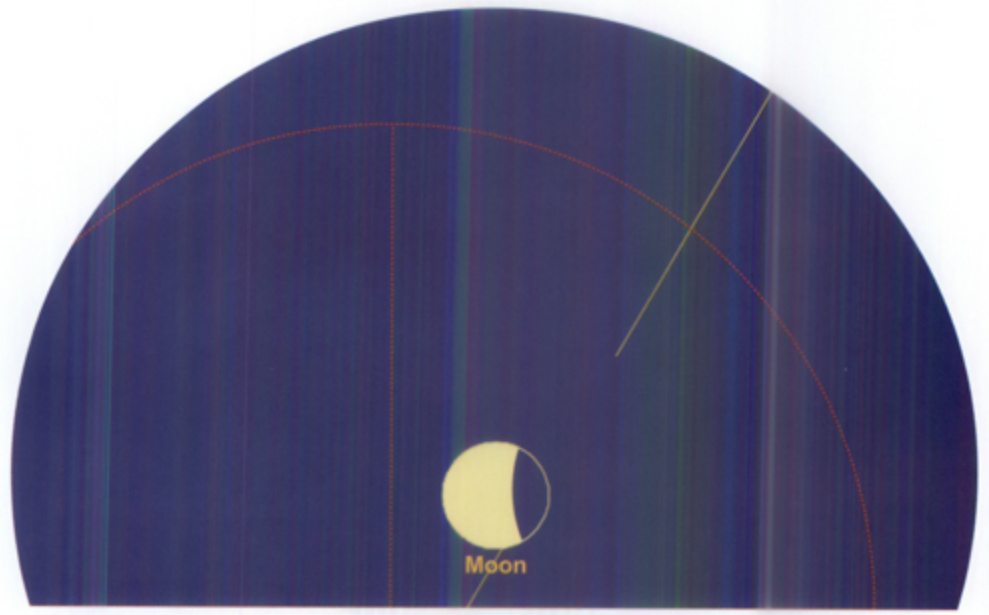
SLIDE 41

The Lunar Eclipse of 12 September 5076 BC - Eclipsed Moon Rising in the East



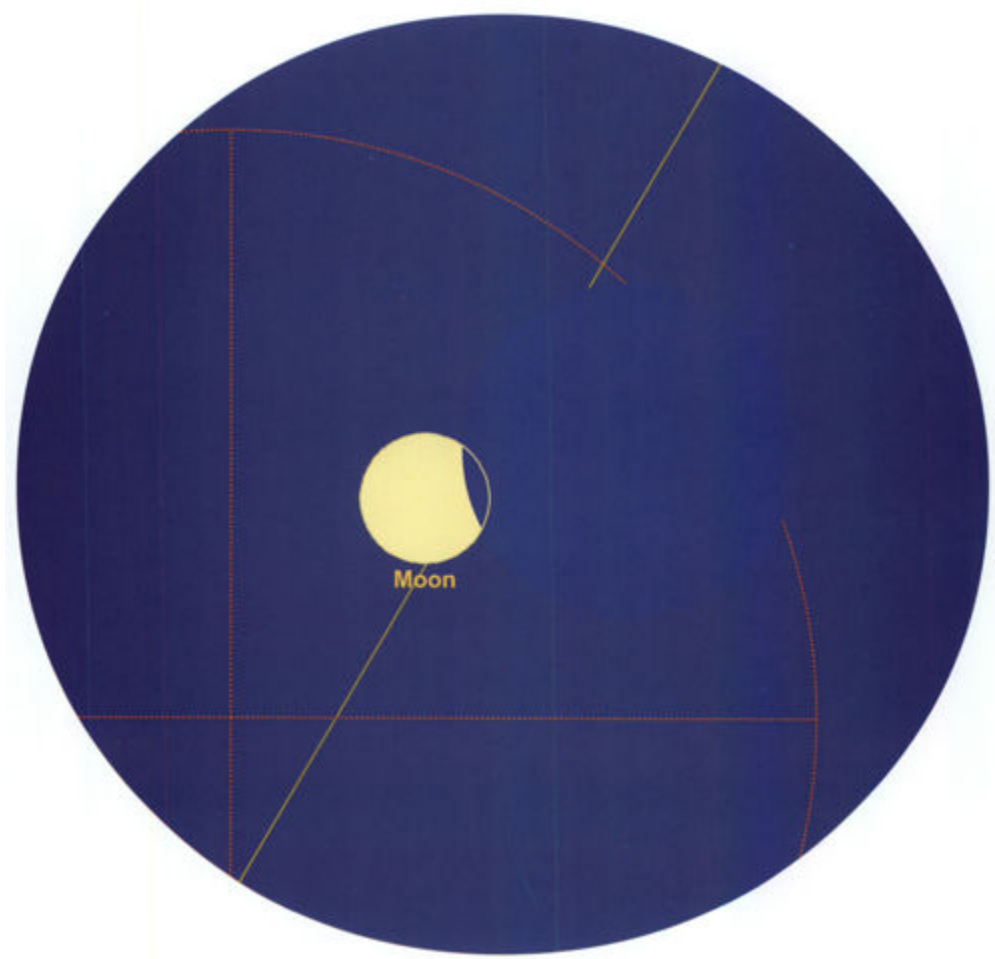
This slide shows the sky view on the evening (6.30 p.m.) of 12 September 5076 BC, the day Hanuman visited Lanka. Note that the eclipsed moon rose in the eastern sky. See the gap in the ecliptic where the moon is located. This was not a total lunar eclipse. The next slide shows the magnified view of moon at this time and it confirms the occurrence of the partial eclipse. Note that the place of observation is Sri Lanka.

SLIDE 42
The Lunar Eclipse of 12 September 5076 BC - Eclipsed Moon
Rising in the East



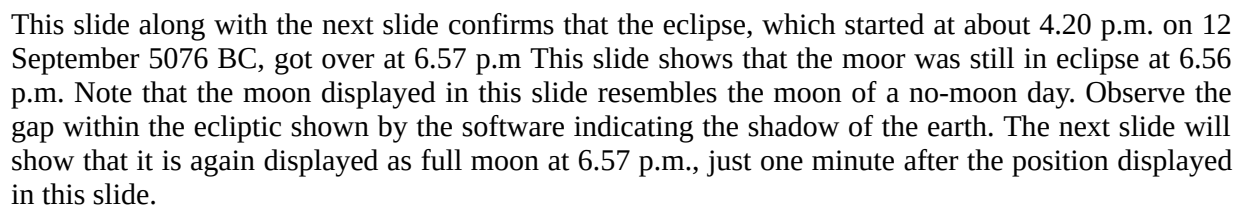
This slide shows the magnified view of the partially eclipsed moon rising in the evening of the full moon day of the month of Margshirsh. The moon was cleared of the eclipse at 6.57 p.m. Precisely the same facts have been mentioned in the Ramayana for the day Hanuman visited Lanka and met Sita. There are numerous references to the 'full moon'; as well as the 'moon released from the captivity of Rahu or eclipse'. This slide confirms that Hanuman visited Lanka on the full moon day of Margshirsh and not of Paush in 5076 BC.

SLIDE 43
The Date of Hanuman's Visit to Lanka - 12 September 5076 BC -
Lunar Eclipse

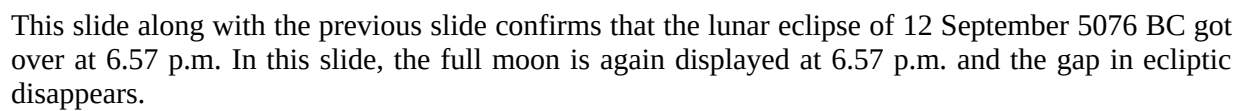


This slide shows the partially eclipsed moon at 6.40 p.m. The eclipse is on the wane and it got cleared at 6.57 p.m. Hanuman had entered the city of Lanka after the darkness of night descended and while the "full moon was shining". Subsequently, he found Sita in Ashok Vatika but at that time Ravan was also present there. Hanuman waited till Ravan departed after failing to persuade her. Sita was relieved of being rid of Ravan. The sequence justifies Hanuman's remark that Sita's face looked like a "full moon released from the captivity of Rahu".

Time when the Lunar Eclipse of 12 September 5076 BC got Over



Time when the Lunar Eclipse of 12 September 5076 BC got Over



The Date and Time of Hanuman's Return from Lanka - 14 September 5076 BC

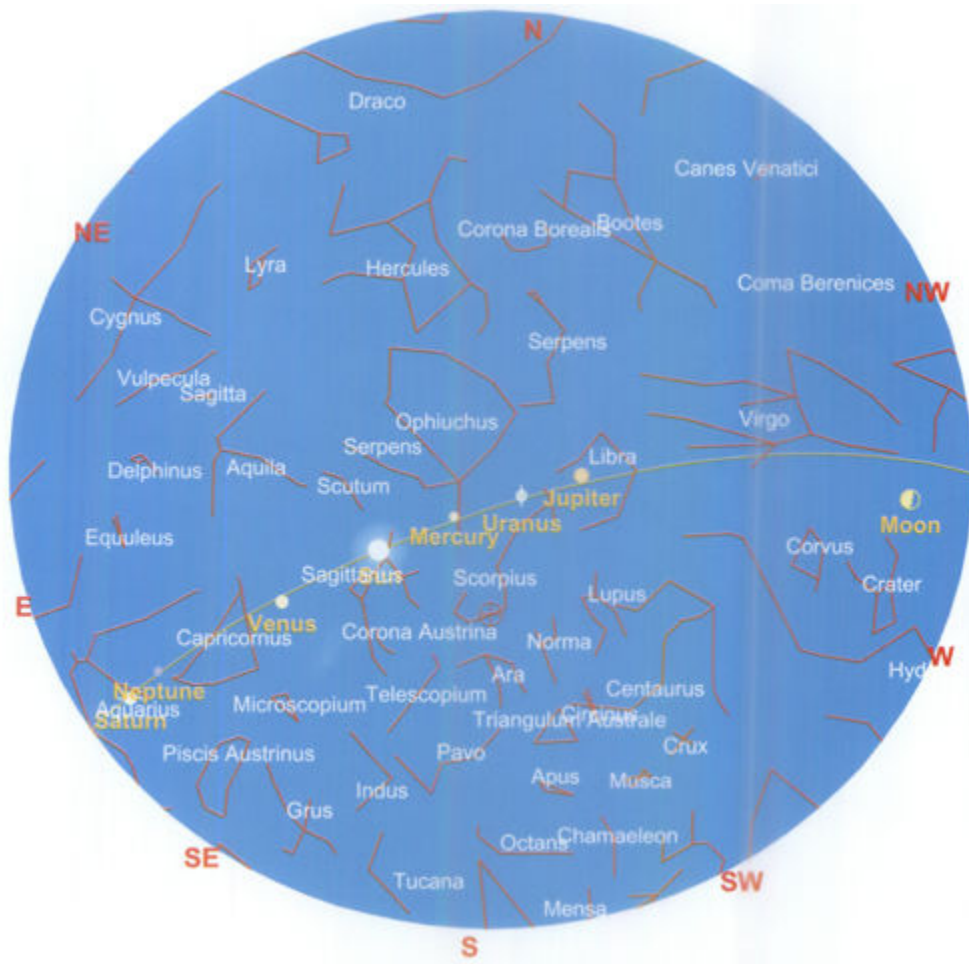


This slide shows the sky at 6.30 a.m. on 14 September 5076 BC from Sri Lanka. As described in the Ramayana, see that the sun and the moon are shining together. Punarvasu nakshatra is shining in the Gemini constellation that resembles a large fish. The moon, Mars and Jupiter are shining. Also seen are Pushya (Cancer zodiac) and Swati (Virgo zodiac). Airavat or the legendary elephant of Indra corresponds to the Scorpio zodiac, which looks like the trunk of an elephant. This should have been the time when Hanuman started his return journey.

This slide shows the sky of 14 September 5076 BC at 10 a.m. from Sri Lanka. While the previous slide showed the time of Hanuman starting his return journey, this slide shows the time when he reached the middle of the sea to rest on a small hill. The only nakshatra, which could not be displayed in the previous slide was the Shrawan in Capricornus zodiac since it rises at about 10 a.m. Note that Capricorn constellation looks like an island if the sky is taken as sea. This slide shows the time when all the objects stated in this context had risen.

SLIDE 48

The Date when Hanuman and others met Lord Ram and they all Proceeded to Lanka



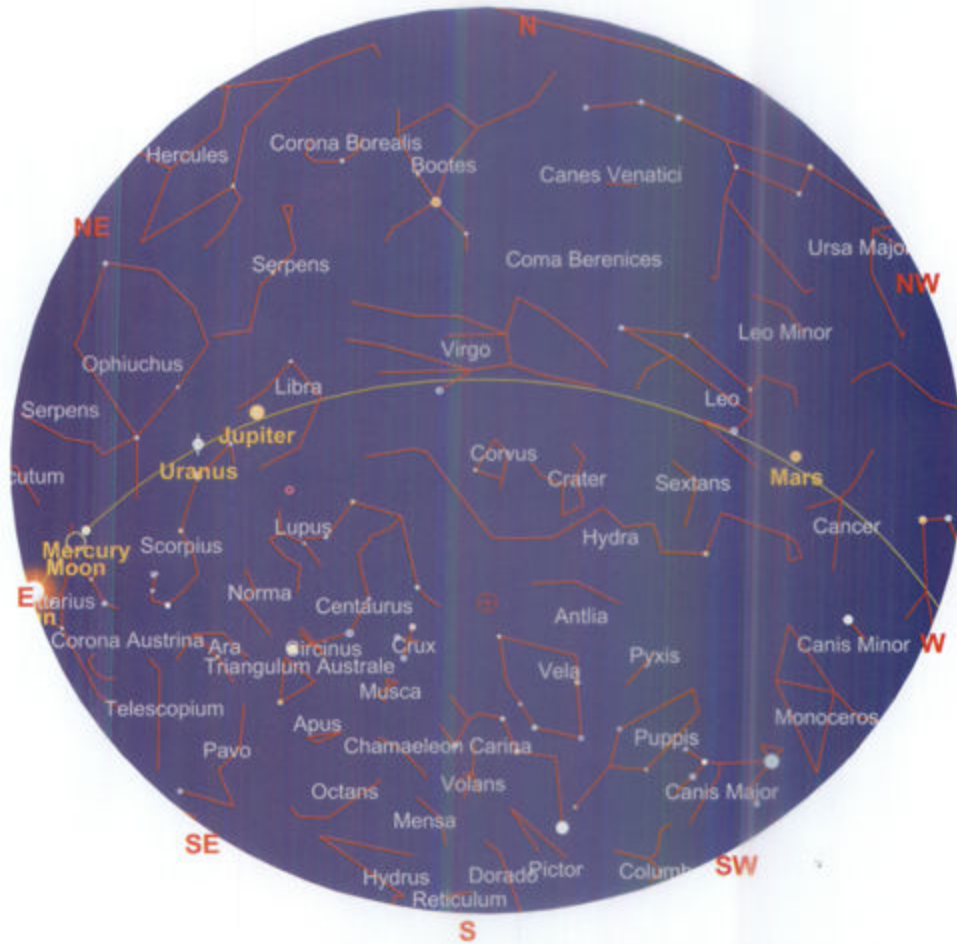
After crossing the sea, Hanuman met his companions on the shore and they all rushed to meet Lord Ram. Lord Ram decides to immediately start the journey to Lanka and said “Today is Uttar-Phalguni day. Sun has already risen above our heads and I want to proceed in this Vijay Mahurt.” This slide shows that on 19 September 5076 BC, the moon was near Uttar Phalguni nakshatra (star p in Virgo) at noon time, while sun was also at zenith. Hence, we may conclude that Hanuman met Lord Ram at around noon of that day.

SLIDE 49
The Description of the Sky by Lakshman on the way to Lanka - 20
September 5076 BC



On their way to Lanka, Lakshman said, "The Venus has gone to the back side, Sapta-rishi (Ursa Major) are visible brightly, Trishanku (Crux) is visible in front. The Moola nakshatra, protector of Rakshas, is inflicted with Dhumra-ketu (node)". See the early morning sky of 20 September 5076 BC above. Ursa Major was clearly seen. Crux, being a small constellation in the southern sky must have been seen in the front as they were moving southwards. Venus was not visible. Next slide will show if the Moola nakshatra was inflicted by Ketu (node) or not?

SLIDE 50
Authenticity of the Statement made by Lakshman - Moola Inflicted by Ketu



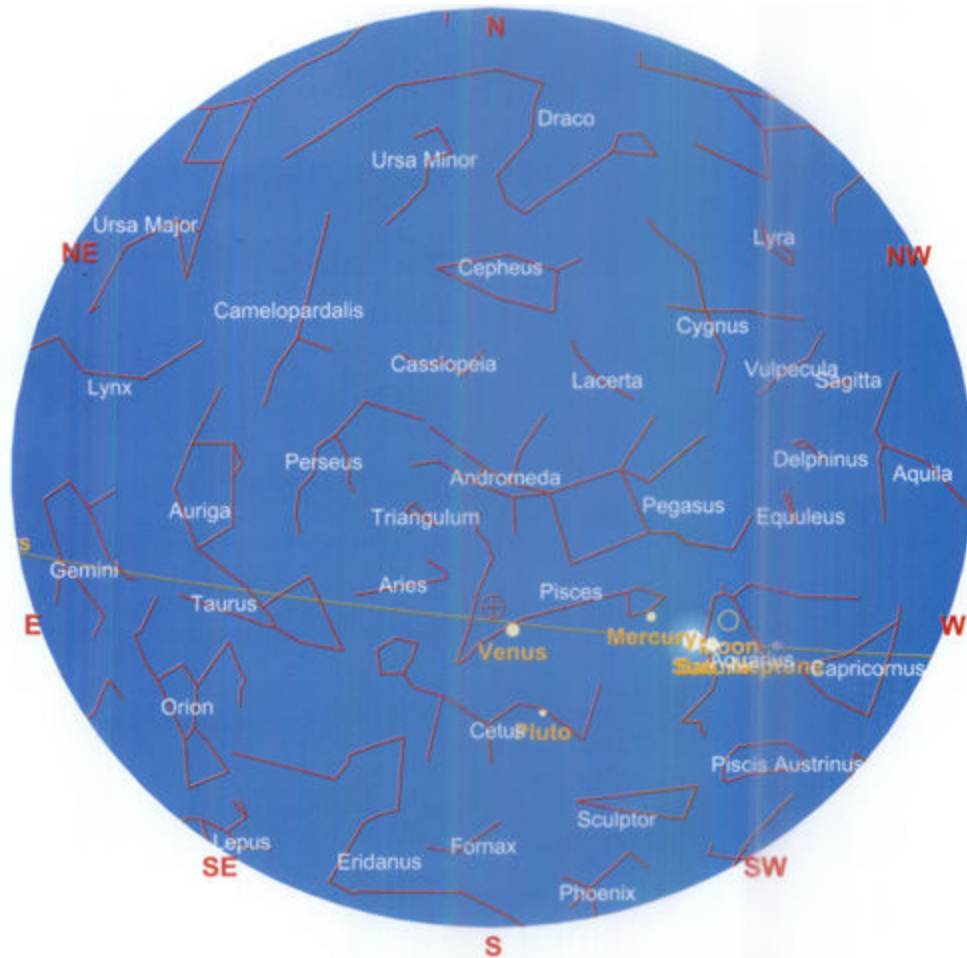
Moola nakshatra (42 6 OPH) corresponds to Sagittarius constellations on the ecliptic. Refer to Sr. No. 19 in Table 1. This slide confirms that the moon cut the ecliptic in Sagittarius during the month when the army moved towards Lanka and the sky was described by Lakshman. This slide, therefore, demonstrates that when Lakshman described the position of planetary bodies in the sky, the Moola nakshatra, was inflicted by Dhumra-Ketu. Hence, all parts of the statement made by Lakshman tallies with the actual view of the sky.

SLIDE 51
The Date of the Army Reaching Ravan's Fort after Crossing the Sea



After crossing the sea, the army proceeded towards the fort of Ravan. At 6/38/19, it is mentioned that when the army reached the fort located on the hill called 'Suvaal', the full moon night occurred. This slide confirms that it was the night of 12 October 5076 BC. Since the full moon occurs at star Pushya in Cancer constellation, this was the end of the Paus month. The battle between Ram and Ravan had not begun so far. From this day, began the fifteen-day period of waning moon (Krishna Paksh) until amavasya, after which it starts increasing again.

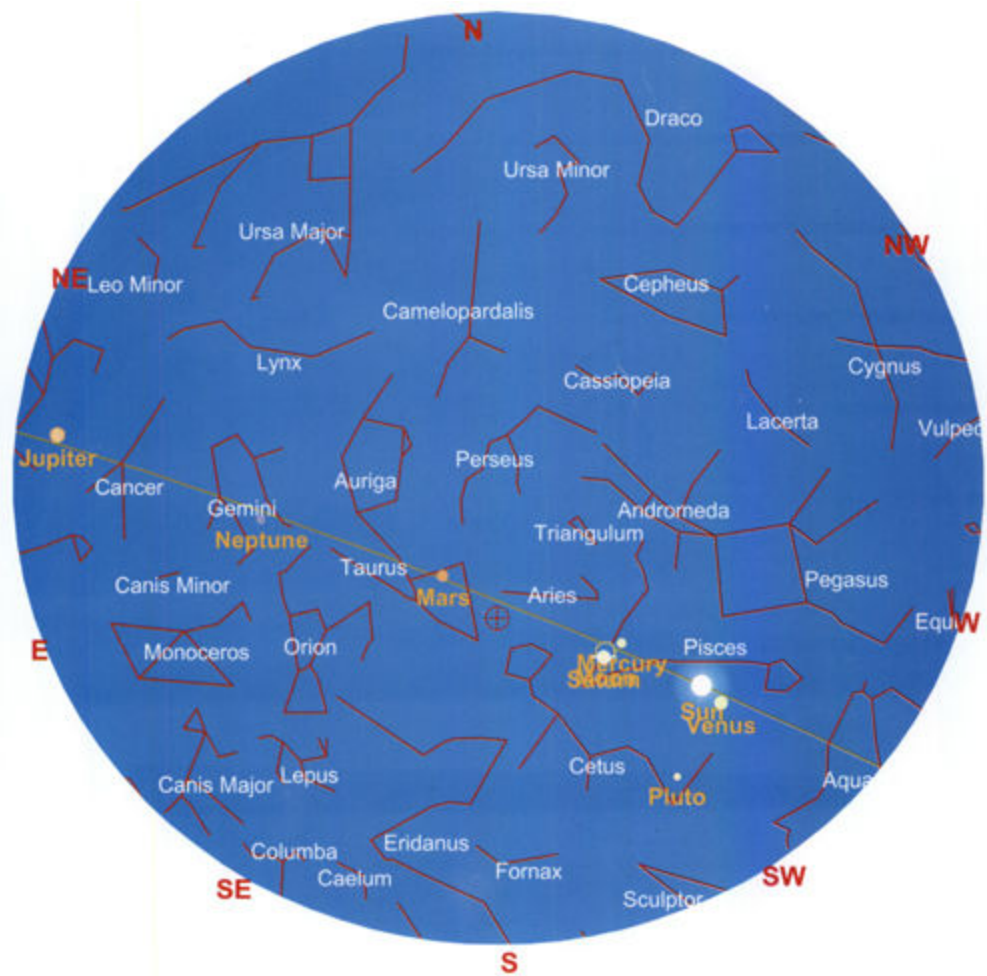
SLIDE 52
Amavasya of 24 November 5076 BC - one Day after Meghnath was Killed



The Ramayana mentions that Lakshman had killed Meghnath, after a battle that continued for three days, on the 14th day of the decreasing phase of moon. The 14th day of the waning moon phase means just one day before the no-moon day. This slide shows that the no-moon day was on 24 November 5076 BC. It implies that Meghnath was killed on 23 November that year. Further, as he was killed after a battle that was fought for three days and three nights, the battle must have begun on 21 November 5076 BC.

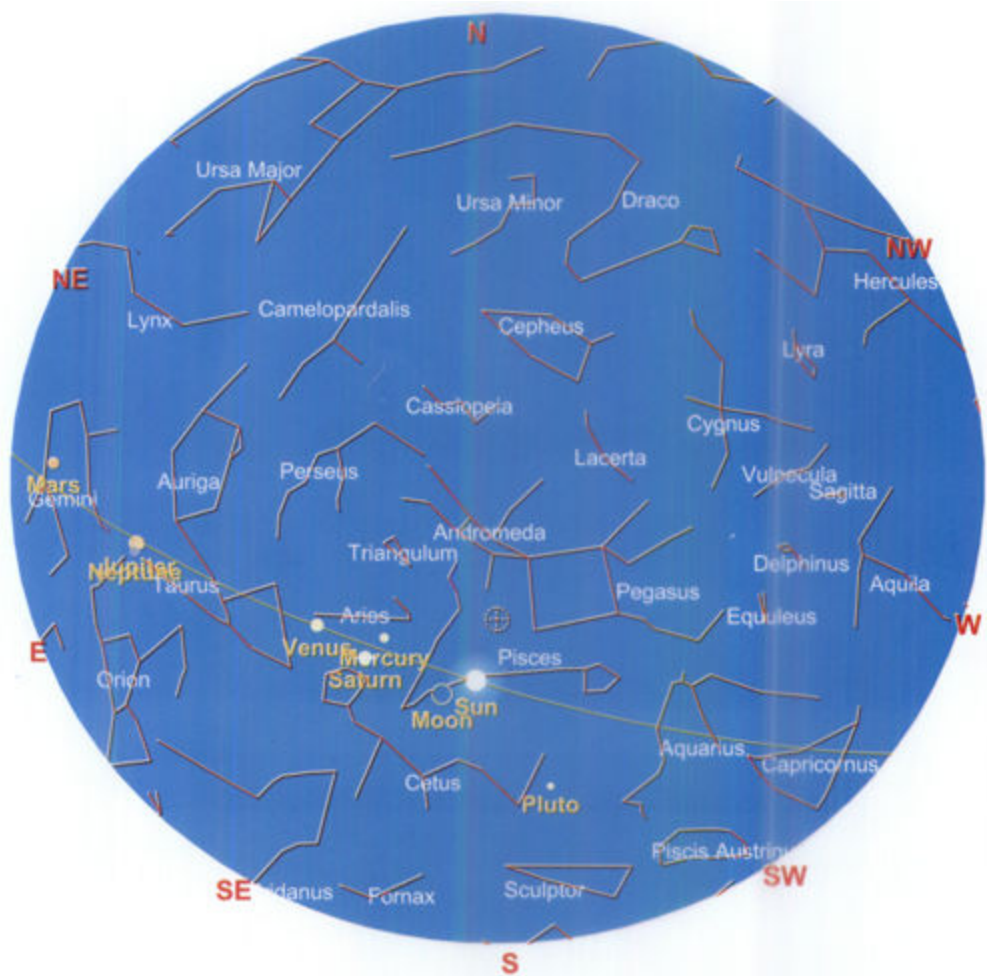
SLIDE 53

The Beginning of the Kali Yuga - Conjunction of Seven Planets at Revati - 1389 BC



The Kali Yuga began when all the seven planets were around the Revati nakshatra (Zeta in Pisces). A conjunction involving the sun and the moon means an amavasya day. This slide shows the sky of 9 February 1389 BC, when the sun and the moon were nearest possible to Revati. A few days later, the sun came closer to Revati, but by then the moon had moved away. At this time only five planets, Sun, Moon, Mercury, Venus and Saturn were in Pisces. Mars was in Taurus and Jupiter was in Leo. Hence, this was not a "conjunction of seven planets".

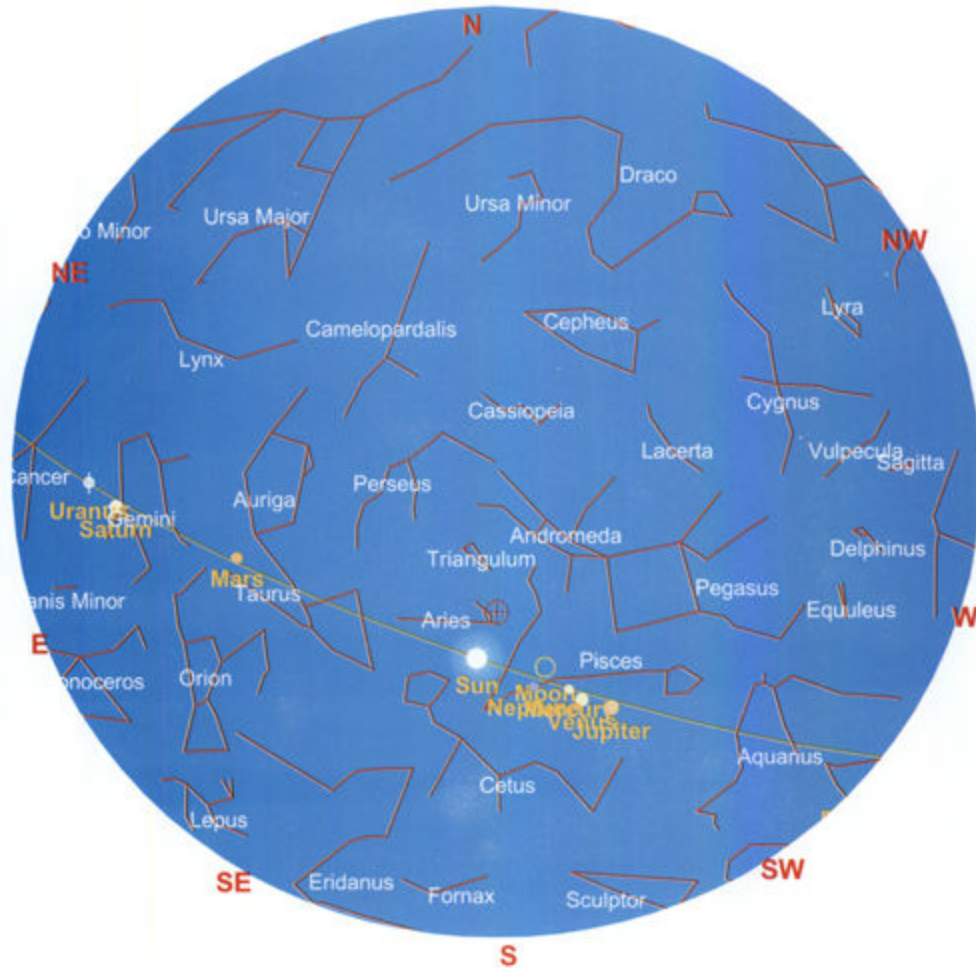
SLIDE 54
The Beginning of the Kali Yuga - Conjunction of Seven Planets at
Revati - 1389 BC



The beginning of the Kali Yuga was marked by the conjunction of all major seven planets on the Revati nakshatra (Zeta in Pisces). This slide shows the sky view on 11 February 1889 BC, which was the no-moon day. As may be noted, it is again a conjunction of only five planets since Jupiter was in Taurus and Mars was seen in Gemini. Obviously, this planetary configuration cannot be classified as a "conjunction of seven planets";.

SLIDE 55

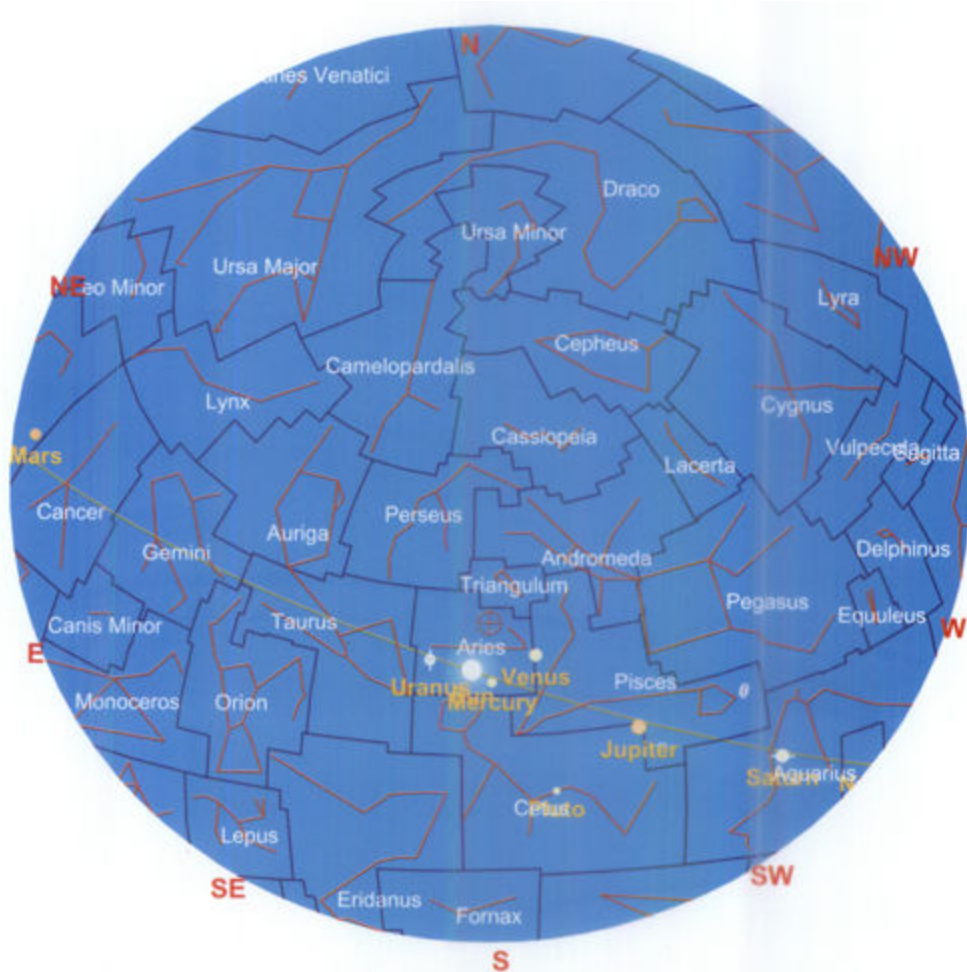
The Beginning of the Kali Yuga - Conjunction of Seven Planets at Revati - 2414 BC



The Kali Yuga began when all the major seven planets were around the Revati nakshatra (Zeta in Piscium). This slide shows the view of the sky on 20 February 2414 BC, the date when the sun could be nearest to the Revati nakshatra on a no-moon day because for the other days when the sun comes very near to Revati, the moon is very far away from the sun. Note that this was also a conjunction of only five planets. Mars was in Taurus and Saturn was in Gemini. Hence, this was also not a "conjunction of seven planets";.

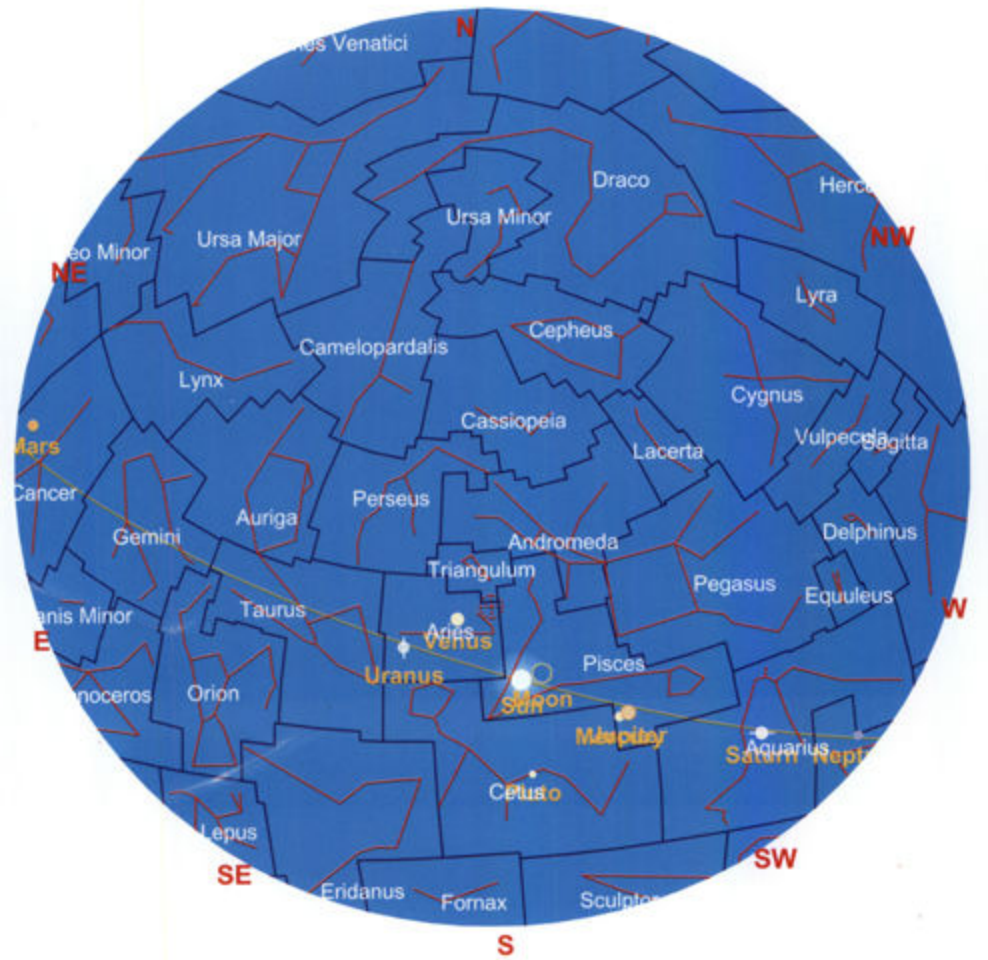
SLIDE 56

The Beginning of the Kali Yuga as per Aryabhatta - 18 February 3102 BC



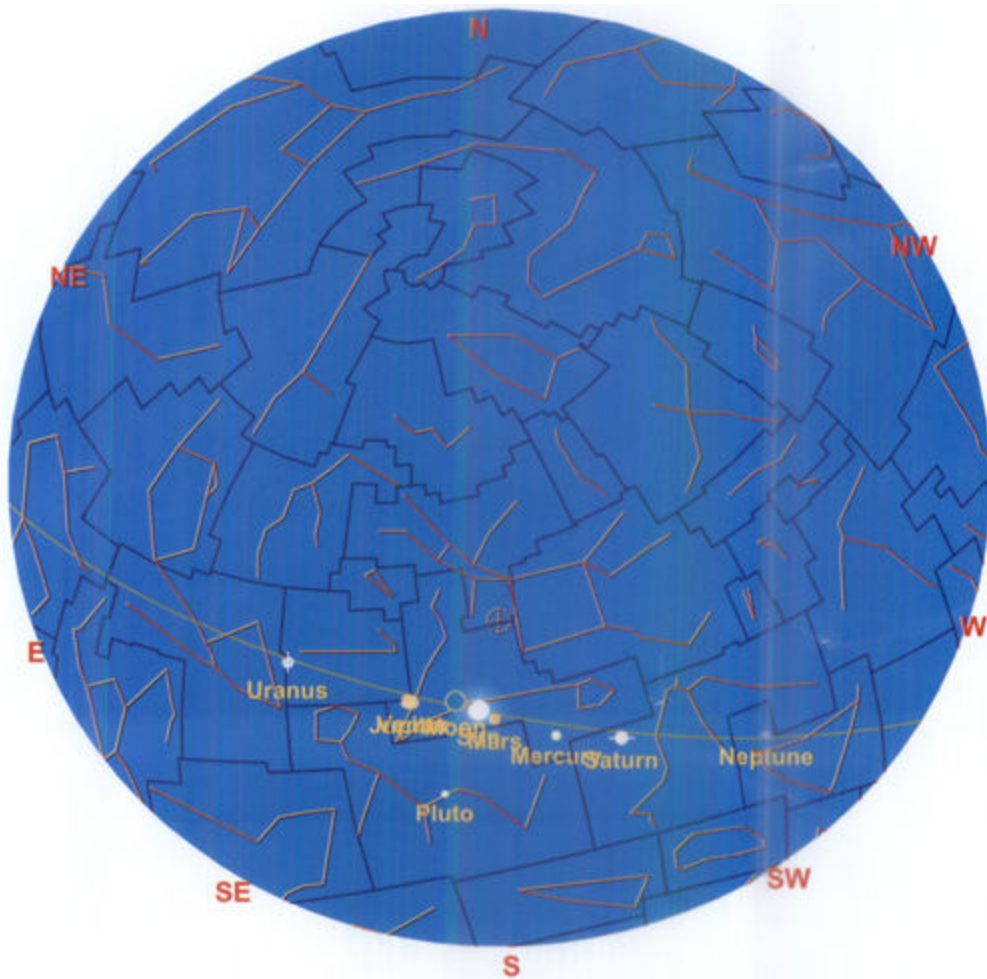
According to the famous astronomer Aryabhatta, Kali Yuga began in 3102 B (generally accepted date being 18 February). This slide shows the position of planets on 18 February 3102 BC. As may be seen, while the sun is in the middle of the constellation of Aries, Saturn is in Aquarius, Mars is too far away in Cancer/Leo and the moon is not even near by. Obviously, it was not an amavasya (no-moon) day, otherwise the moon would have been near the sun. Actually, the no-moon day, that year was on 1 February as shown in the next slide.

SLIDE 57
The Beginning of the Kali Yuga as per Aryabhatta - 1 February
3102 BC



The date of the beginning of Kali Yuga as worked out by the famous astronomer Aryabhatta was 3102 BC. The previous slide showed that the generally accepted date of 18 February was not an amavasya night. This slide shows the position of sky on 1 February 3102 BC, which was the no-moon day. Though very far apart, six planets can still be termed as "around Revati",. However, Mars is clearly very far away in Cancer constellation and therefore, there will be a hesitation in accepting this as a "conjunction of all the main seven planets".

SLIDE 58
The Beginning of the Kali Yuga - Conjunction of Seven Planets at
Revati - 1389 BC



This slide shows the sky on the no-moon day of 3101 BC when the sun was at Revati. This is one year before 3102 BC, which was the beginning of the Kali Yuga as per Aryabhata. Note that all the seven planets are located around the Revati nakshatra. Of all the dates examined so far, this appears to be the best candidate for a seven-planet conjunction at Revati. We may conclude that the no-moon day, when the sun and the moon are at Revati implies amavasya of Chaitra, and that marks the end of the lunar year.

SLIDE 59

Confirmation that the Sun and the Moon on Revati means End of Chaitra Month



For the no-moon (amavasya) day of 22 January 3101 BC, the corresponding full-moon day comes on 6 February 3101 BC, since the full-moon occurs at the Chitra nakshatra on this day as shown in this slide. This implies that the month, which ended on 6 February 3101 BC was the month of Chaitra and confirms that the no-moon day of 22 January 3101 BC was the amavasya of Chaitra, which is also the last day of the lunar year. If this was the end of a yuga, it means that the yuga was a cycle for counting years as both these time periods end together.

Sage Valmiki, a contemporary of Lord Ram, mentioned in his magnum opus *Ramayana* that when Lord Ram was born, the sun was located in Aries, Saturn was in Libra, Jupiter and the moon were shining in Cancer, Venus was seen in Pisces while Mars was located in Capricorn. Moreover, it was the ninth day of the increasing phase of the moon in the lunar month of Chaitra.



The two illustrations on the cover display, with the help of a powerful software, that these unique astral conditions were present in the sky on January 10, 5114 BC. This path-breaking book employs the scientific advancement of the West to prove the antiquity of the East. Not only the date of birth, the book also discovers the dates of a number of events in the life of Lord Ram by using the Planetarium software.

Rupa & Co